
Internet of Things (IoT) - Demystified

Keynote – ICIT 2017
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Talk - Outline

- Motivations for IoT
- Selected Components of IoT
- Selected Applications of IoT
- Driving Technologies of IoT
- Challenges and Research in IoT
- IoT Design Flow
- Tools and Solutions for IoT
- Related Buzzwords of IoT
- Conclusions and Future Directions

Population Trend – Urban Migration

“India is to be found not in its few cities, but in its 700,000 villages.”
- Mahatma Gandhi

- 2025: 60% of world population will be urban
- 2050: 70% of world population will be urban



Source: <http://www.urbangateway.org>

Human Migration Problem

- Uncontrolled growth of urban population
- Limited natural and man-made resources



Source: <https://humanitycollege.org>

Smart Cities - A Solution

- Smart Cities: For effective management of limited resource to serve largest possible population to improve:
 - Livability
 - Workability
 - Sustainability

“Cities around the world could spend as much as \$41 trillion on smart tech over the next 20 years.”

Source: <http://www.cnn.com/2016/10/25/spending-on-smart-cities-around-the-world-could-reach-41-trillion.html>



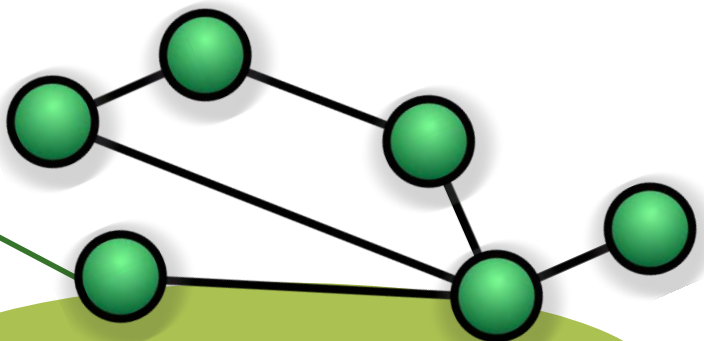
Smart Cities - 3 Is

Instrumentation

The 3Is are provided by the Internet of Things (IoT).

Smart Cities

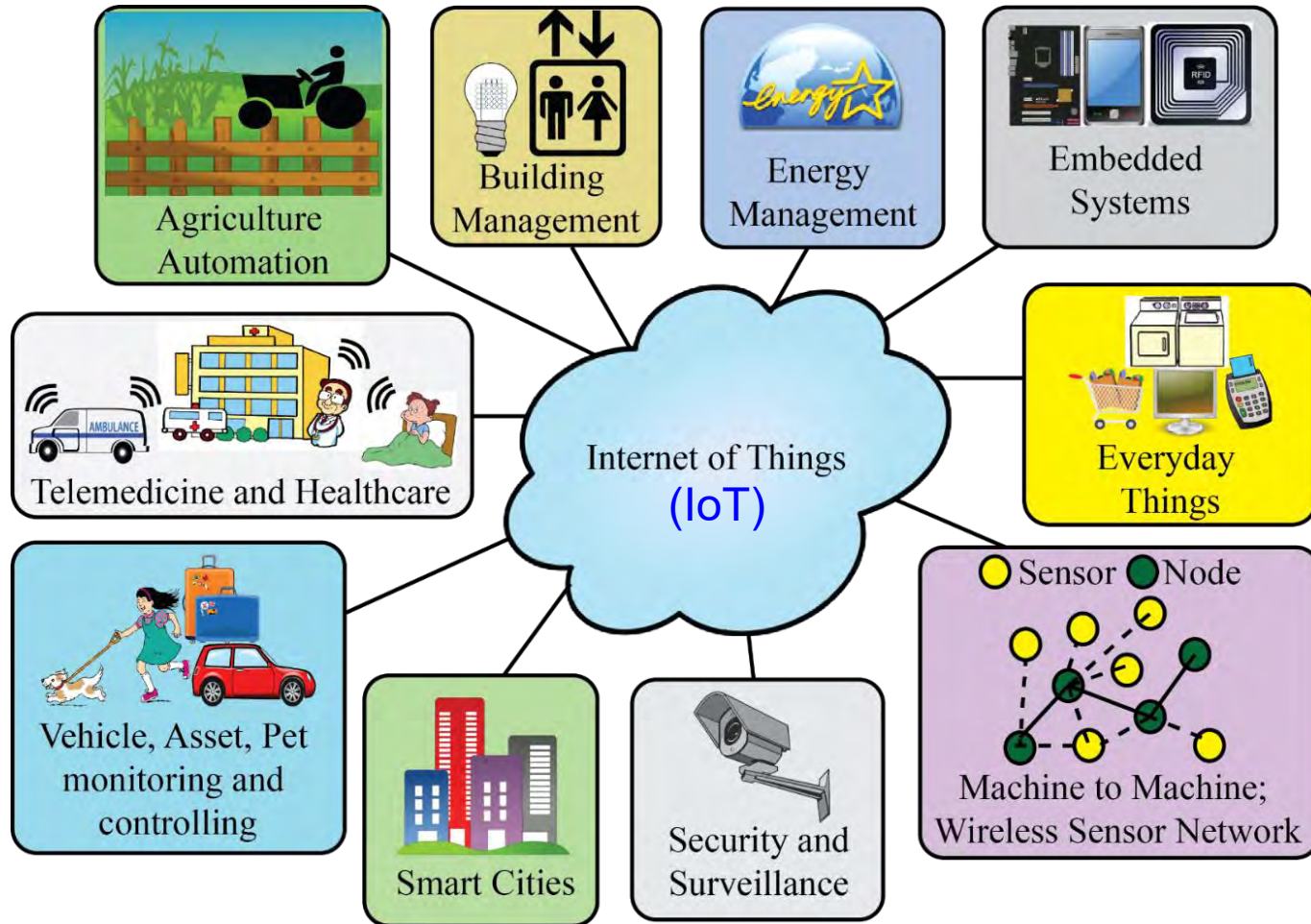
Intelligence



Interconnection

Source: Mohanty 2016, EuroSimE 2016 Keynote Presentation

IoT is the Backbone Smart Cities



Source: Mohanty 2016, CE Magazine July 2016

Internet of Things (IoT) - History



1969

The Internet Emerges

The first nodes of what would eventually become known as ARPANET, the precursor to today's Internet, are established at UCLA and Stanford universities.



1982

TCP/IP Takes Shape

Internet Protocol (TCP/IP) becomes a standard, ushering in a worldwide network of fully interconnected networks called the Internet.



1990

A Thing Is Born

John Romkey and Simon Hackett create the world's first connected device (other than a computer): a toaster powered through the Internet.



1999

The IoT Gets a Name

Kevin Ashton coins the term "Internet of things" and establishes MIT's Auto-ID Center, a global research network of academic laboratories focused on RFID and the IoT.



2005

Getting Global Attention

The United Nations first mentions IoT in an International Telecommunications Union report. Three years later, the first international IoT conference takes place in Zurich.



2008

Connections Count

The IPSO Alliance is formed to promote IP connections across networks of "smart objects." The alliance now boasts more than 50 member firms.



2011

IPv6 Launches

The protocol expands the number of objects that can connect to the Internet by introducing 340 undecillion IP addresses (2¹²⁸).



2013

Google Raises the Glass

Google Glass, controlled through voice recognition software and a touchpad built into the device, is released to developers.



2014

Apple Takes a Bite

Apple announces HealthKit and HomeKit, two health and home automation developments. The firm's iBeacon advances context and geolocation services.

Source: <http://events.linuxfoundation.org/sites/events/files/slides/Design%20-%20End-to-End%20IoT%20Solution%20-%20Shivakumar%20Mathapathi.pdf>

Components



IoT – Definition - IoT European Research Cluster (IERC)

A dynamic global network infrastructure

with self configuring capabilities

based on standard and interoperable communication protocols

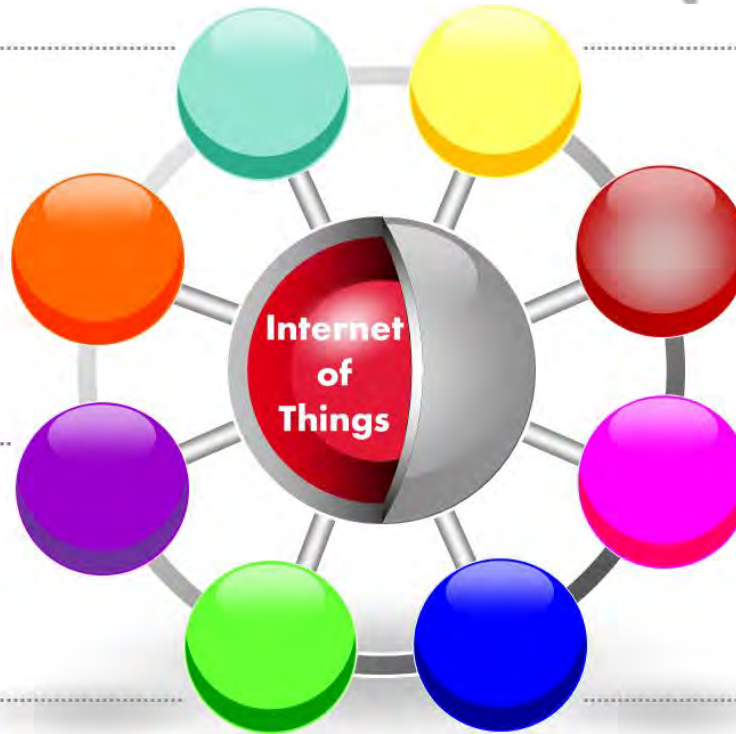
where physical and virtual “things”

have identities, physical attributes, and virtual personalities and

use intelligent interfaces,

and are seamlessly integrated

into the information network.

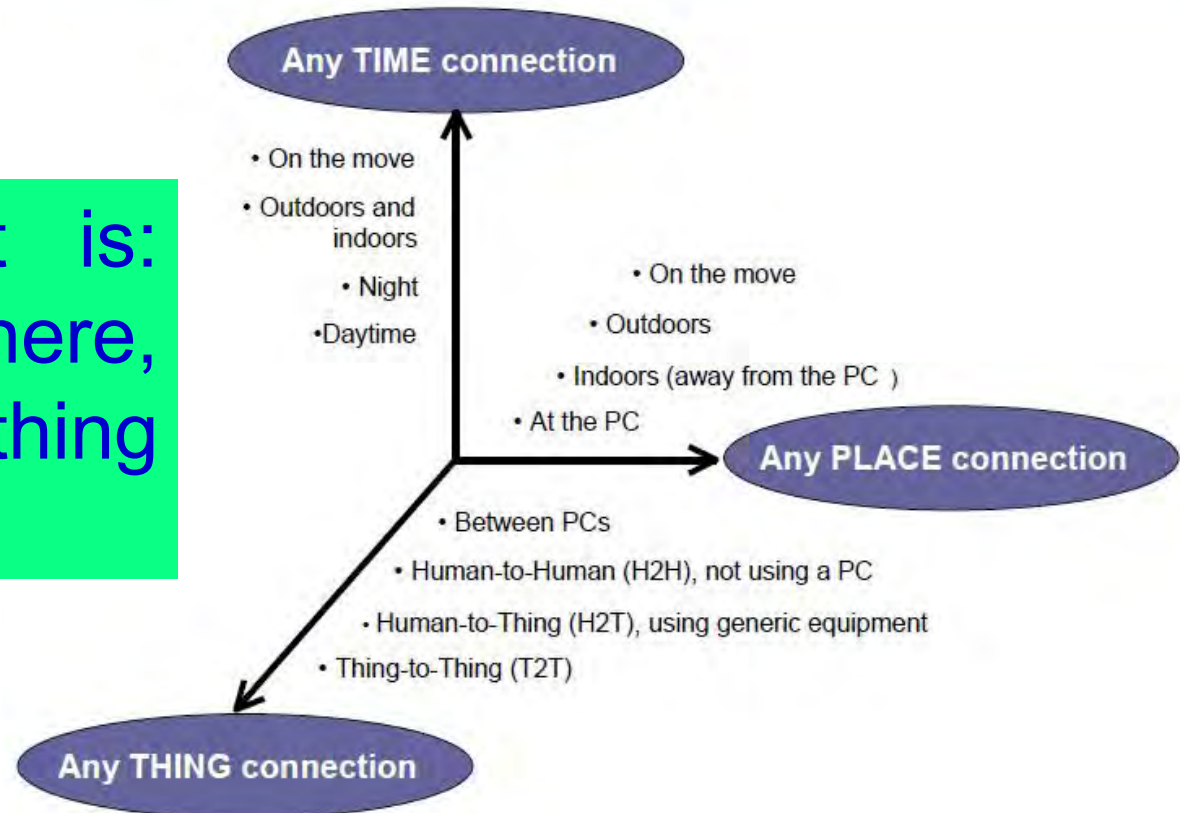


Source: http://iot.ieee.org/images/files/pdf/IEEE_IoT_Towards_Definition_Internet_of_Things_Revision1_27MAY15.pdf

IEEE also provides a formal, comprehensive definition of IoT.

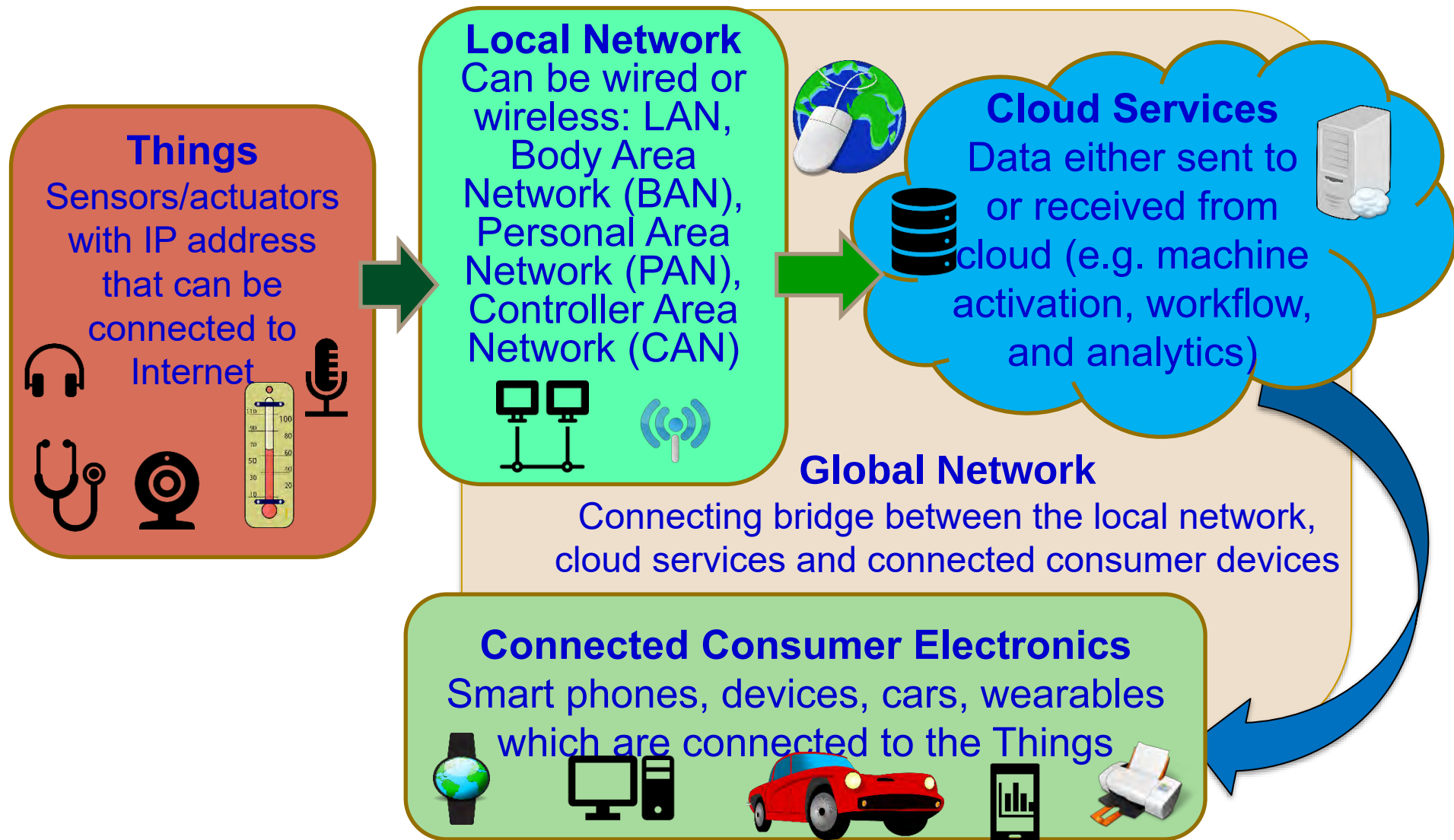
IoT – Definition - International Telecommunication Union (ITU)

A network that is:
“Available anywhere,
anytime, by anything
and anyone.”

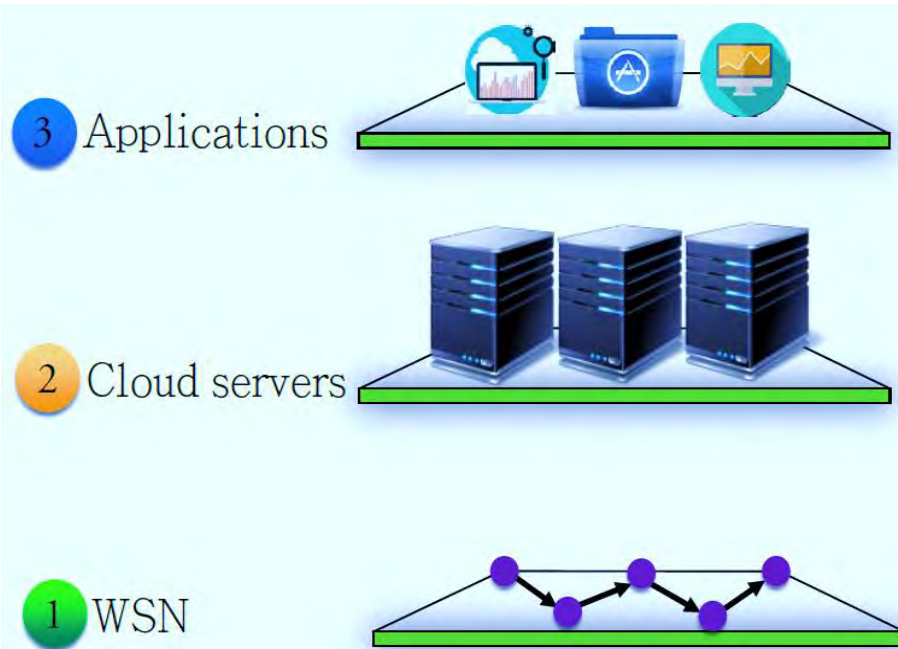


Source: http://iot.ieee.org/images/files/pdf/IEEE_IoT_Towards_Definition_Internet_of_Things_Revision1_27MAY15.pdf

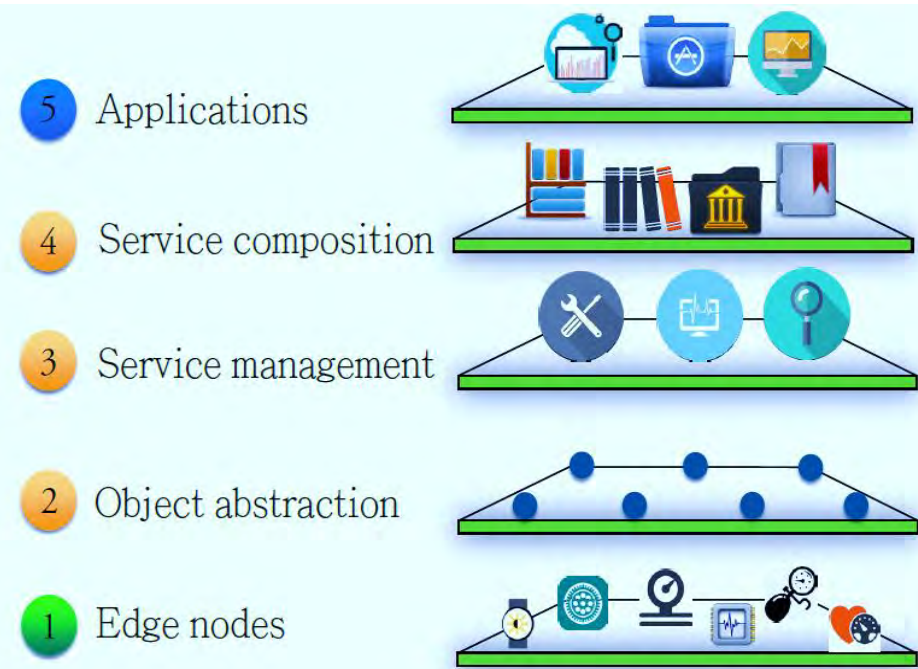
Internet of Things (IoT) – Concept



IoT Architecture - 3 & 5 Level Model



Three Level Model

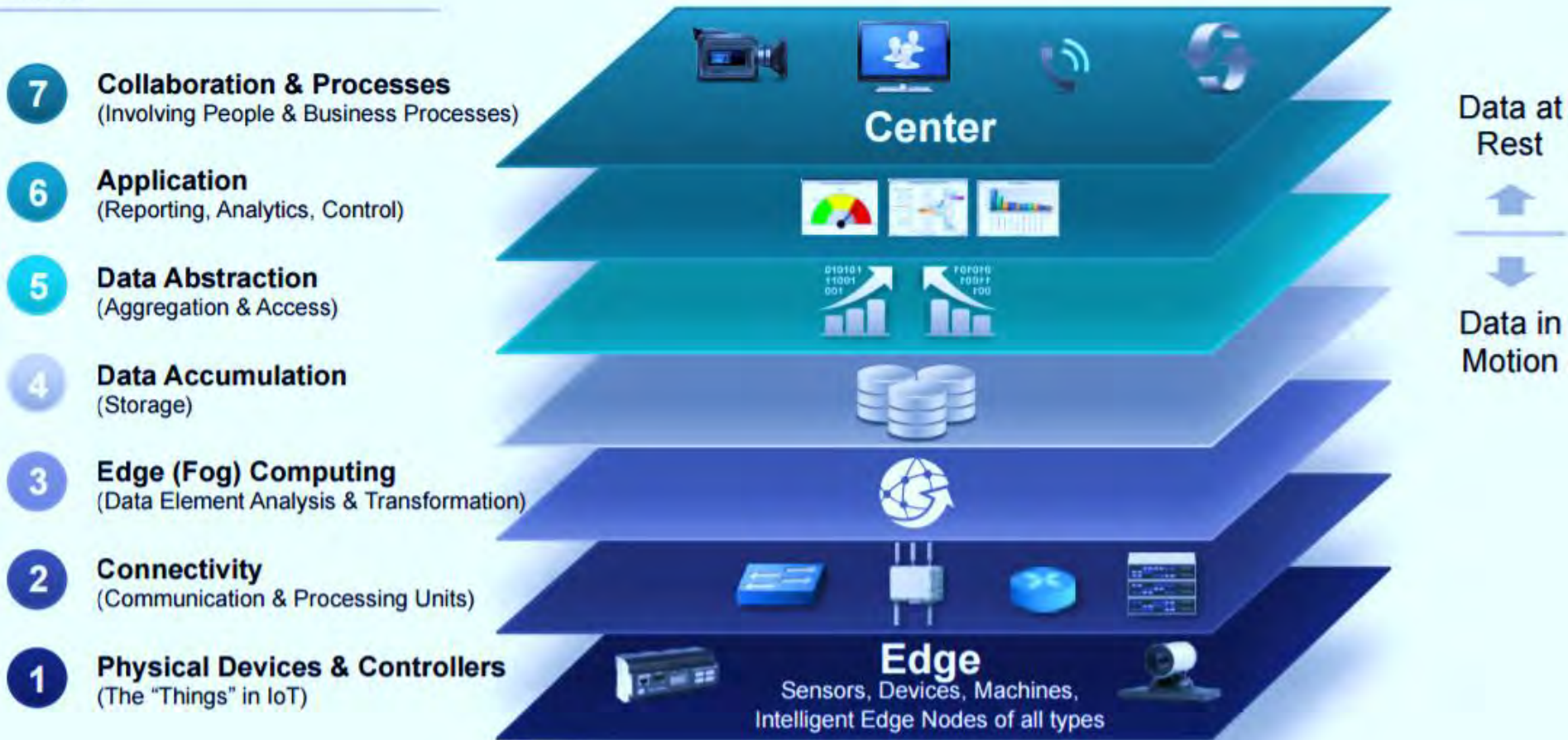


Five Level Model

Source: Nia 2017, IEEE TETC 2017

IoT Architecture - 7 Level Model

Levels

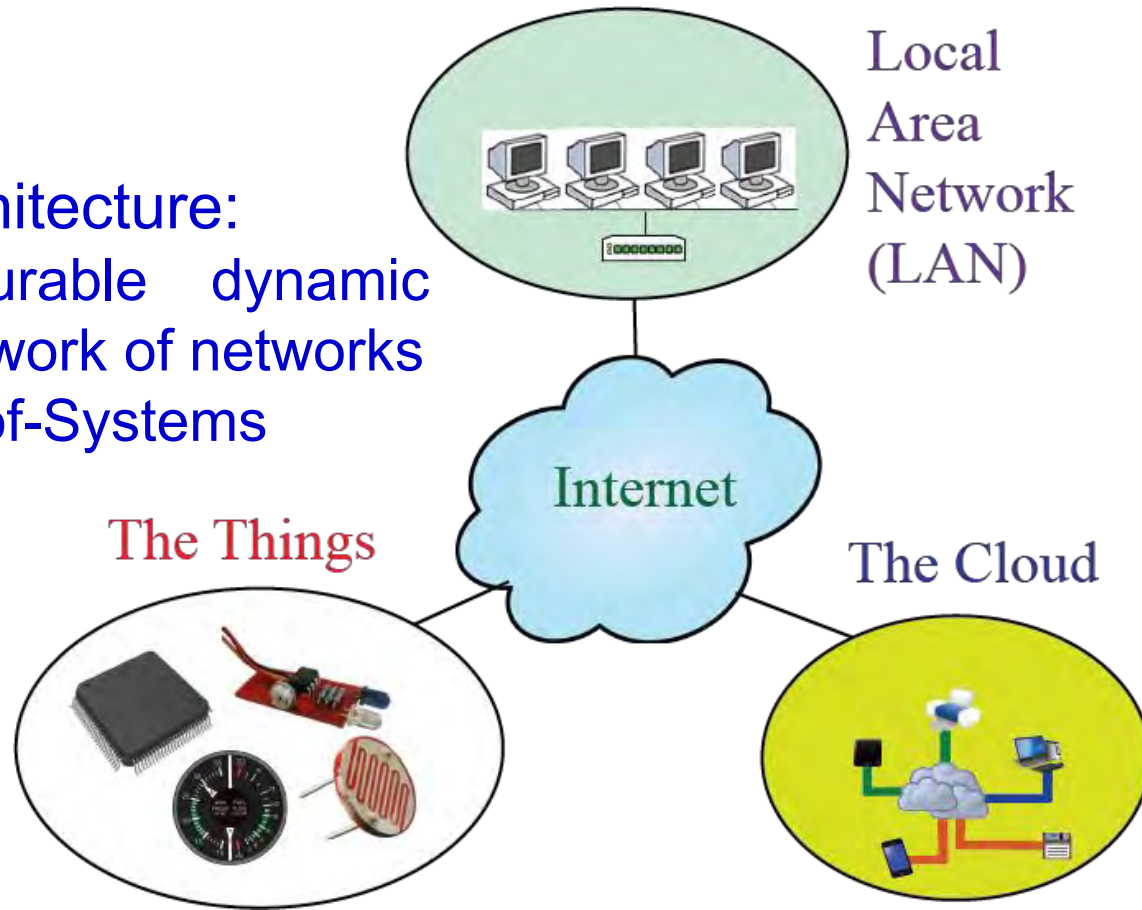


Source: http://cdn.iotwf.com/resources/71/IoT_Reference_Model_White_Paper_June_4_2014.pdf

IoT - Architecture

Overall architecture:

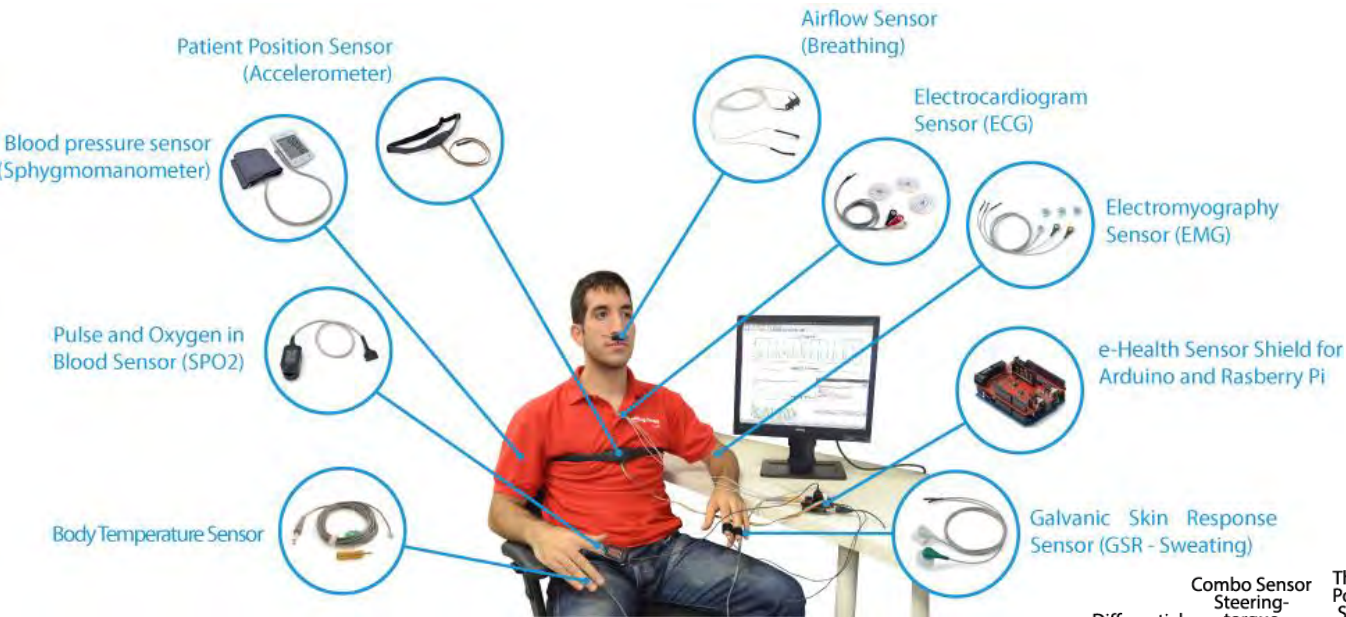
- ❖ A configurable dynamic global network of networks
- ❖ Systems-of-Systems



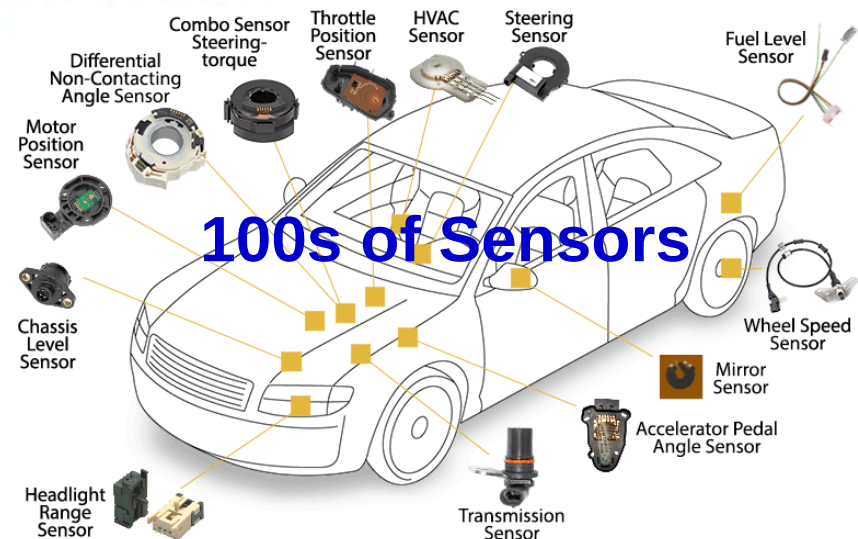
Four Main Components of IoT.

Source: Mohanty 2016, EuroSimE 2016 Keynote Presentation

IoT – Sensors

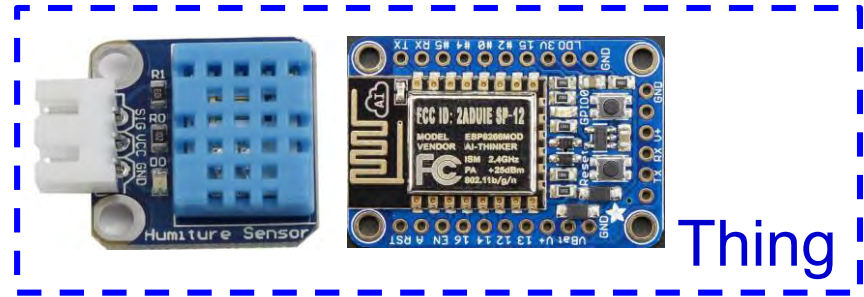
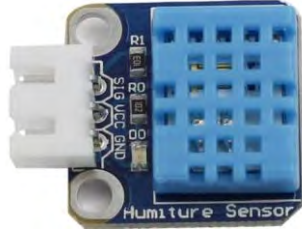


Source: <http://www.libelium.com/e-health-low-cost-sensors-for-early-detection-of-childhood-disease-inspire-project-hope/>



IoT – Things

Sensor



Thing



Sensors + Device with its own IP address → Things

IP Address for Internet Connection

The “Things” refer to any physical object with a device that has its own IP address and can connect and send/receive data via network.

IoT - Communications

Selected IoT Communications Technology

Bluetooth Low-Energy (BLE)



Zigbee



Z-Wave



6LoWPAN

6LoWPAN

Thread



WiFi



Cellular



NFC



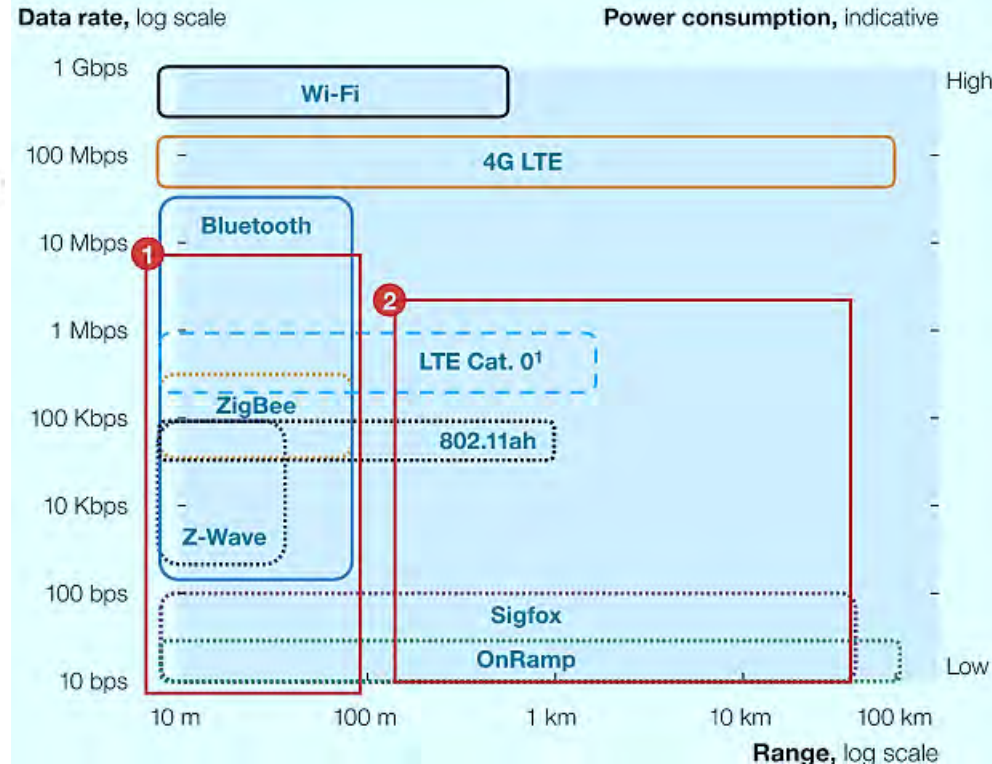
Sigfox



Neul



LoRaWAN



Source: <https://www.postscapes.com/internet-of-things-protocols/>

Source: <https://www.rs-online.com/designspark/eleven-internet-of-things-protocols-you-need-to-know-about>

IoT - Cloud

Servers

Virtual Desktop

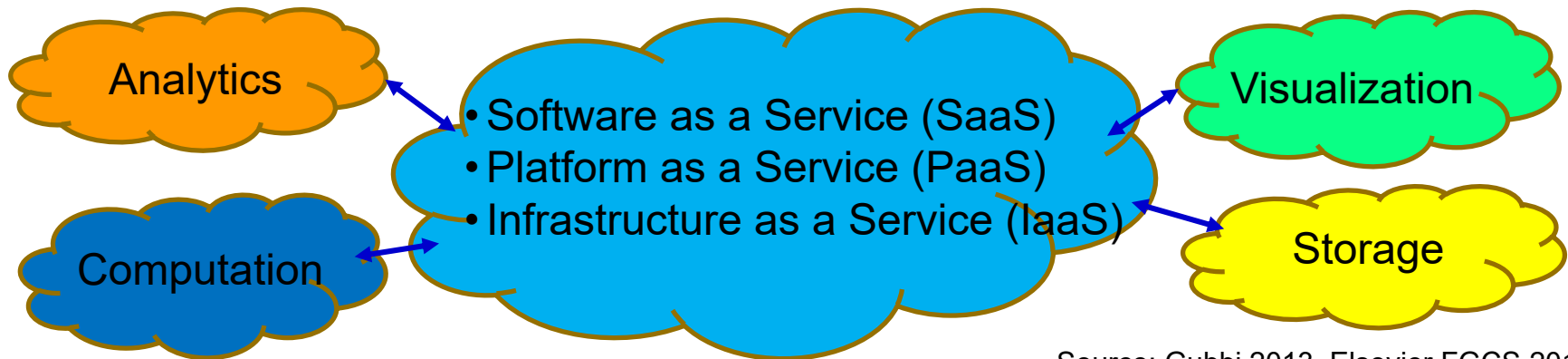
Software Platform

Applications

Storage / Data



Source: https://www.livewireindia.com/cloud_computing_training.php

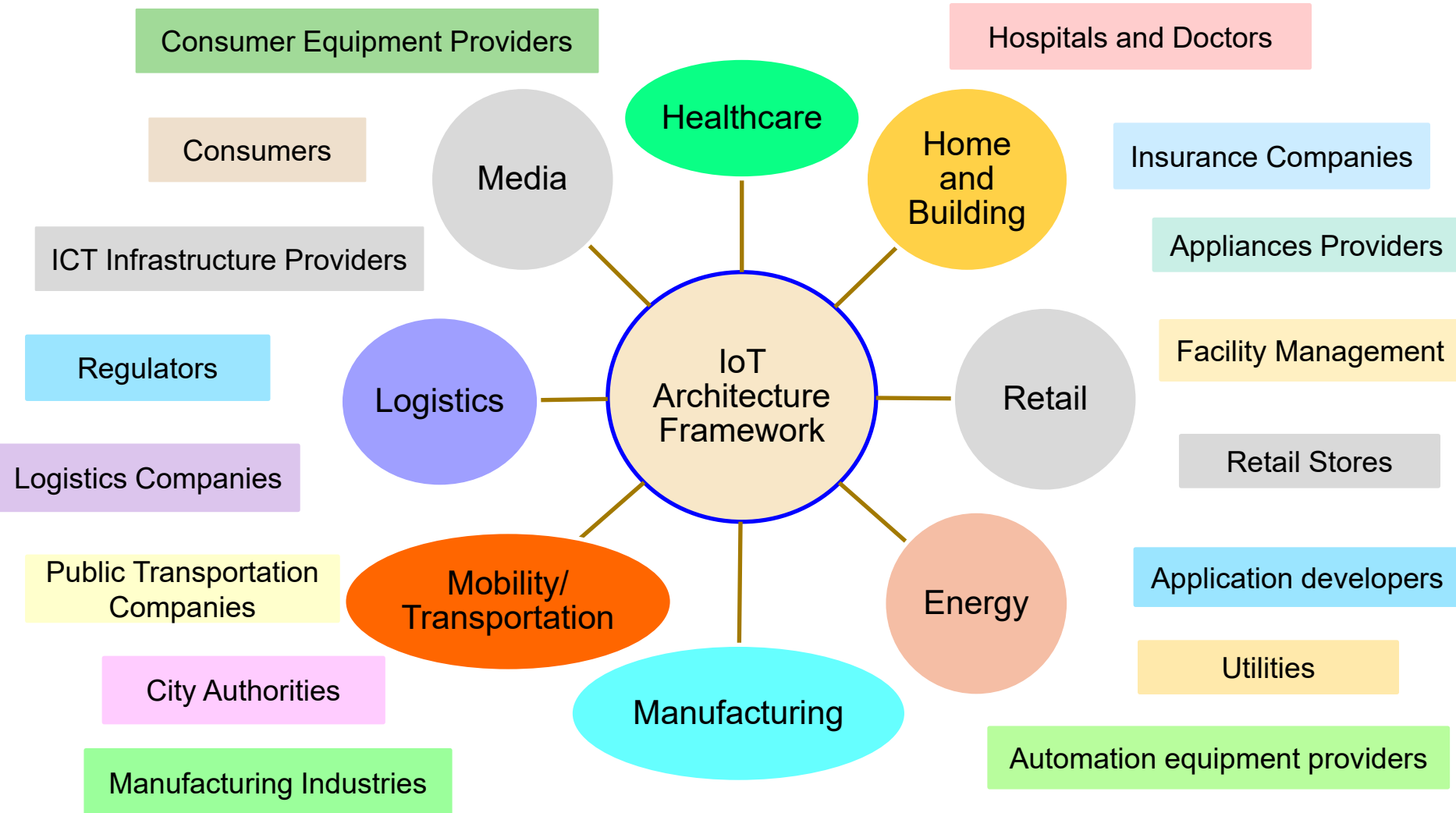


Source: Gubbi 2013, Elsevier FGCS 2013

IoT - Applications



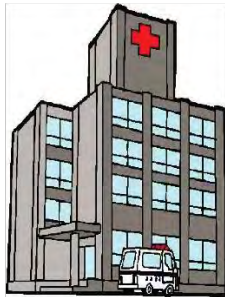
IoT - Markets and Stakeholders



Source: http://iot.ieee.org/images/files/pdf/IEEE_IoT_Towards_Definition_Internet_of_Things_Revision1_27MAY15.pdf

IoT in Smart Healthcare

Smart Hospital



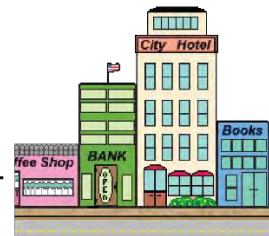
Emergency Response



Smart Home



Smart Infrastructure



Nurse



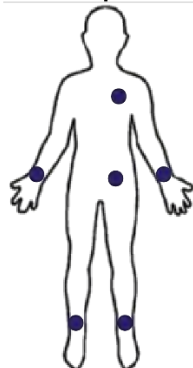
Doctor



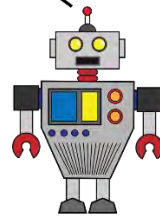
Technician



On-body Sensors



Robots



Smart Gadgets



Fitness Trackers

IoT Role Includes:

- Real-time monitoring
- Better emergency response
- Easy access of patient data
- Connectivity among stake holders
- Remote access to healthcare

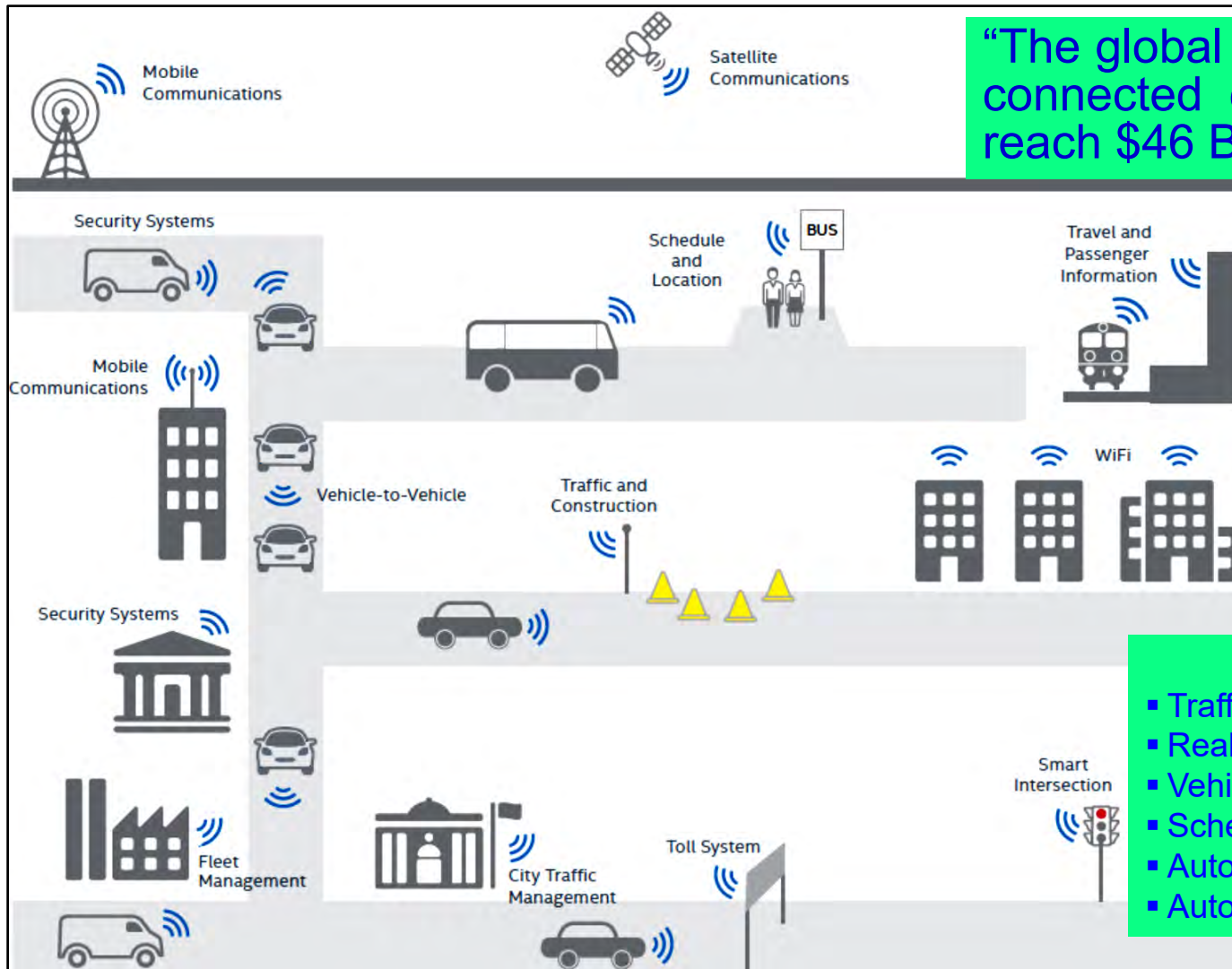
"\$117 Billion Market For IoT in Healthcare By 2020."

<https://www.forbes.com/sites/tjmccue/2015/04/22/117-billion-market-for-internet-of-things-in-healthcare-by-2020/>

Quality and sustainable healthcare with limited resources, anywhere, anytime.

Source: Mohanty 2016, CE Magazine July 2016

IoT in Smart Transportation



“The global market of IoT based connected cars is expected to reach \$46 Billion by 2020.”

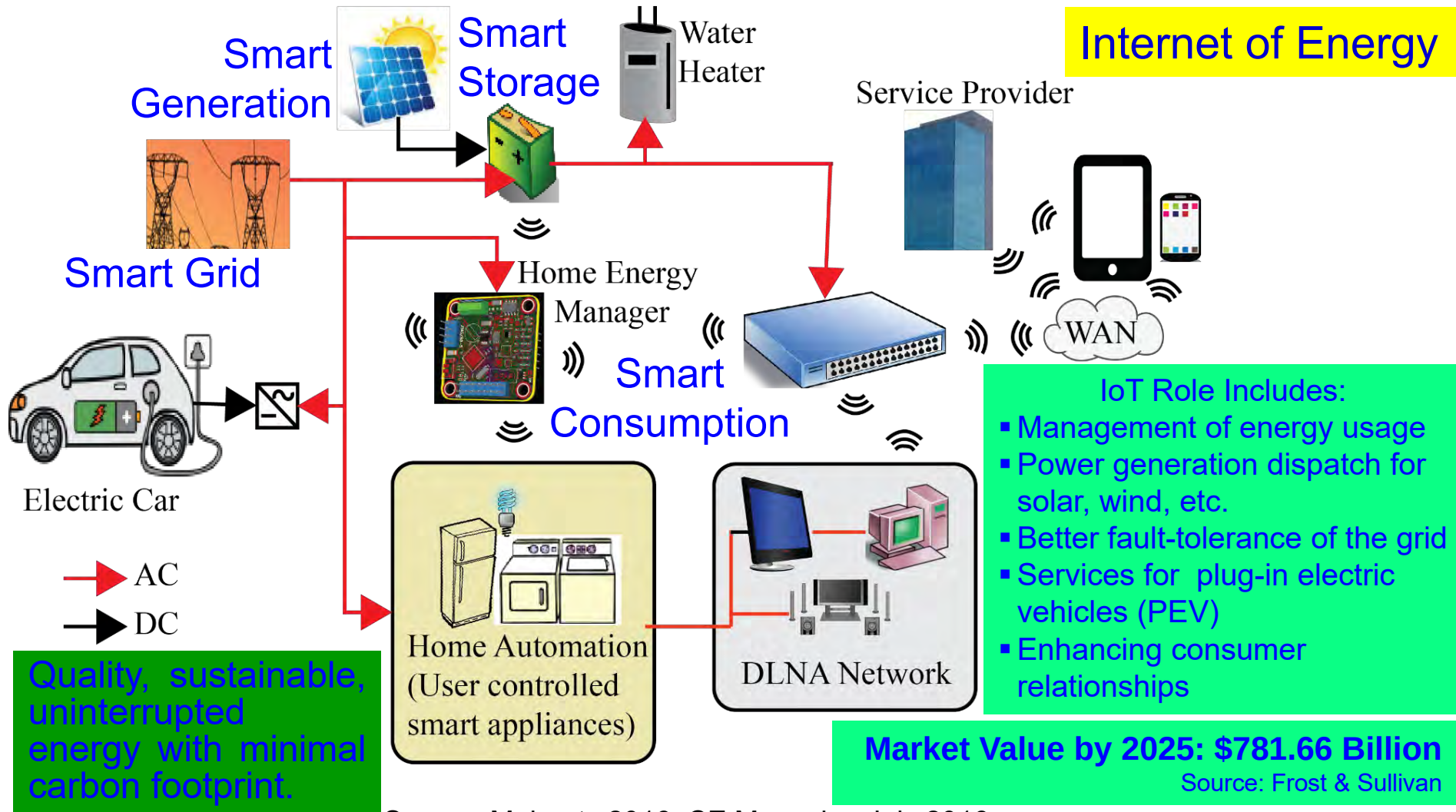
Source: Datta 2017,
CE Magazine Oct 2017

IoT Role Includes:

- Traffic management
- Real-time vehicle tracking
- Vehicle-to-Vehicle communication
- Scheduling of train, aircraft
- Automatic payment/ticket system
- Automatic toll collection

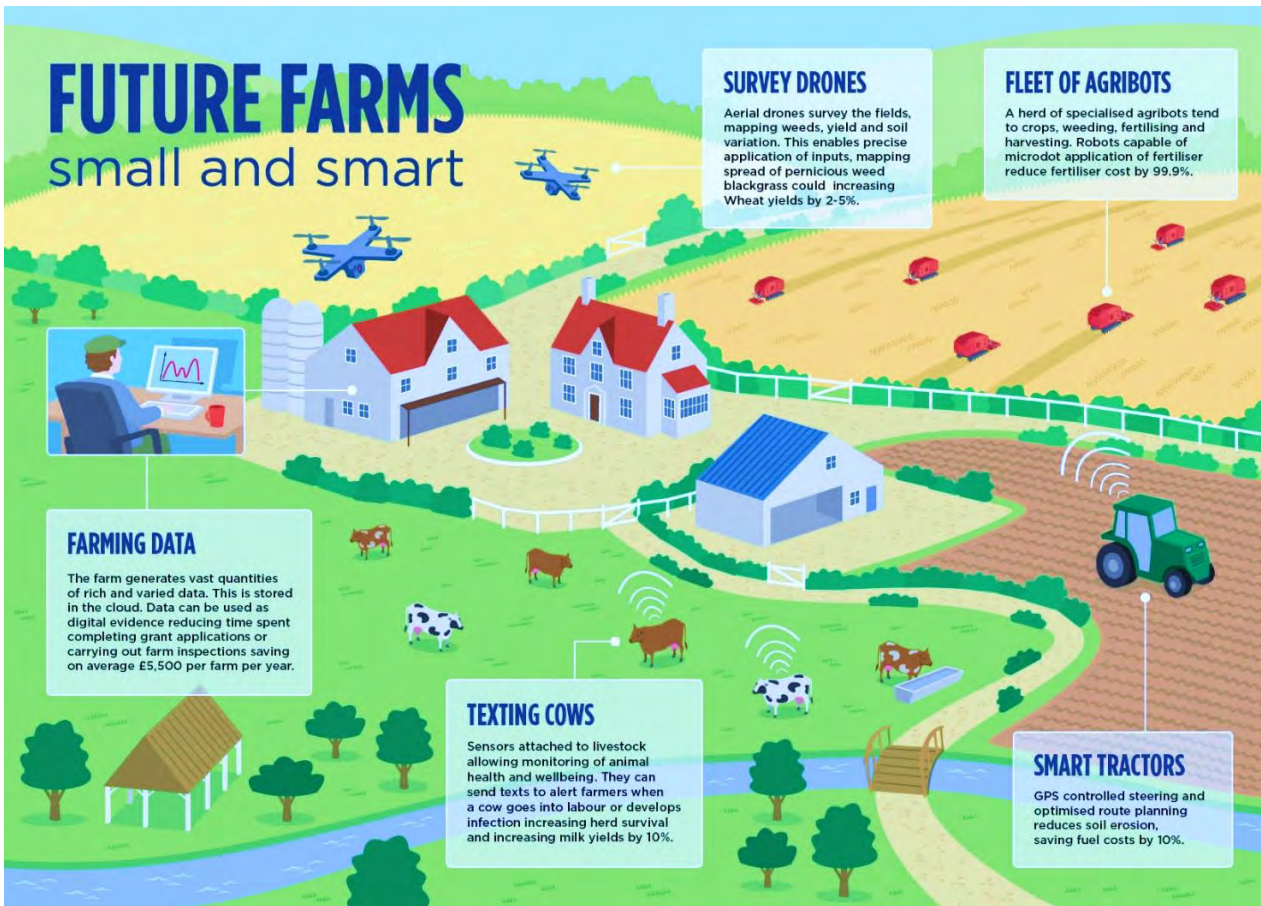
Source: <https://www.mcafee.com/us/resources/white-papers/wp-automotive-security.pdf>

IoT in Smart Energy



Source: Mohanty 2016, CE Magazine July 2016

IoT in Smart Agriculture



Source: <http://www.nesta.org.uk/blog/precision-agriculture-almost-20-increase-income-possible-smart-farming>

Smart Agriculture/Farming Market Worth \$18.21 Billion By 2025

Sources: <http://www.grandviewresearch.com/press-release/global-smart-agriculture-farming-market>

Climate-Smart Agriculture Objectives:

- Increasing agricultural productivity
- Resilience to climate change
- Reducing greenhouse gas

<http://www.fao.org>

Automatic Irrigation System

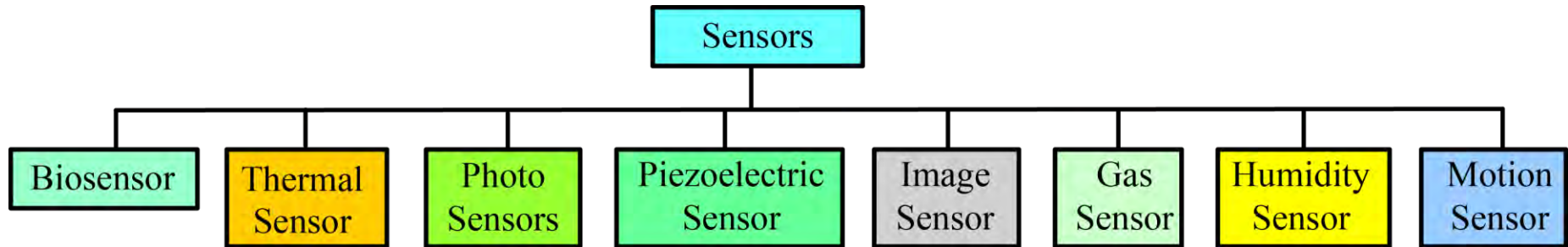


Source: Maurya 2017, CE Magazine July 2017

Driving Technologies



Cheap and Compact Sensor Technology



Source: Mohanty 2015, McGraw-Hill 2015



Gas Sensor



Temperature Sensor



Air Quality Sensor



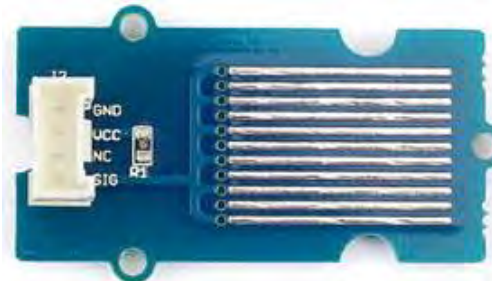
Humidity and Temperature Sensor



Light Sensor



Barometer Sensor



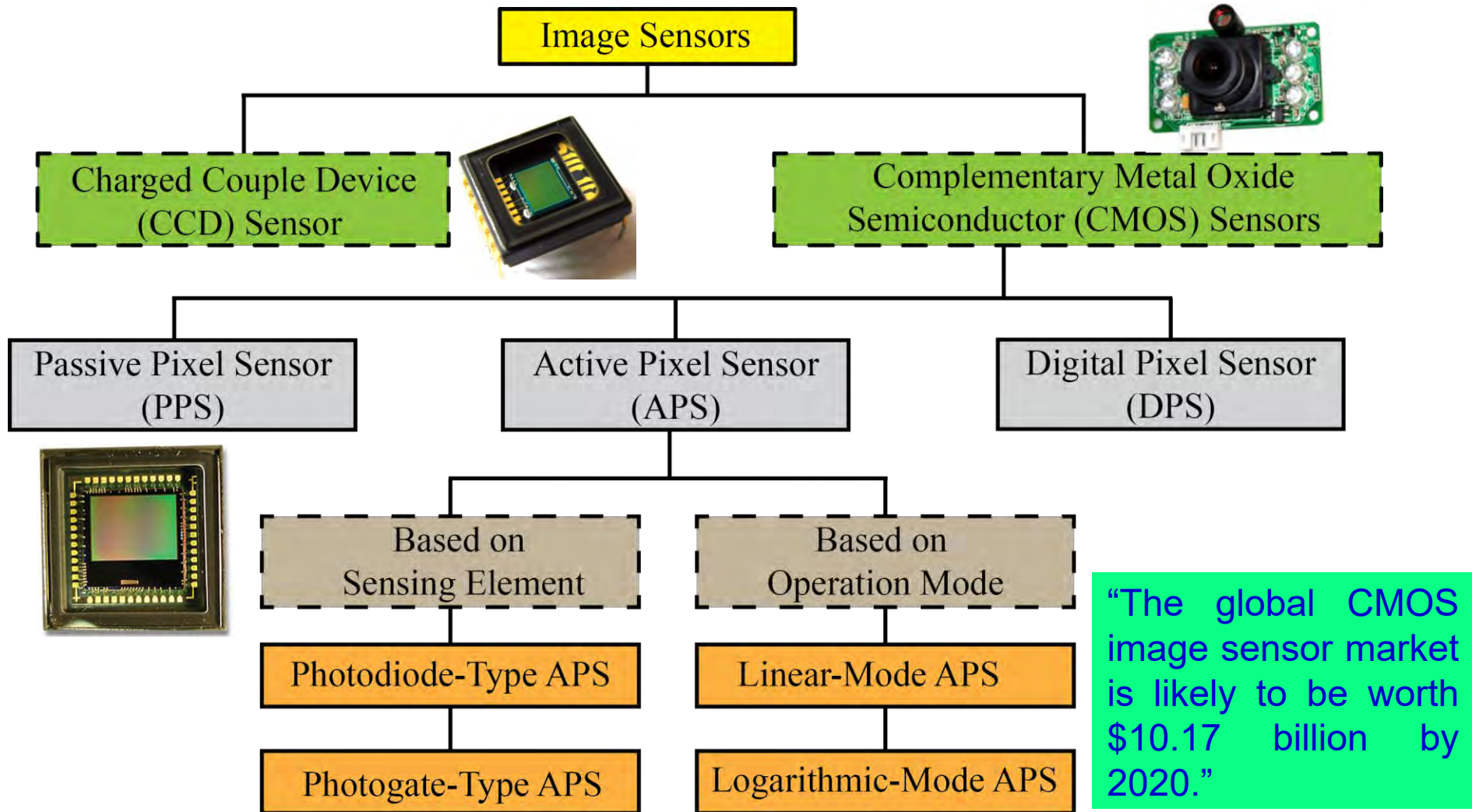
Water Sensor



Dust Sensor

Source: <http://wiki.seeed.cc/Sensor/>

Better Imaging Sensor Technology



“The global CMOS image sensor market is likely to be worth \$10.17 billion by 2020.”

Source: Mohanty 2015, McGraw-Hill 2015 Source: <http://www.grandviewresearch.com/press-release/global-cmos-image-sensors-market>

Visible Light Communications (VLC)

- ❑ LEDs can switch their light intensity at a rate that is imperceptible to human eye.
- ❑ This property can be used for the value added services based on Visible Light Communication (VLC).

Characteristic	LiFi	WiFi
Bandwidth	Huge	Limited
Requires Line of Sight	Yes	No
EMI + Hazard Concerns	Low	High
Susceptibility to Eavesdropping	Low	High
Range	Short	Medium
Data Density	High	Limited



Source: VLCS-2014

Source: Ribeiro 2017, CE Magazine October 2017

Efficient Media Compression – Better Portable Graphics (BPG)

- ❑ BPG compression instead of JPEG?
- Attributes that differentiate BPG from JPEG and make it an excellent choice include:
 - Meeting modern display requirements: high quality and lower size.
 - BPG compression is based on the High Efficiency Video Coding (HEVC), which is considered a major advance in compression techniques.
 - Supported by most web browsers with a small Javascript decoder.



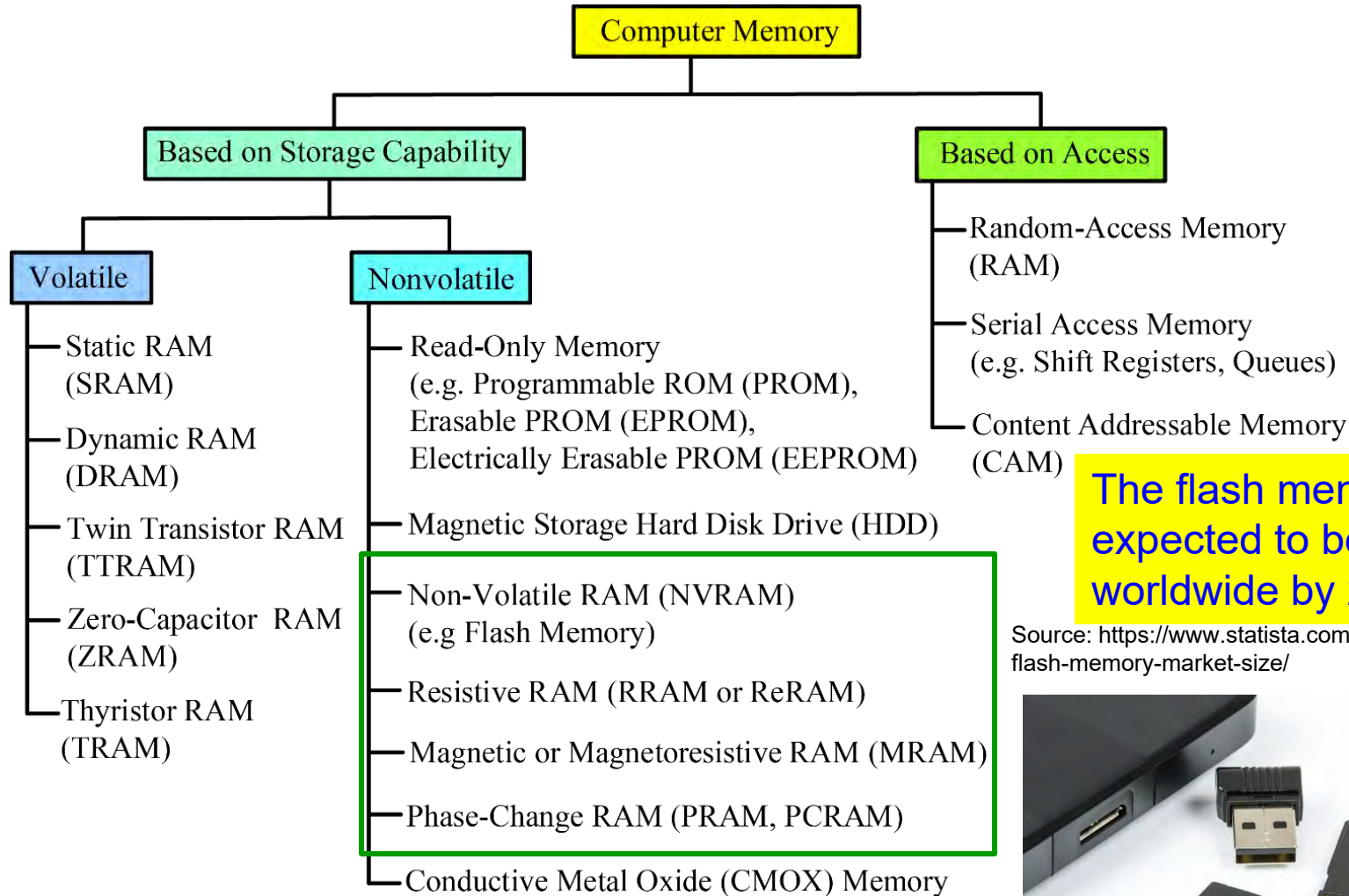
JPEG Compression



BPG Compression

Source: Mohanty 2016, IEEE Access 2016

Variety of Computer Memory



The flash memory market is expected to be worth \$37.6 worldwide by 2020.

Source: <https://www.statista.com/statistics/553556/worldwide-flash-memory-market-size/>



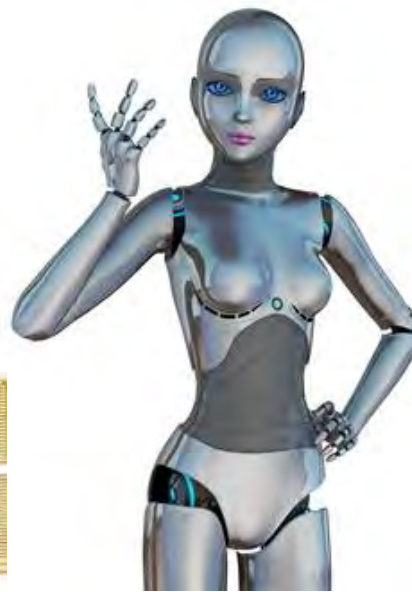
Source: Mohanty 2015, McGraw-Hill 2015

Machine Learning Technology

Artificial Intelligence



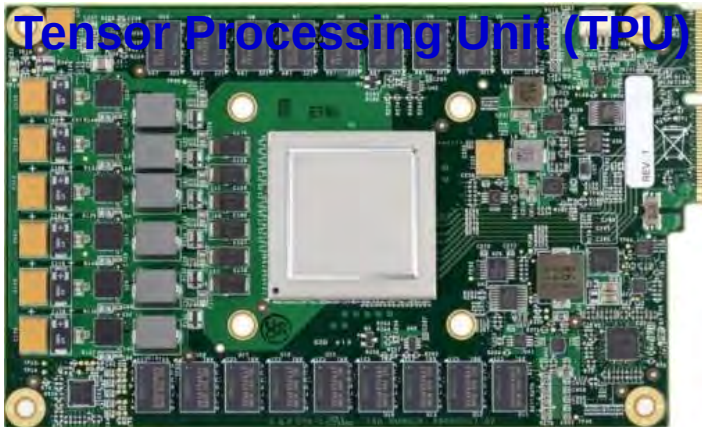
Source: <http://transmitter.ieee.org/impact-aimachine-learning-iot-various-industries/>



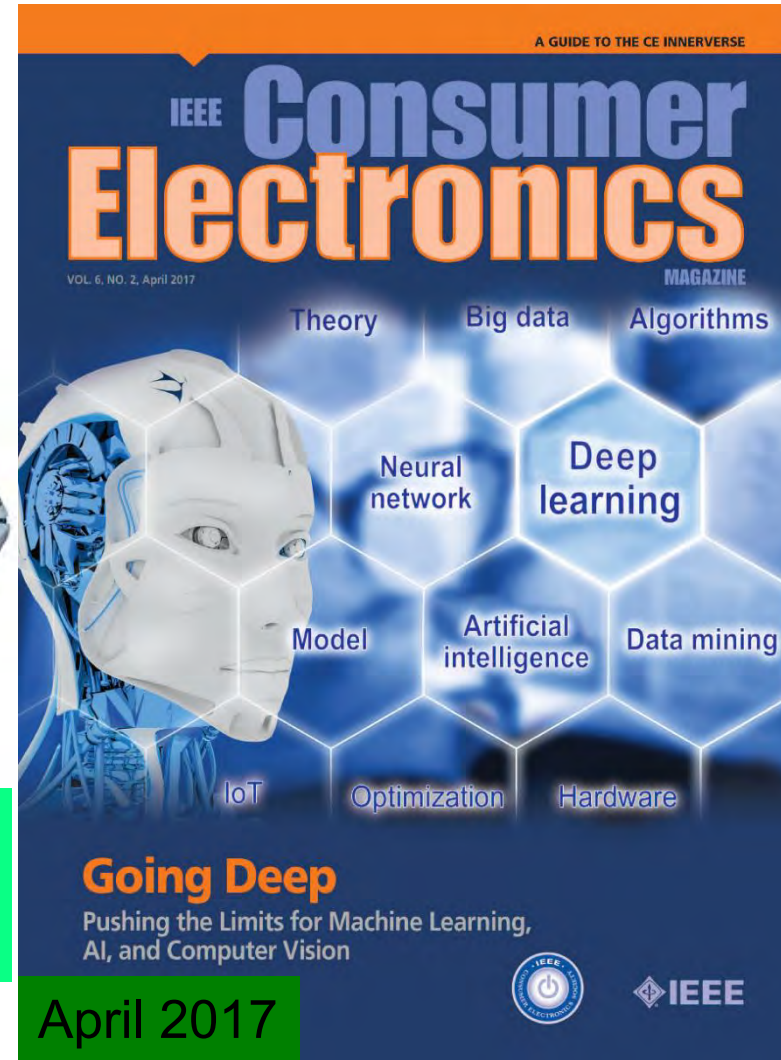
IoT Use:

- Better decision
- Faster response

Tensor Processing Unit (TPU)

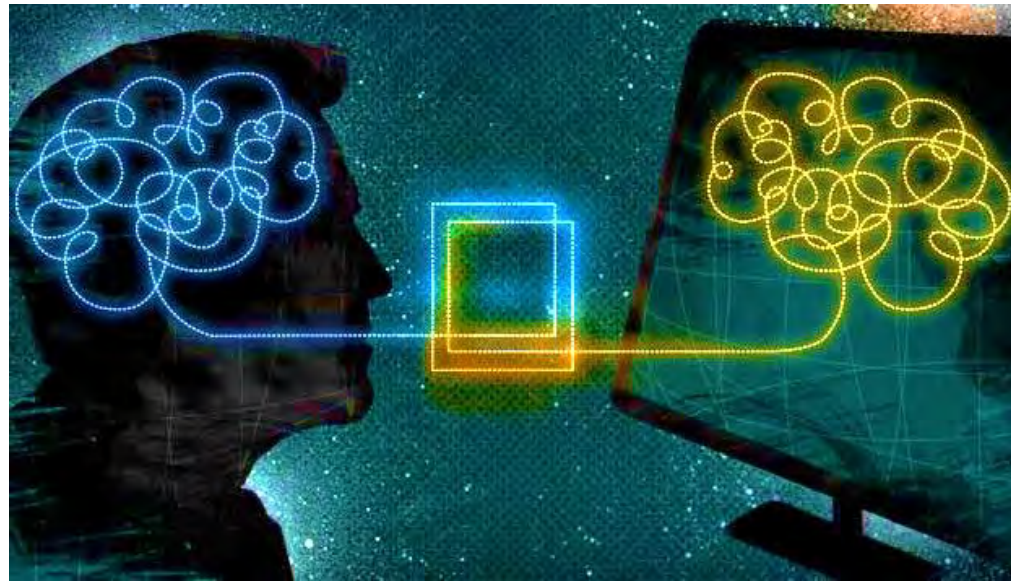


Source: <https://fossbytes.com/googles-home-made-ai-processor-is-30x-faster-than-cpus-and-gpus/>

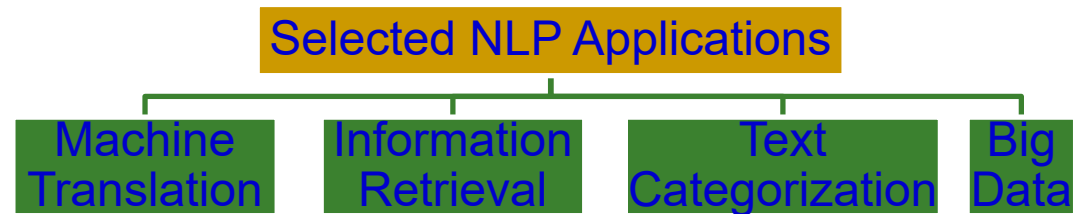
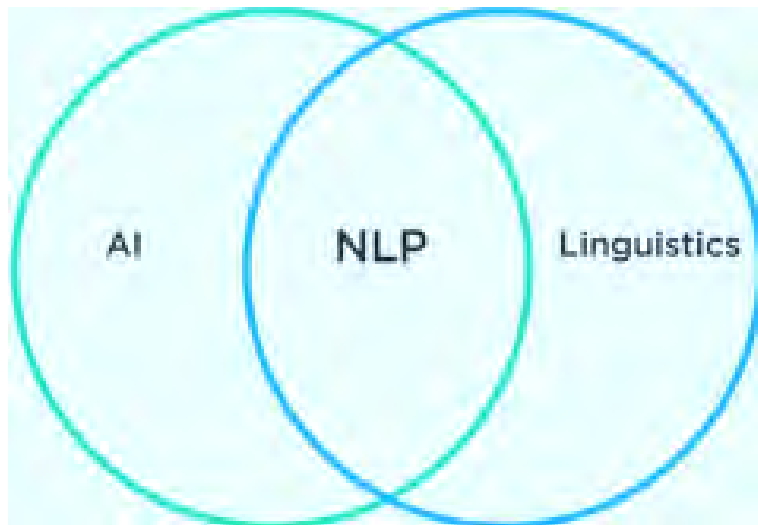


Natural Language Processing (NLP)

- NLP is the computer method to analyze, understand, and derive meaning from human language.
- Enables user to address computers as if they are communicating with a person.

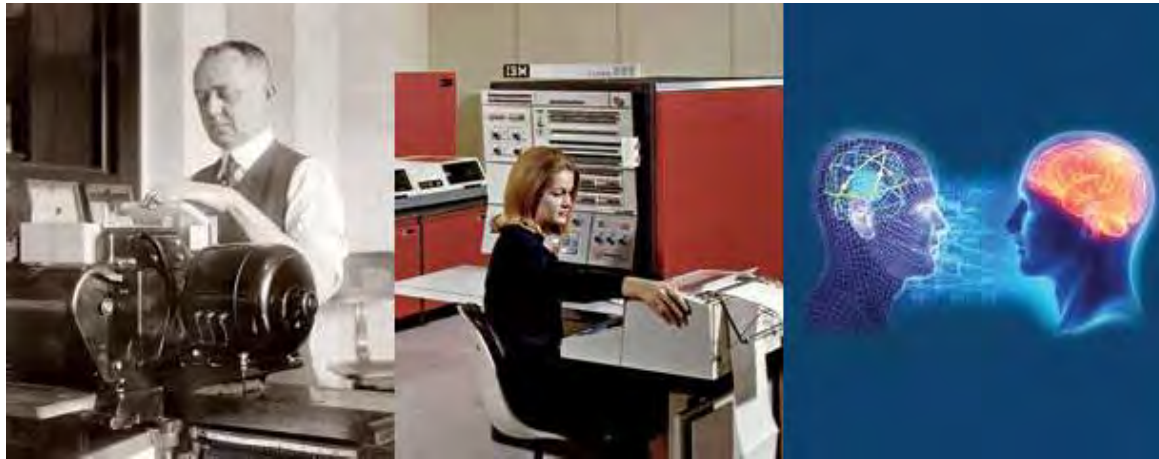


Source: <https://www.linkedin.com/pulse/natural-language-processing-2016-global-market-forecasts-rane>



Source: <http://blog.algorithmia.com/introduction-natural-language-processing-nlp/>

Cognitive Computing



The Tabulating Era
(1900s – 1940s)

The Programming Era
(1950s – present)

The Cognitive Era
(2011 –)

Cognitive Computing: Not just “right” or “wrong” anymore but “probably”.

- ❑ Systems that learn at scale, reason with purpose and interact with humans naturally.
- ❑ Learn and reason from their interactions with humans and from their experiences with their environment; not programmed.

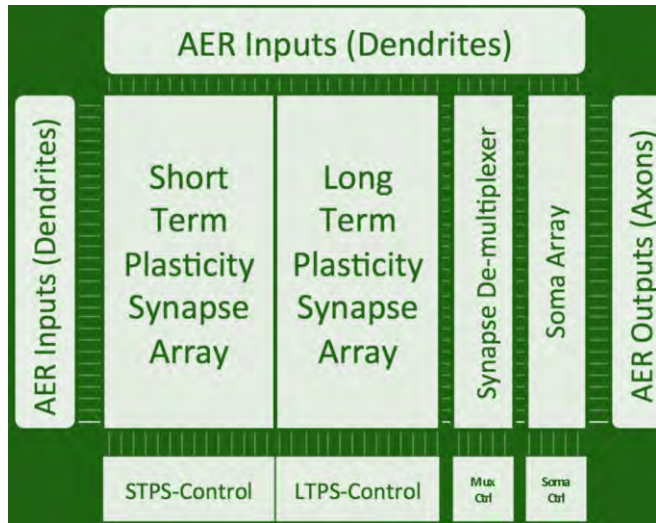
Usage:

- AI applications
- Expert systems
- Natural language processing
- Robotics
- Virtual reality

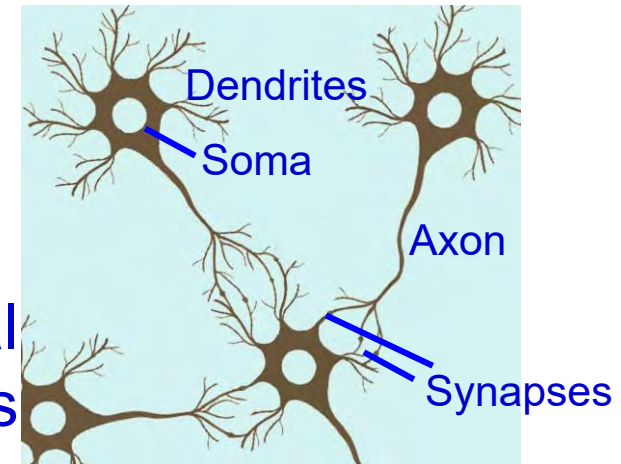
Source: http://www.research.ibm.com/software/IBMResearch/multimedia/Computing_Cognition_WhitePaper.pdf

Neuromorphic Computing or Brain-Inspired Computing

Neuromorphic Architecture

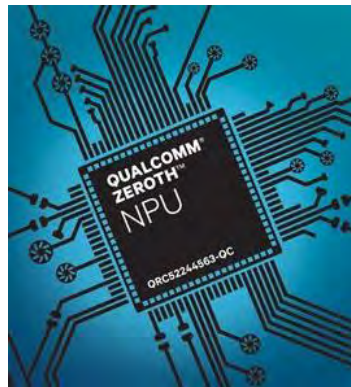


Neuronal Circuits



Processing Powers

MIT Technical Review



Types of Chips	Functions	Applications
Traditional Chips (von Neumann Architecture)	Reliably make precision calculations	Any numerical problem, Complex problems require more amount of energy
Neuromorphic Chips	Detect and Predict Patterns in complex data using minimal energy	Applications with significant visual/ auditory data requiring a system to adjust its behavior as it interacts with the world

Source: <https://www.qualcomm.com/news/onq/2013/10/10/introducing-qualcomm-zeroth-processors-brain-inspired-computing>

Neuromorphic Computing or Brain-Inspired Computing



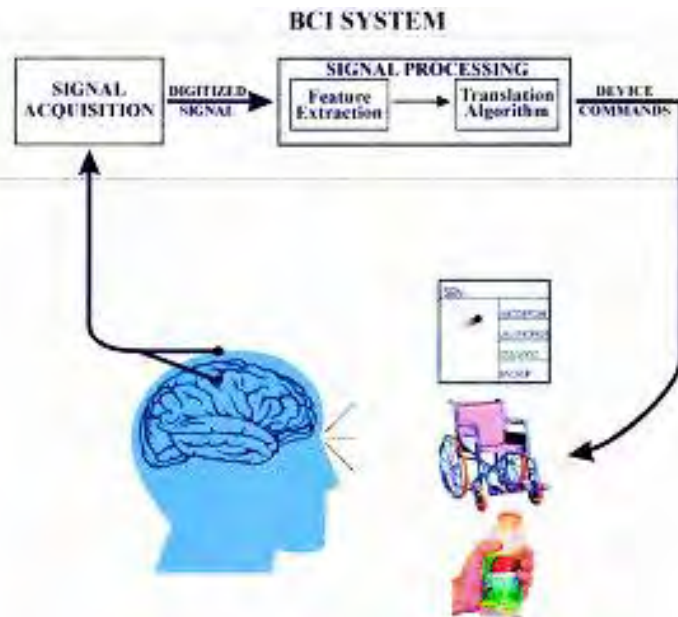
Application 1: Integrate into assistive glasses for visually impaired people for navigating through complex environments, even without the need for a WiFi connection.



Application 2: Neuromorphic-based, solar-powered “sensor leaves” equipped with sensors for sight, smell or sound can help to monitor natural disasters.

Source: <https://blogs.scientificamerican.com/observations/brain-inspired-computing-reaches-a-new-milestone/>

Brain Computer Interface (BCI)



Source: <http://brainpedia.org/brain-computer-interface-allows-paralysis-als-patients-type-much-faster/>

Brain-Computer Interface Allows paralysis patients to Type Faster

“Currently, people interact with their devices by thumb-typing on their phones. A high-bandwidth interface to the brain would help achieve a symbiosis between human and machine intelligence and could make humans more useful in an AI-driven world.”

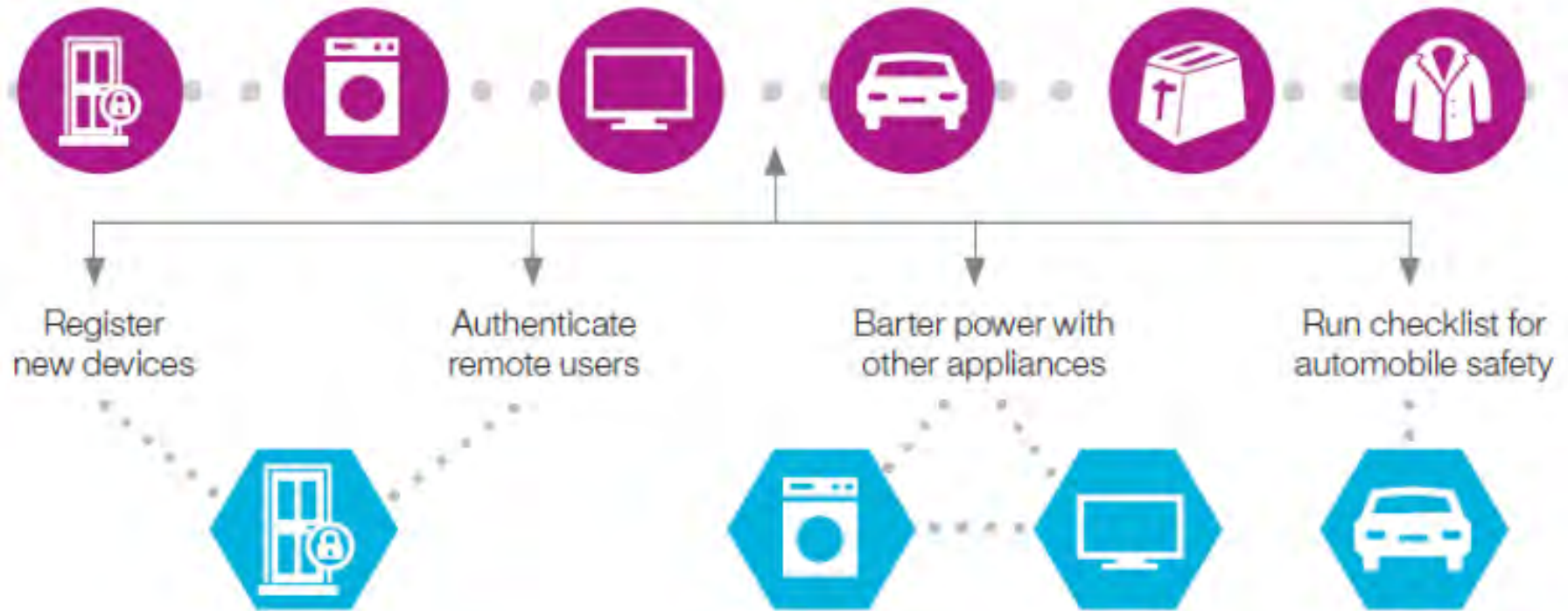
-- Neuralink - neurotechnology company - Elon Musk.

Sources: <http://brainpedia.org/elon-musk-wants-merge-human-brain-ai-launches-neuralink/>

The Blockchain



Universal digital ledger



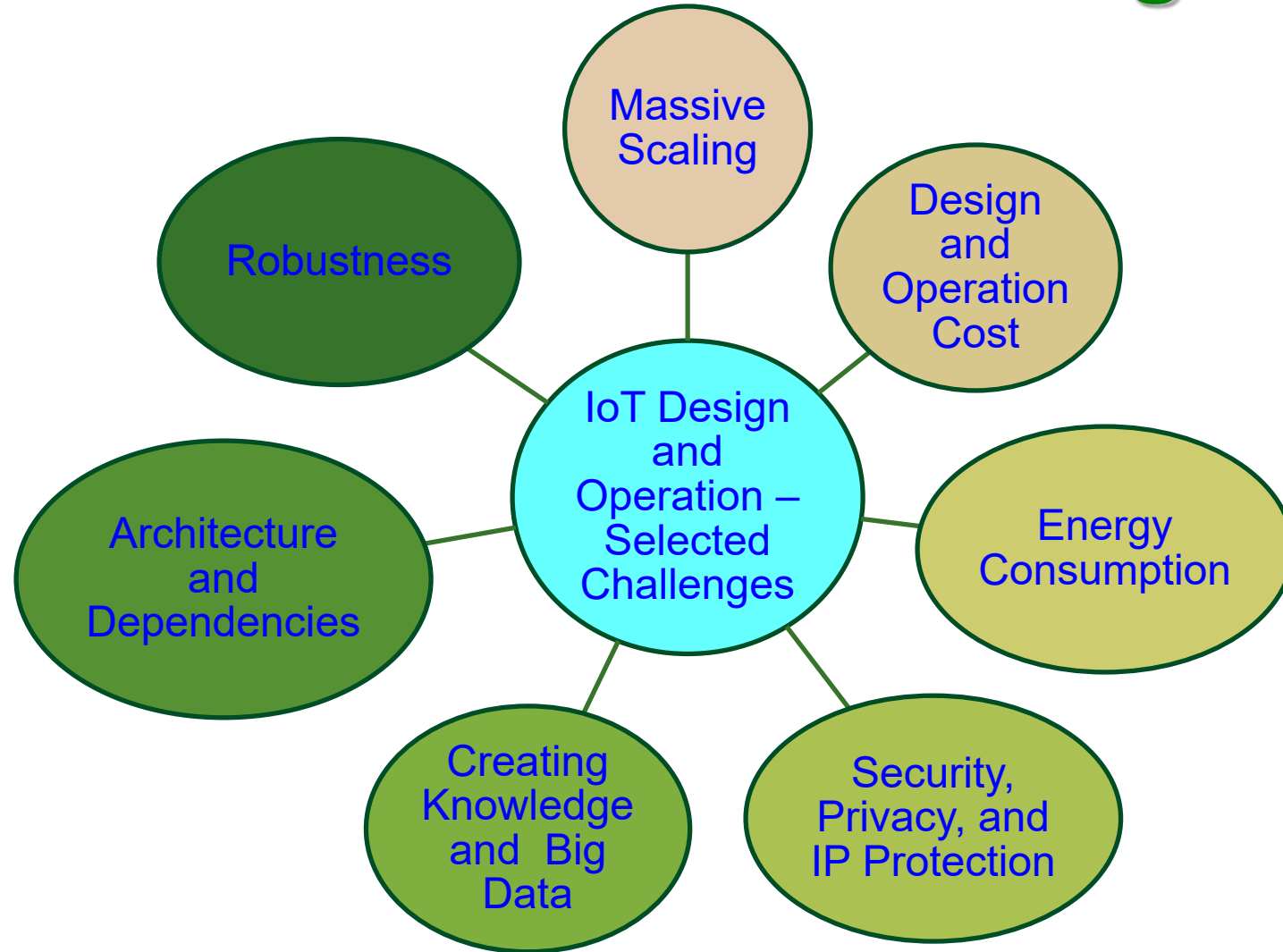
- Think of it as cloud based peer to peer ledger.
- A Blockchain is a cloud based database shared by every participant in a system.
- The Blockchain contains the complete transaction or other record keeping.

Source: <https://www.linkedin.com/pulse/securing-internet-things-iot-blockchain-ahmed-banafa>
Stay Tuned to: Puthal, Mohanty 2018, CE Magazine March 2018

Challenges and Research

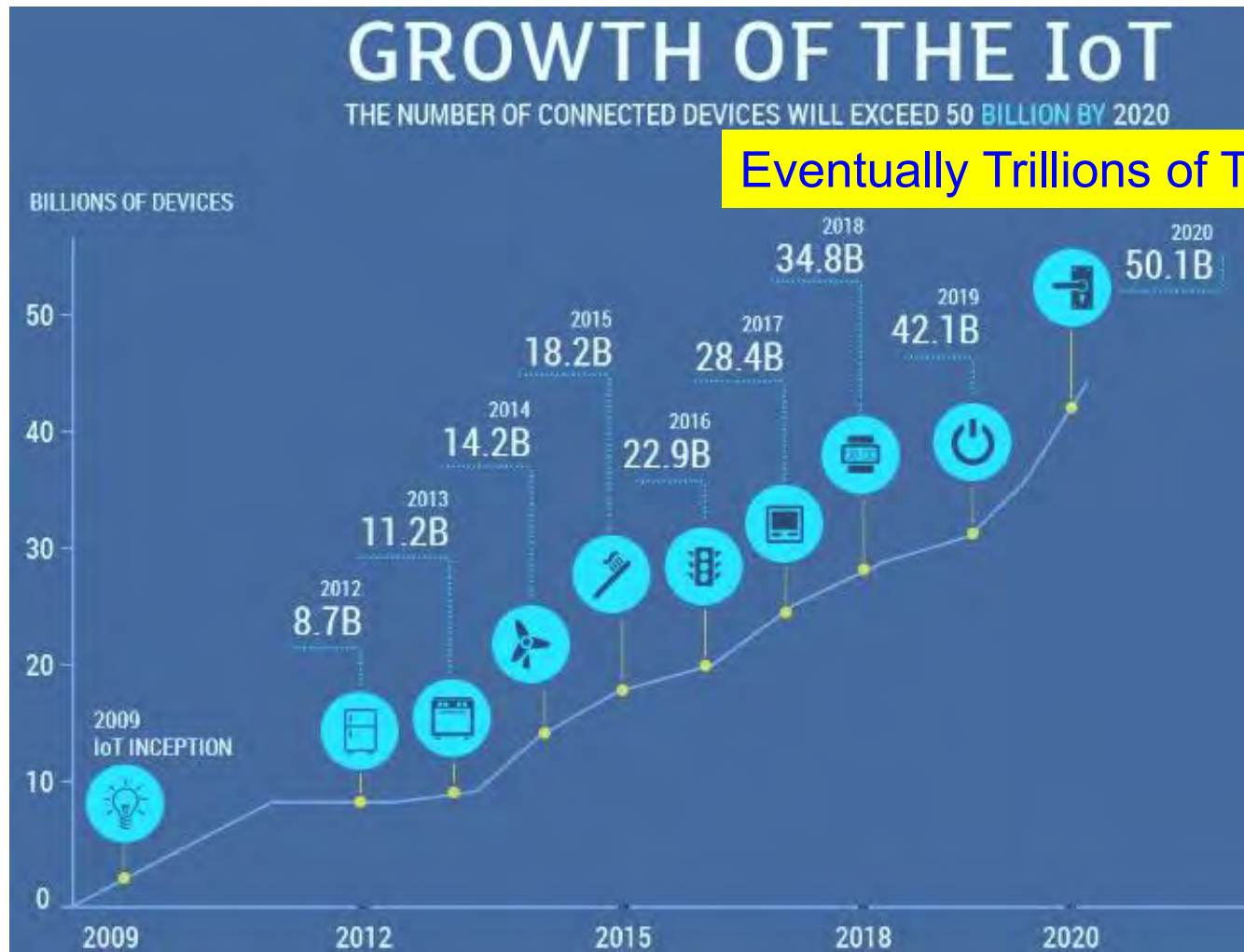


IoT – Selected Challenges



Source: Mohanty 2016, EuroSimE 2016 Keynote Presentation

Massive Scaling



Source: <https://www.linkedin.com/pulse/history-iot-industrial-internet-sensors-data-lakes-0-downtime>

High Design and Operation Cost

- The design cost is a one-time cost.
- Design cost needs to be small to make a IoT realization possible.
- The operations cost is that required to maintain the IoT.
- A small operations cost will make it easier to operate in the long run with minimal burden on the budget of application in which IoT is deployed.



Source: <http://www.industrialisation-produits-electroniques.fr>



“Cities around the world could spend as much as \$41 trillion on smart tech over the next 20 years.”

Source: <http://www.cnn.com/2016/10/25/spending-on-smart-cities-around-the-world-could-reach-41-trillion.html>

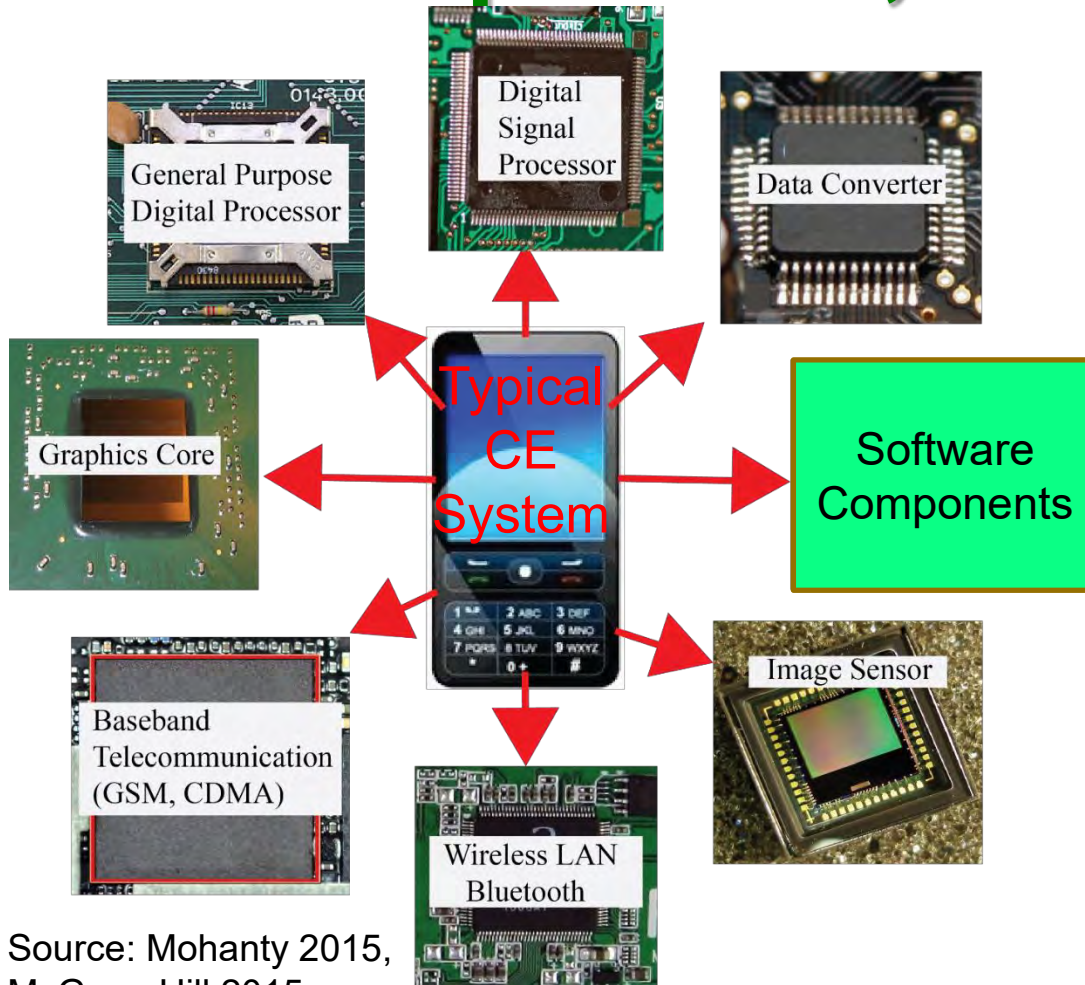
Communication Latency and Energy Consumption

- Connected cars require latency of ms to communicate and avoid impending crash.
 - Faster connection
 - Low latency
 - Lower power
- 5G for connected world: This enables all devices to be connected seamlessly.
- How about 5G, WiFi working together more effectively?

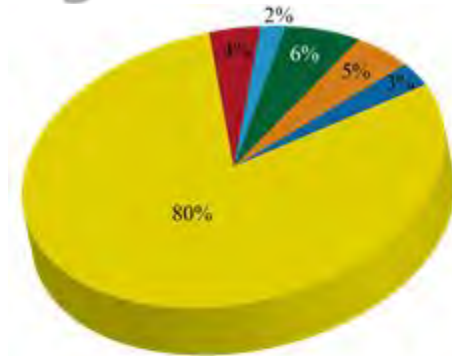


Source: <https://www.linkedin.com/pulse/key-technologies-connected-world-cloud-computing-ioe-balakrishnan>

Energy Consumption of Sensors, Components, and Systems

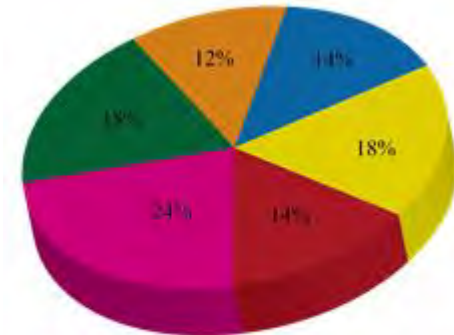


Source: Mohanty 2015, McGraw-Hill 2015



Legend: GSM (Yellow), CPU (Red), RAM (Blue), Graphics (Green), LCD (Orange), Others (Dark Blue)

During GSM Communications



Legend: GSM (Yellow), CPU (Red), WiFi (Pink), Graphics (Green), LCD (Orange), Others (Blue)

During WiFi Communications

Battery-Less IoT

Battery less operations can lead to reduction of size and weight of the edge devices.

Go Battery-Less

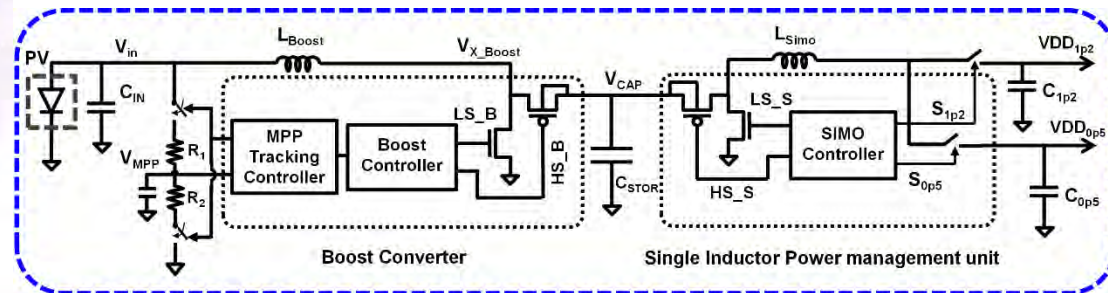


Source: <http://newscenter.ti.com/2015-02-25-TI-makes-battery-less-IoT-connectivity-possible-with-the-industrys-first-multi-standard-wireless-microcontroller-platform>



Batter-Less SoC

Source: <https://www.technologyreview.com/s/529206/a-batteryless-sensor-chip-for-the-internet-of-things/>



Energy Harvesting and Power Management

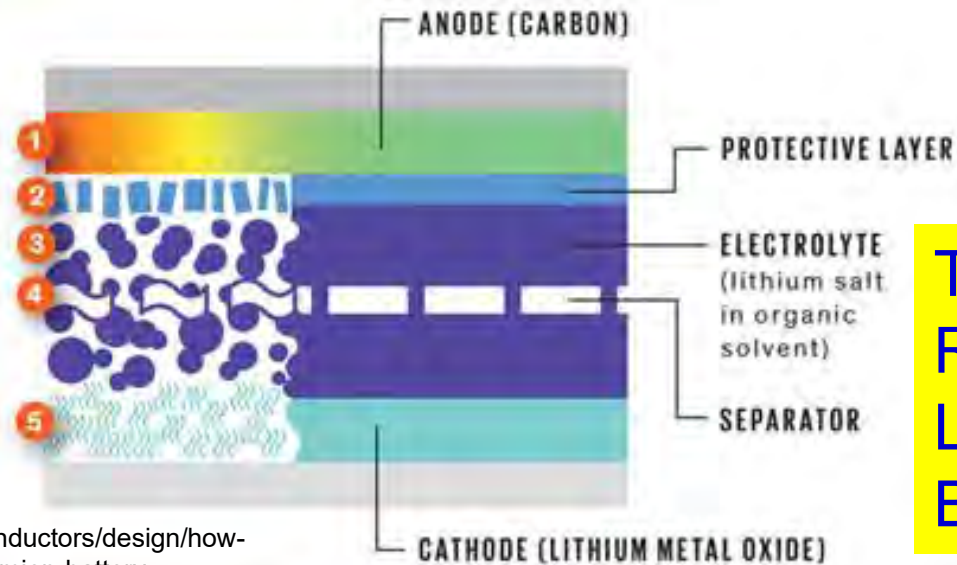
Source: <http://rlpvlsi.ece.virginia.edu/node/368>

Safety of Electronics



Smartphone Battery

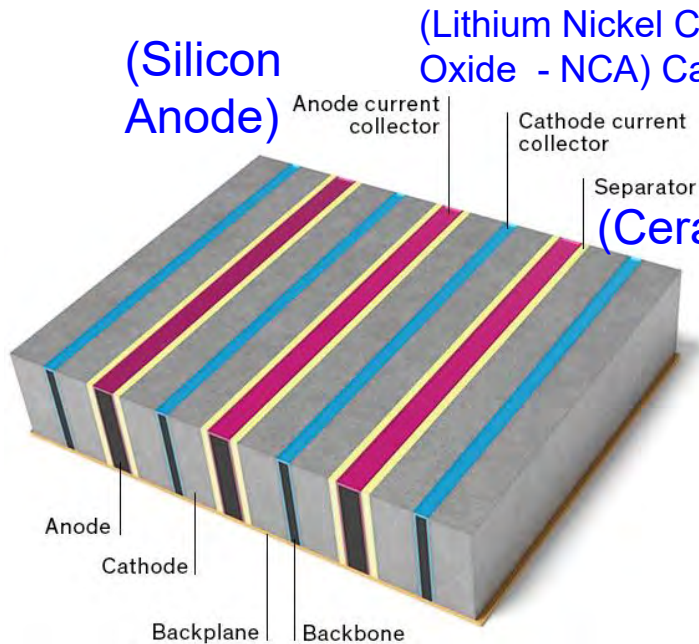
1. Heating starts.
2. Protective layer breaks down.
3. Electrolyte breaks down into flammable gases.
4. Separator melts, possibly causing a short circuit.
5. Cathode breaks down, generating oxygen.



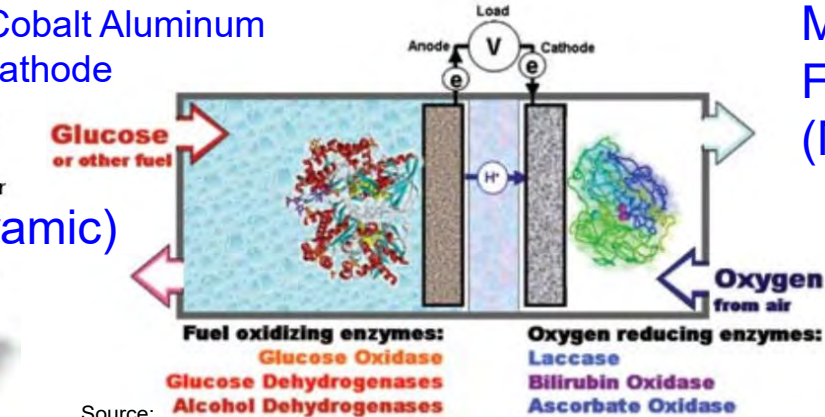
Thermal
Runaway in a
Lithium-Ion
Battery

Source: <http://spectrum.ieee.org/semiconductors/design/how-to-build-a-safer-more-energydense-lithiumion-battery>

Energy Storage - High Capacity and Safer Needed



Source: <http://spectrum.ieee.org/semiconductors/design/how-to-build-a-safer-more-energydense-lithiumion-battery>

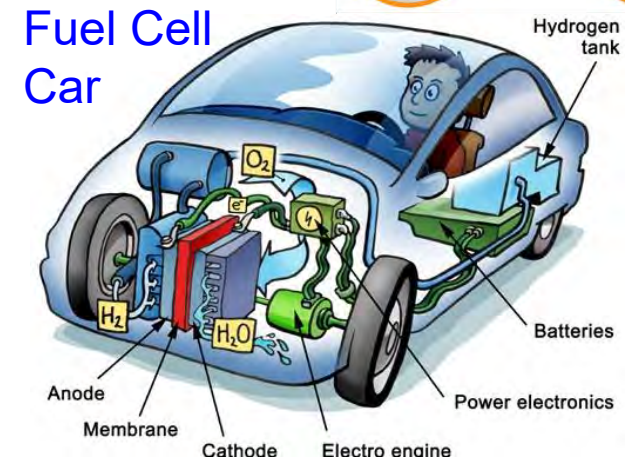
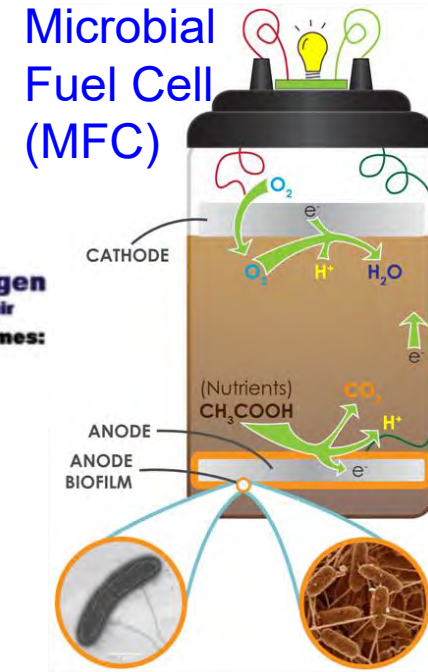


Source: https://www.electrochem.org/dl/interface/sum/sum07/su07_p28_31.pdf



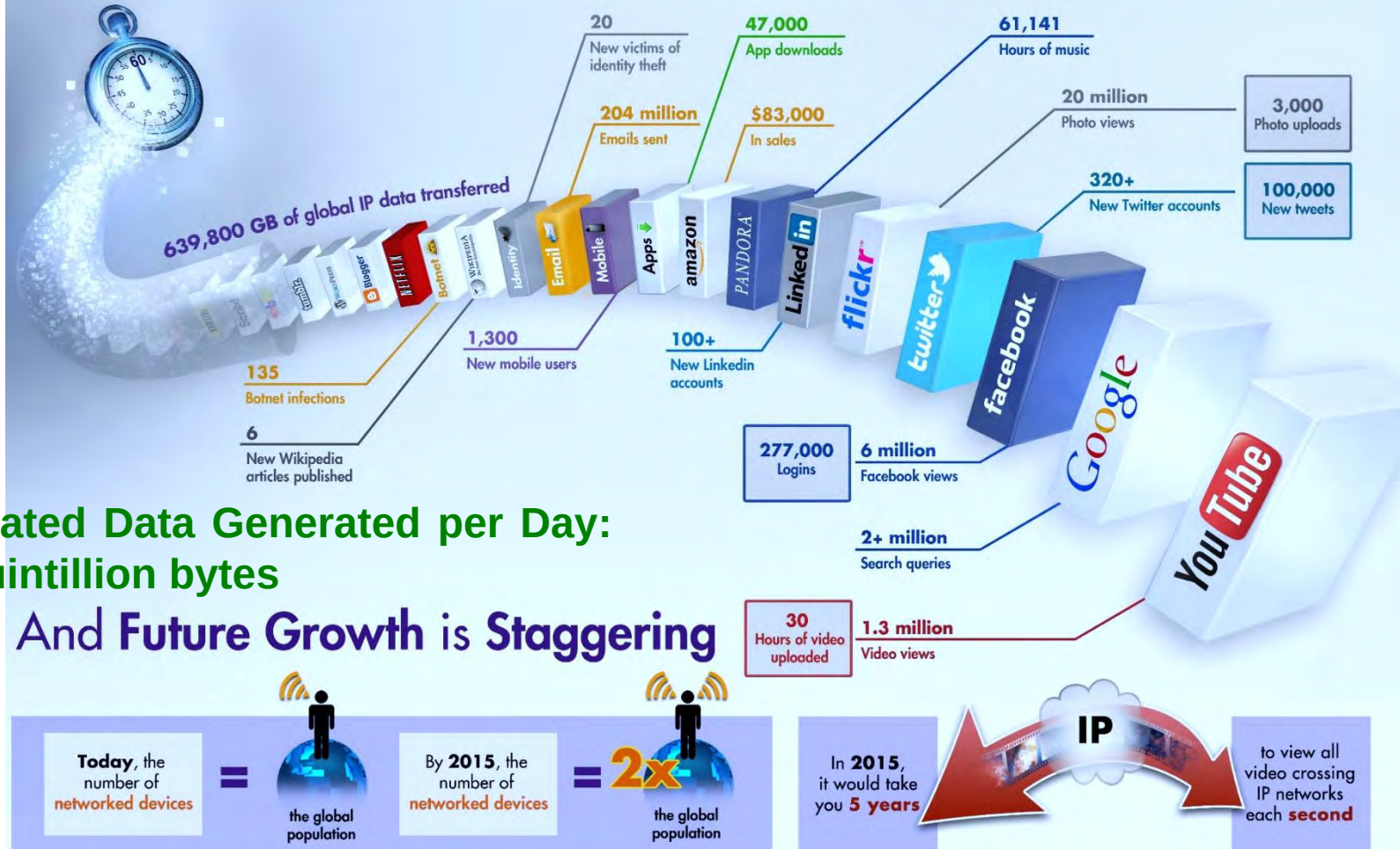
Solid Polymer Lithium Metal Battery

Source: <https://www.nytimes.com/2016/12/11/technology/designing-a-safer-battery-for-smartphones-that-wont-catch-fire.html>



Huge Amount of Data

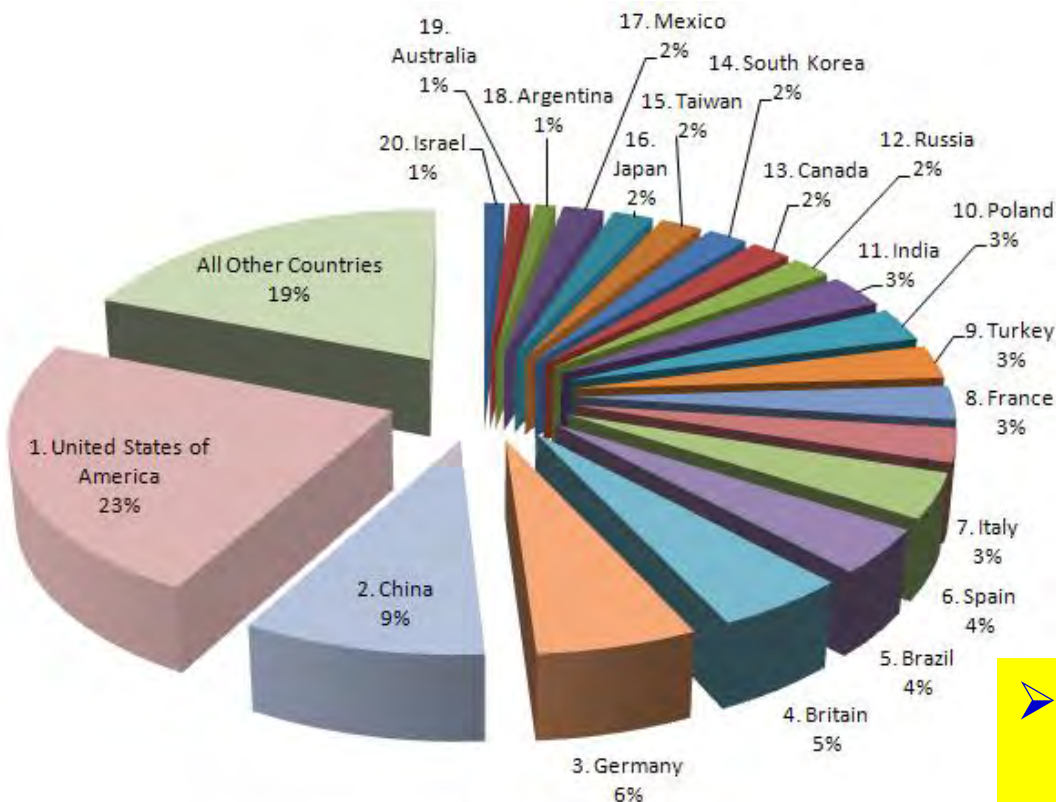
What Happens in an Internet Minute?



Security, Privacy, and Copyright



Security - Information, System



Cybercrime: Top 20 Countries

Source: <https://www.enigmasoftware.com/top-20-countries-the-most-cybercrime/>



- Cybercrime damage costs to hit \$6 trillion annually by 2021
- Cybersecurity spending to exceed \$1 trillion from 2017 to 2021

Source: <http://www.csoonline.com/article/3153707/security/top-5-cybersecurity-facts-figures-and-statistics-for-2017.html>

Security in Communications Technology

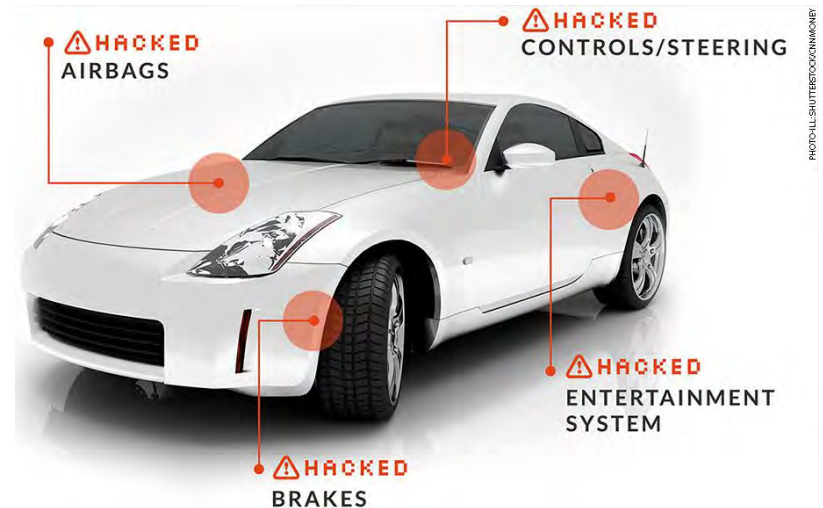


Security - Systems ...

Power Grid Attack



Source:
<http://www.csoonline.com/article/3177209/security/why-the-ukraine-power-grid-attacks-should-raise-alarm.html>

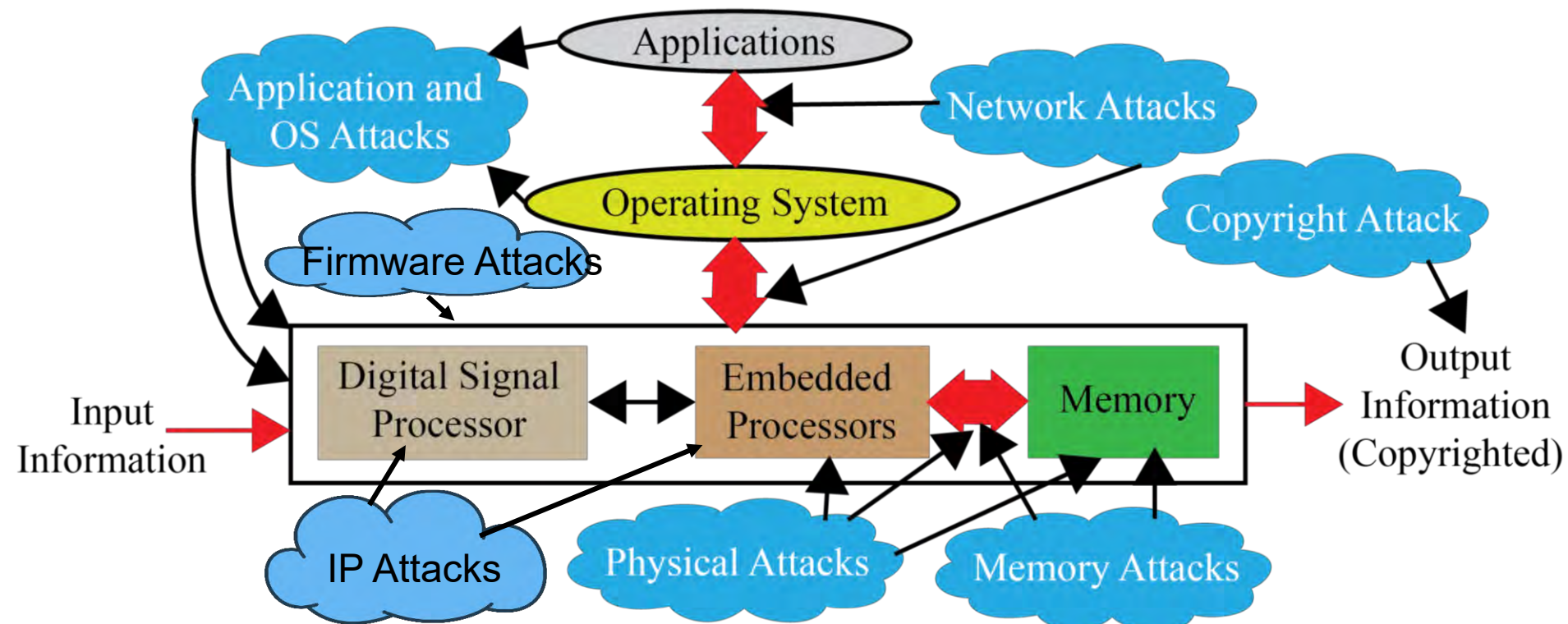


Source: <http://money.cnn.com/2014/06/01/technology/security/car-hack/>



Source: <http://politicalblindspot.com/u-s-drone-hacked-and-hijacked-with-ease/>

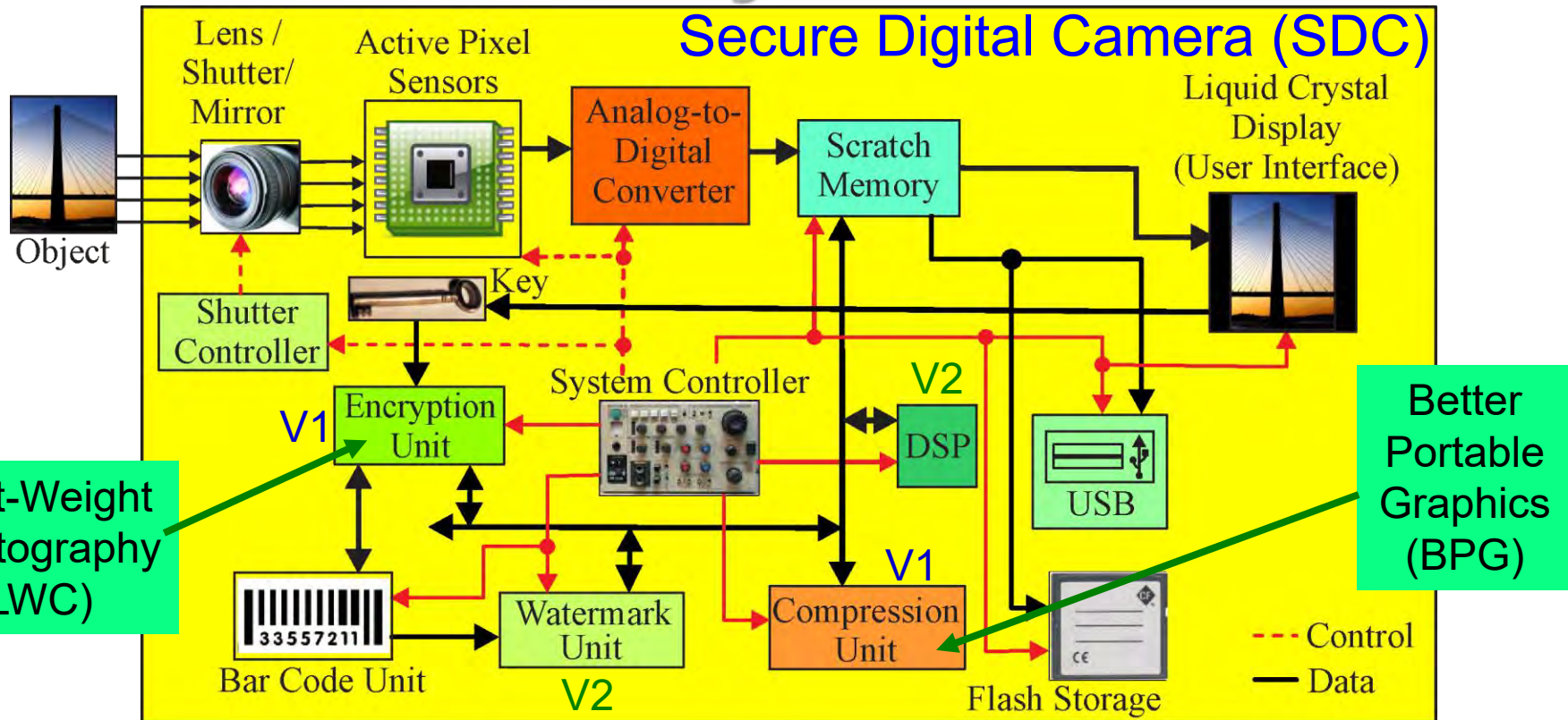
Selected Attacks on a Typical CE System – Security, Privacy, IP Right



Source: Mohanty 2015, McGraw-Hill 2015

Diverse forms of Attacks, following are not same: System Security, Information Security, Information Privacy, System Trustworthiness, Hardware IP protection, Information Copyright Protection.

CE System Energy & Security Tradeoff – System Level



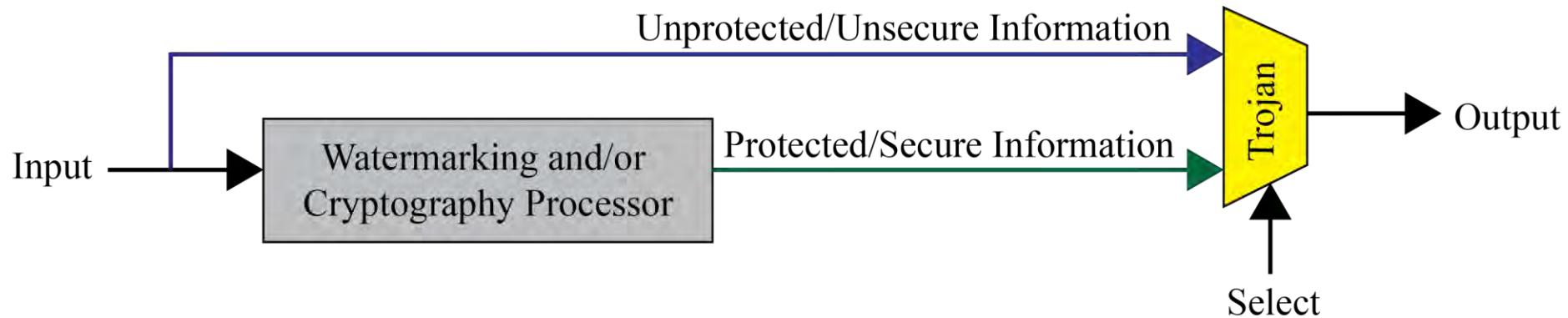
Include additional hardware components, but perform DVFS like technology for energy and performance optimization.

Source: Mohanty 2006, TCAS-II May 2006; Mohanty 2009, JSA Oct 2009; Mohanty 2016, Access 2016

Malicious Design Modifications Issue

Information may bypass giving a non-watermarked or non-encrypted output.

Hardware Trojans



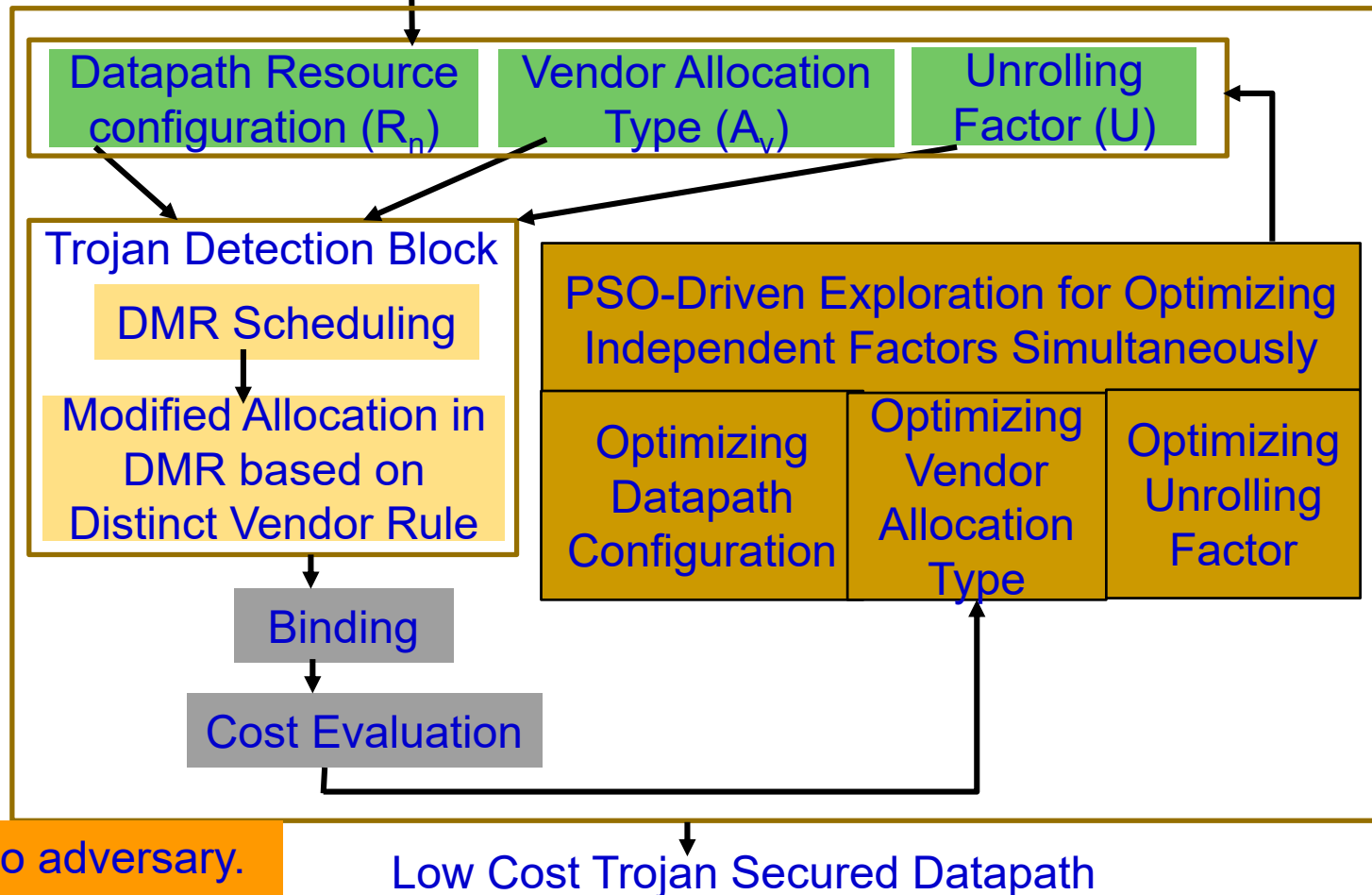
Source: Mohanty 2015, McGraw-Hill 2015



Provide backdoor to adversary.
Chip fails during critical needs.

Trojan Secure Digital Hardware Synthesis

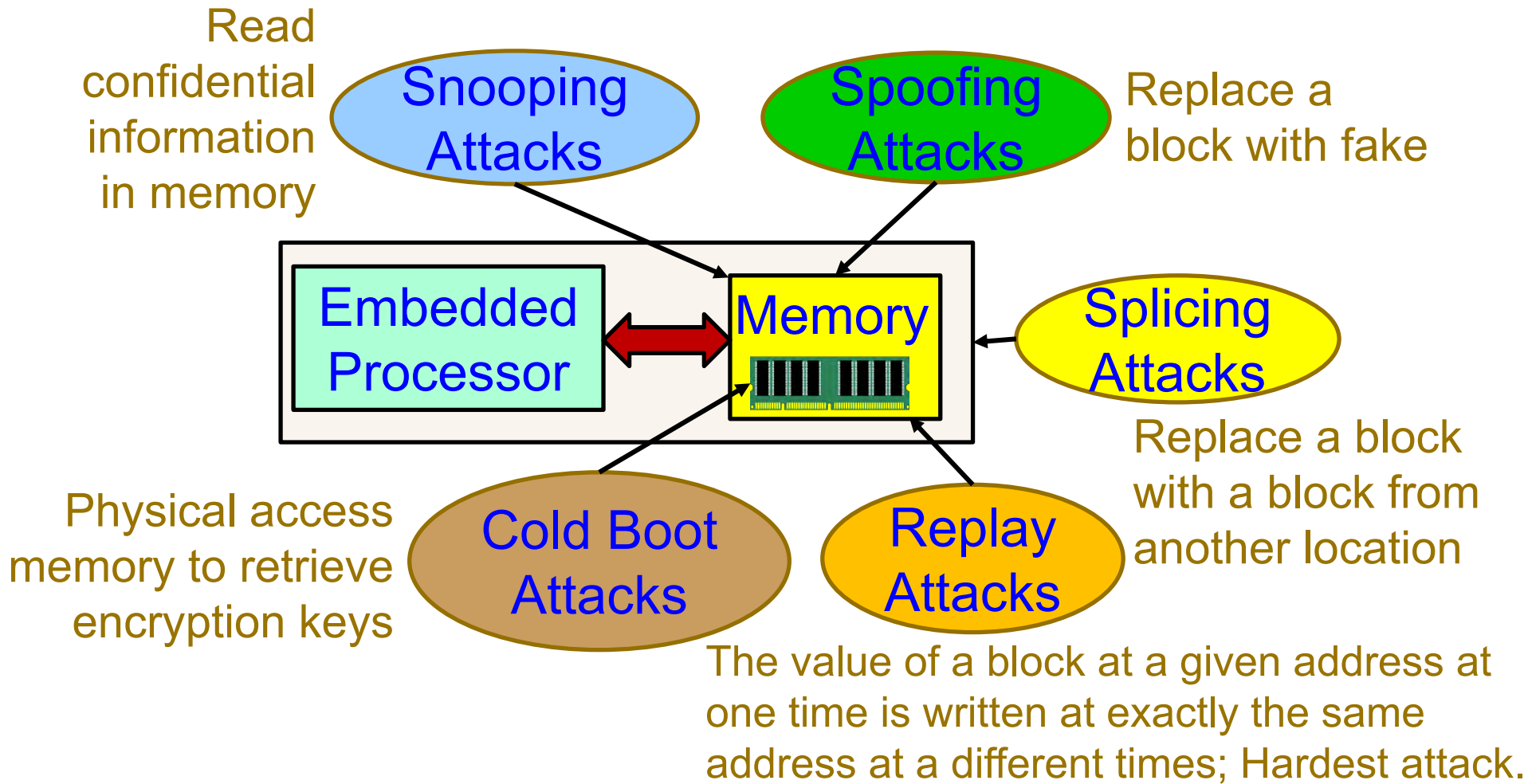
HLS Library Comprising of Module info from Two Vendors



Provide backdoor to adversary.
Chip fails during critical needs.

Source: Sengupta, Mohanty 2017: TCAD April 2017

Memory Attacks



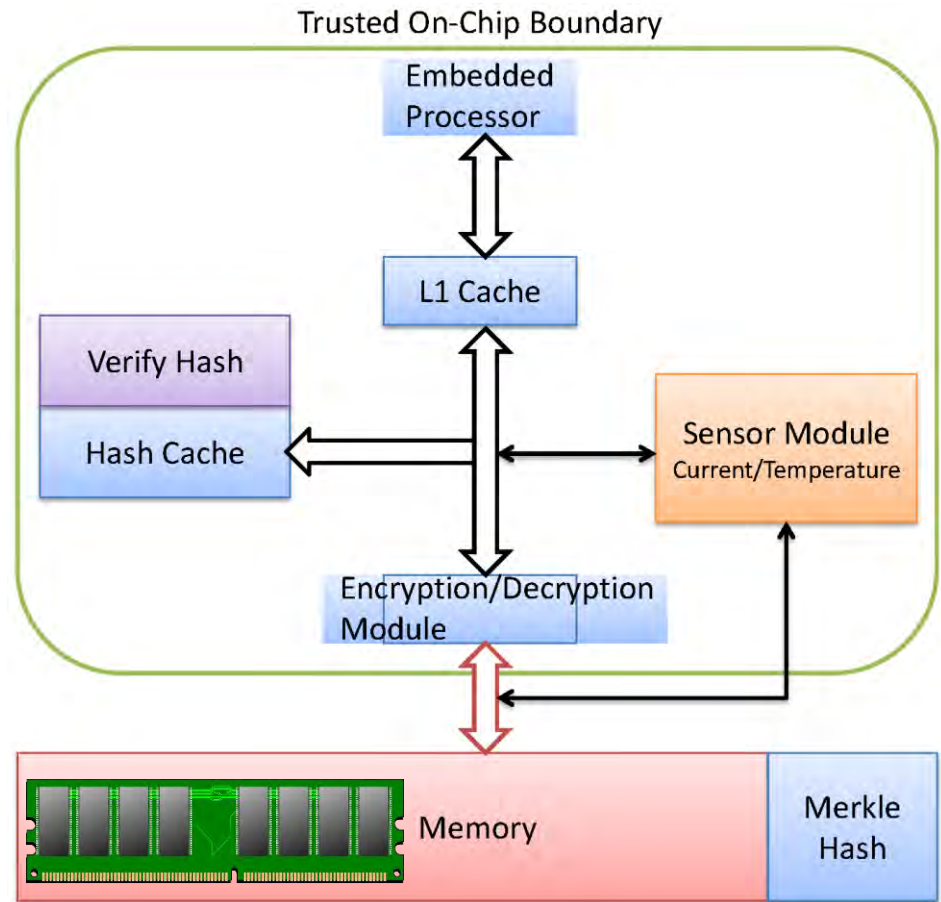
Source: Mohanty 2013, Springer CSSP Dec 2013

Memory Security and Protection



Nonvolatile Storage

Source: <http://datalocker.com>



On-Chip/On-Board Memory Protection

Source: Mohanty 2013, Springer CSSP Dec 2013

RFID Security - Attacks



Selected
RFID
Attacks



Physical
RFID
Threats

Disabling Tags

Tag Modification

Cloning Tags

Reverse Engineering and Physical Exploration

RFID
Channel
Threats

Eavesdropping

Snooping

Skimming

Replay Attack

Relay Attacks

Electromagnetic Interference

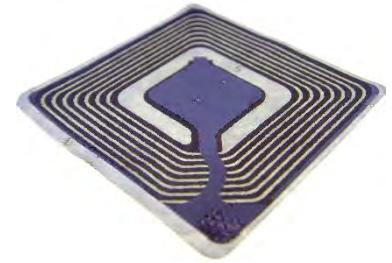
System
Threats

Counterfeiting and Spoofing Attacks

Tracing and Tracking

Password Decoding

Denial of Service (DoS) Attacks



Numerous Applications

Source: Khattab 2017; Springer 2017 RFID Security

RFID Security - Solutions

Selected RFID Security Methods

Killing Tags

Sleeping
Tags

Faraday
Cage

Blocker
Tags

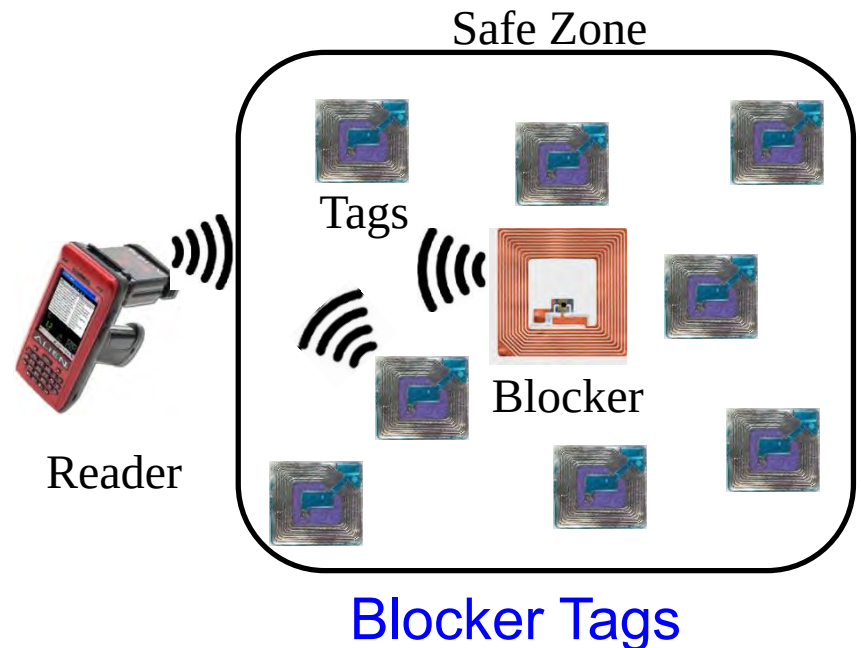
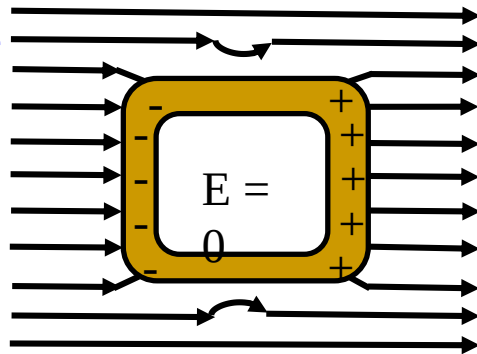
Tag
Relabeling

Minimalist
Cryptography

Proxy
Privacy
Devices



Faraday Cage



Source: Khattab 2017, Springer 2017 RFID Security

NFC Security - Attacks

Selected NFC Attacks

Eavesdropping

Data
Modification

Relay
Attacks

Data
Corruption

Spoofing

Interception
Attacks

Theft



Source: <http://www.idigitaltimes.com/new-android-nfc-attack-could-steal-money-credit-cards-anytime-your-phone-near-445497>

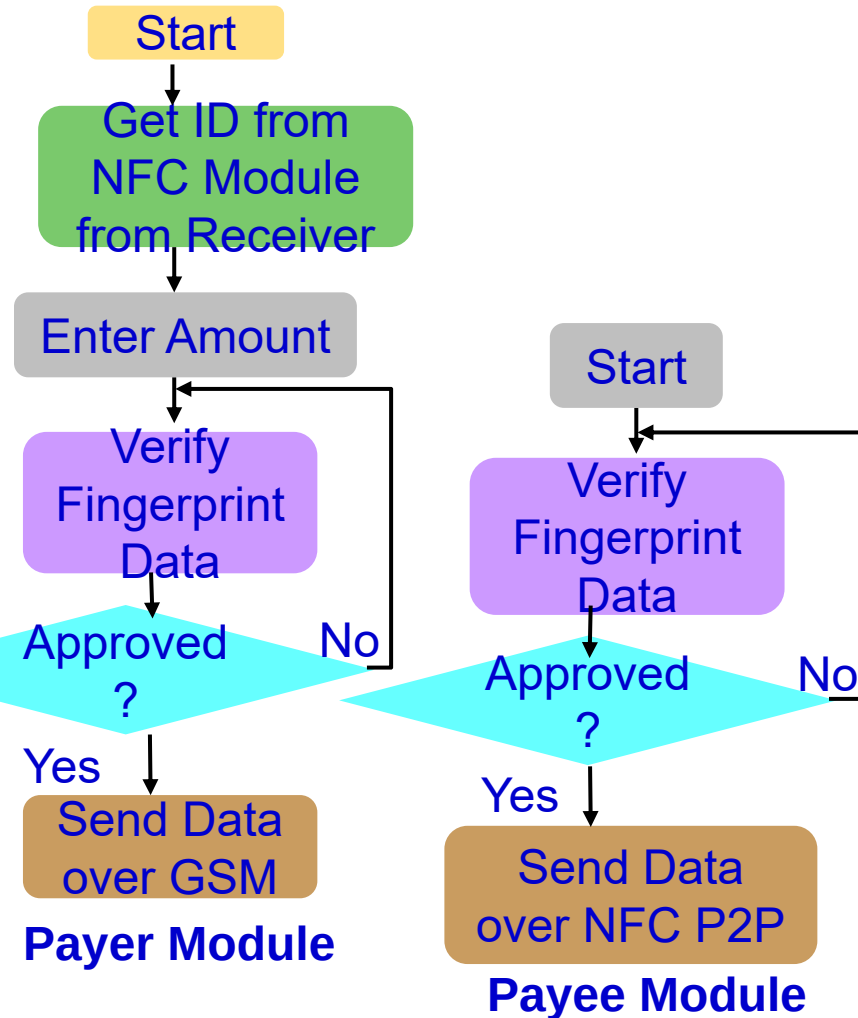
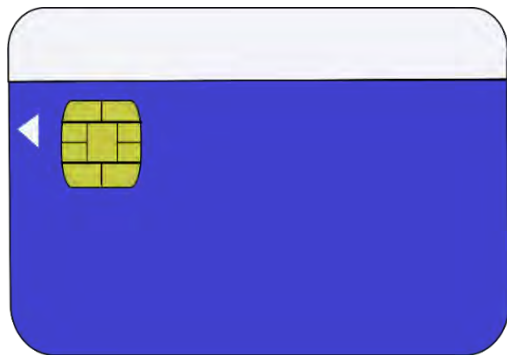
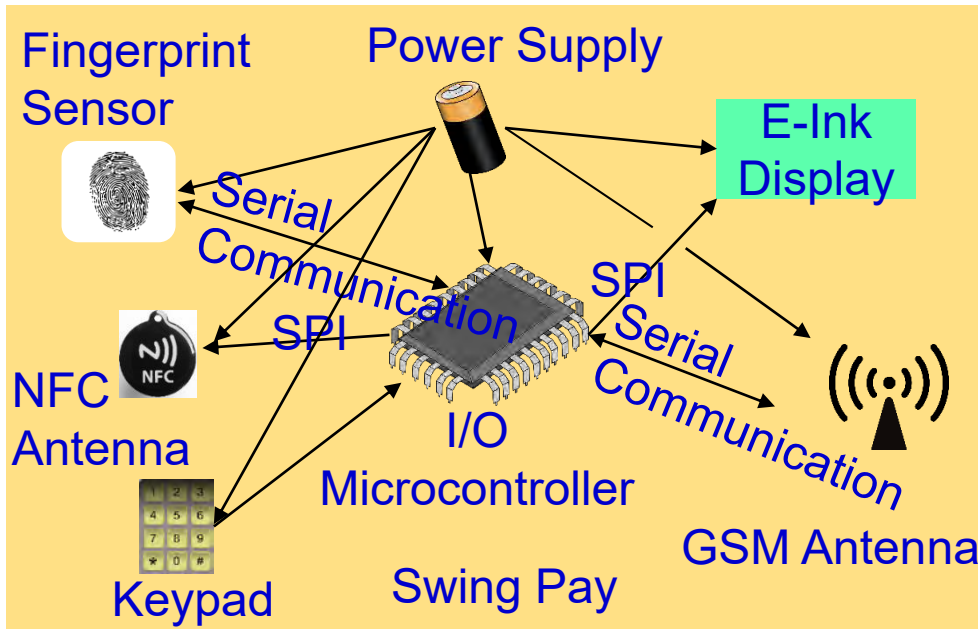


Source: <http://resources.infosecinstitute.com/near-field-communication-nfc-technology-vulnerabilities-and-principal-attack-schema/>



Source: <https://www.slideshare.net/cgvwzq/on-relaying-nfc-payment-transactions-using-android-devices>

NFC Security - Solution



Source: Mohanty 2017, CE Magazine Jan 2017

Autonomous Car – Security Vulnerability

Selected Attacks on Autonomous Cars

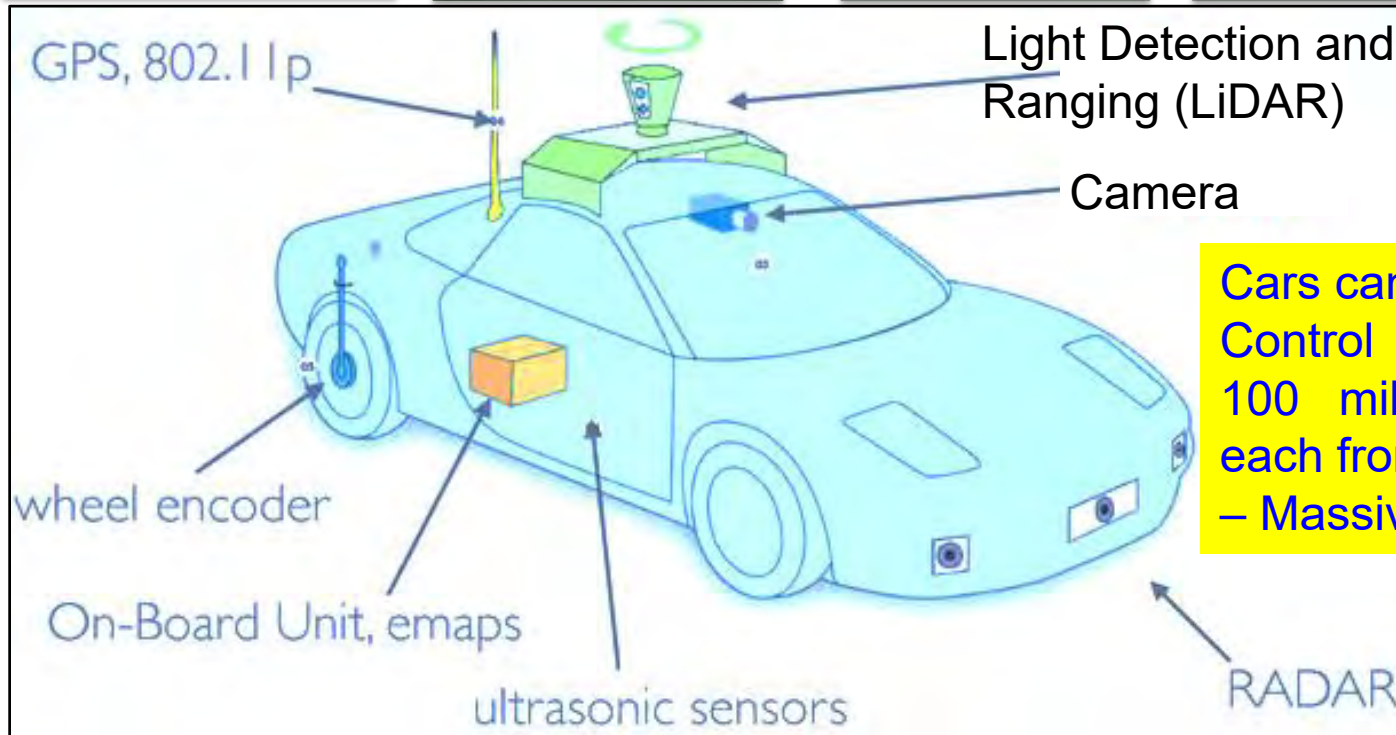
Replay

Relay

Jamming

Spoofing

Tracking



Cars can have 100 Electronic Control Units (ECUs) and 100 million lines of code, each from different vendors – Massive security issues.

Source: <http://www.computerworld.com/article/3005436/cybercrime-hacking/black-hat-europe-it-s-easy-and-costs-only-60-to-hack-self-driving-car-sensors.html>

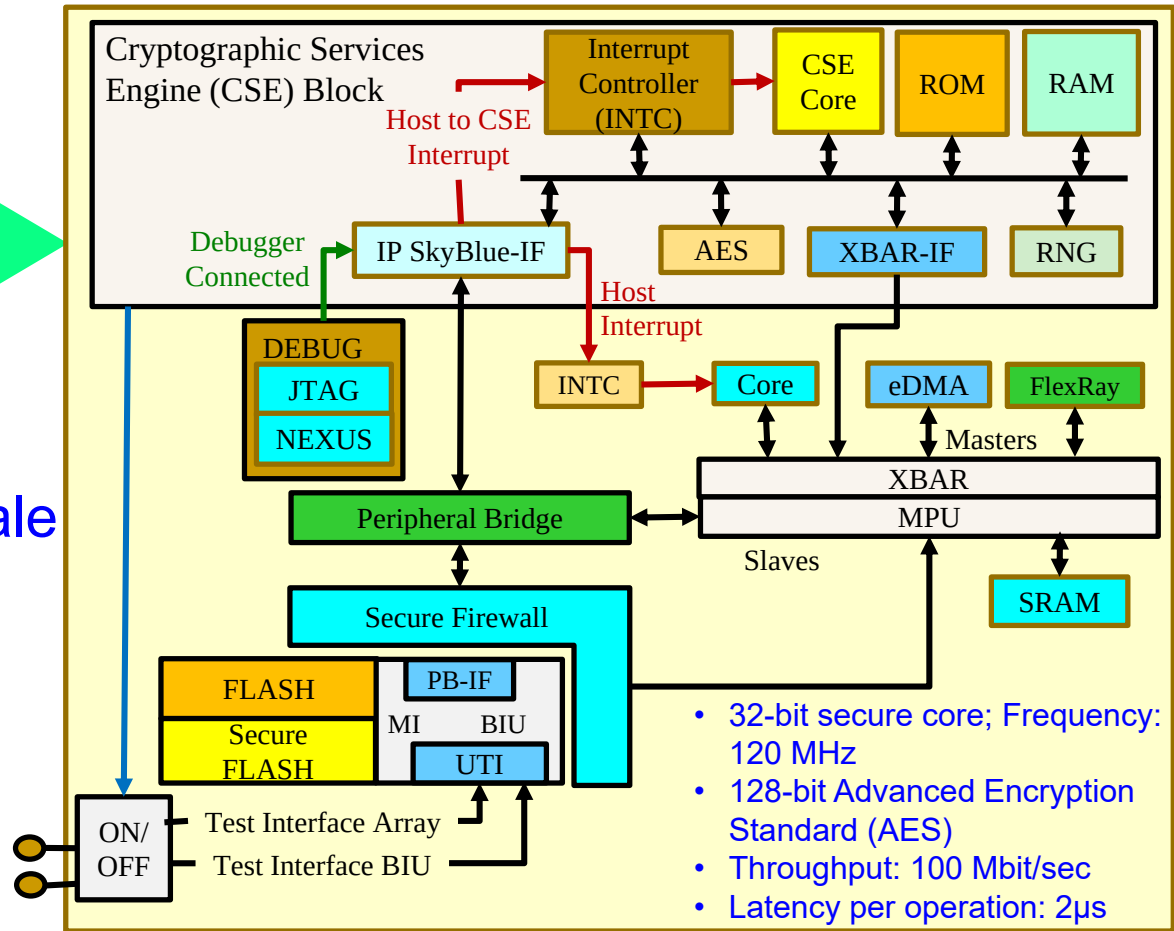
Source: <https://www.mcafee.com/us/resources/white-papers/wp-automotive-security.pdf>

Source: Petit 2015: IEEE-TITS Apr 2015

Autonomous Car Security – Cryptographic Hardware

Cryptographic Services Engine (CSE) Block

Qorivva MPC564xB/C Family from NXP/Freescale



Source: http://www.nxp.com/assets/documents/data/en/supporting-information/DWF13_AMF_AUT_T0112_Detroit.pdf

Smart Healthcare - Security and Privacy Issue



Selected Smart Healthcare Security/Privacy Challenges

Data Eavesdropping

Data Confidentiality

Data Privacy

Location Privacy

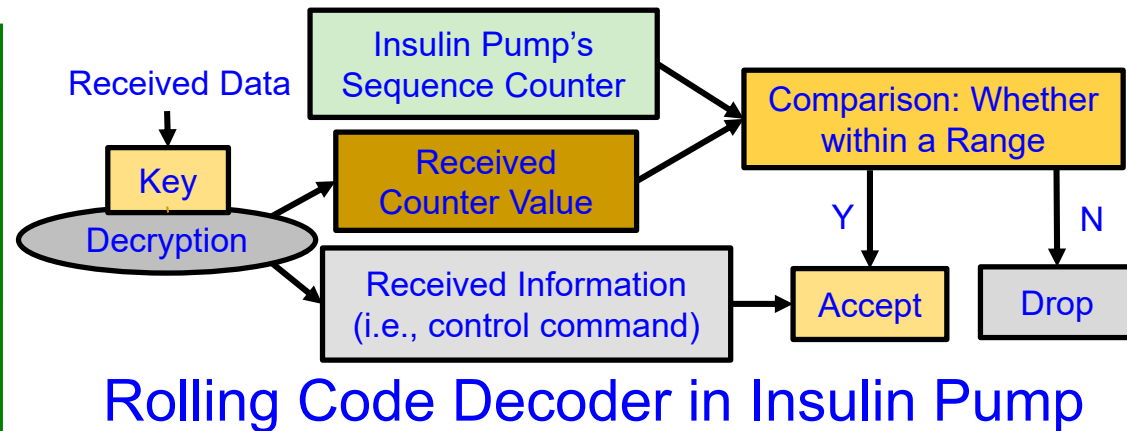
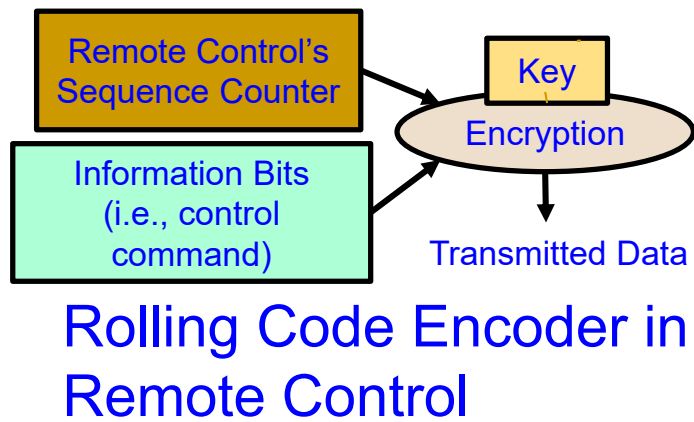
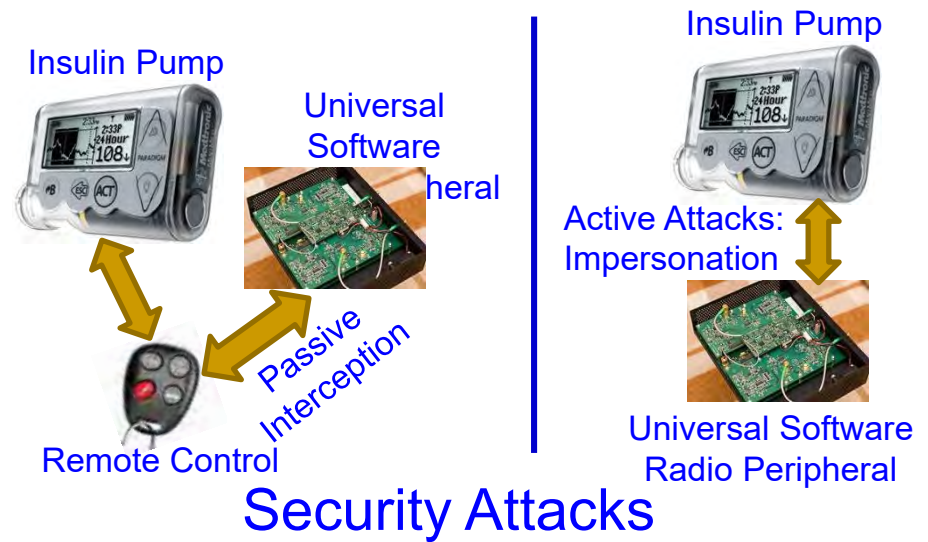
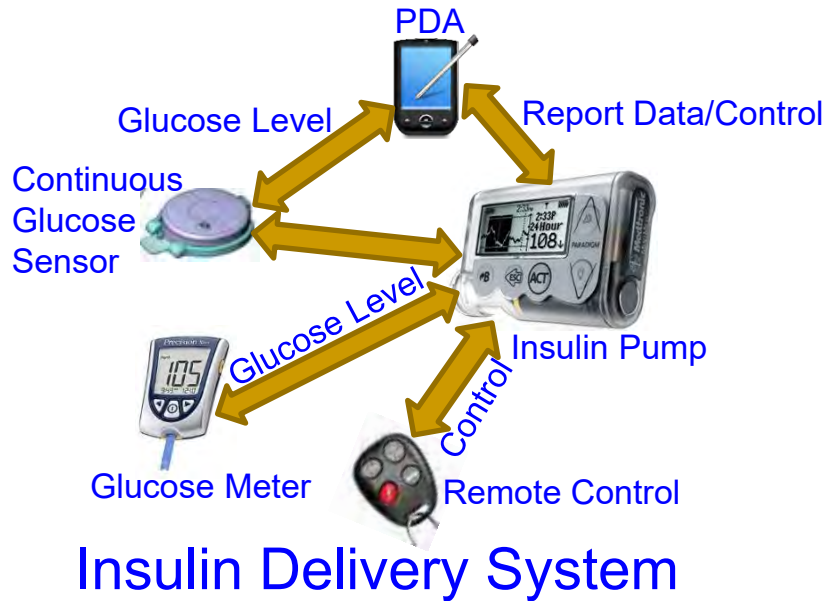
Identity Threats

Access Control

Unique Identification

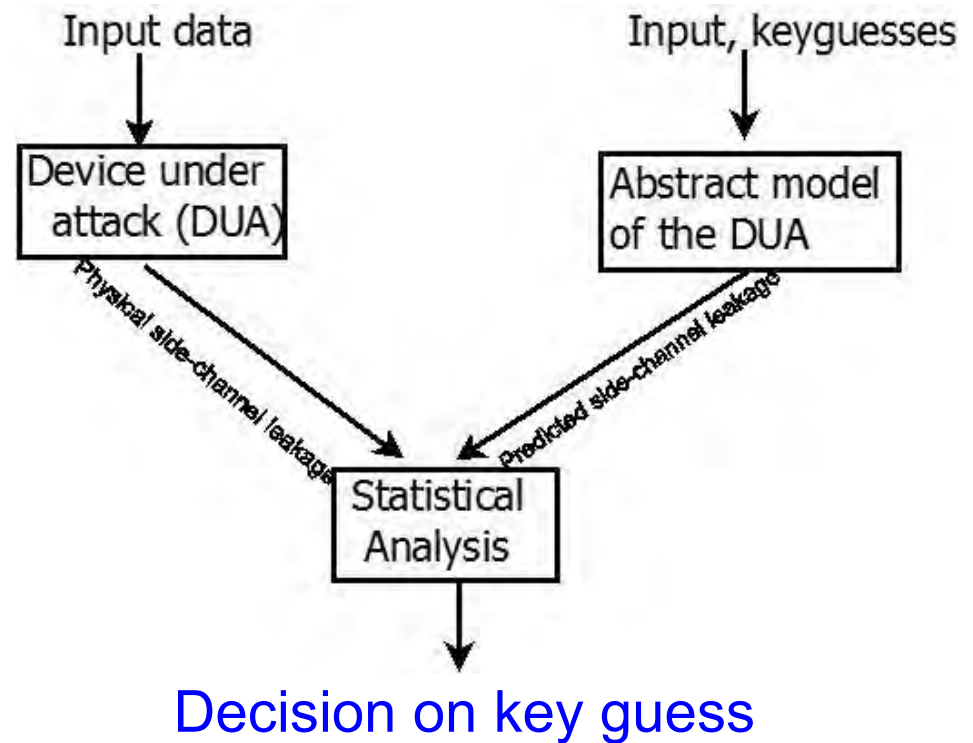
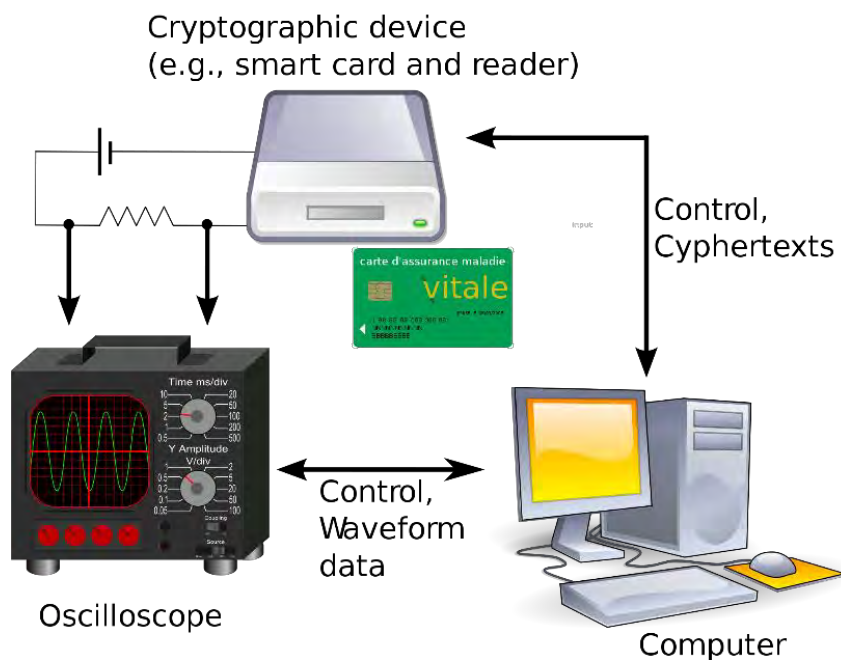
Data Integrity

Smart Healthcare Security

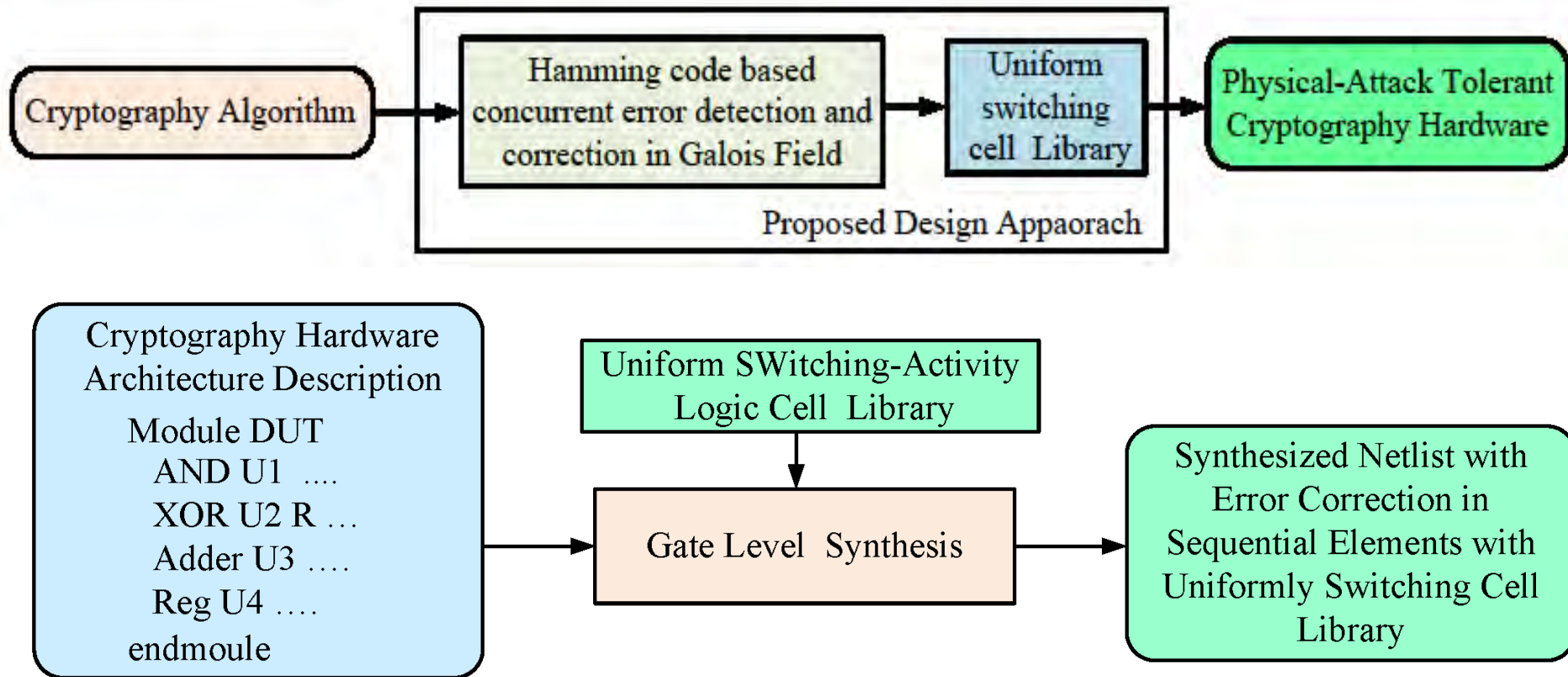


Source: Li 2011, e-Health 2011

Side Channel Attacks – Differential and Correlation Power Analysis (DPA/CDA)

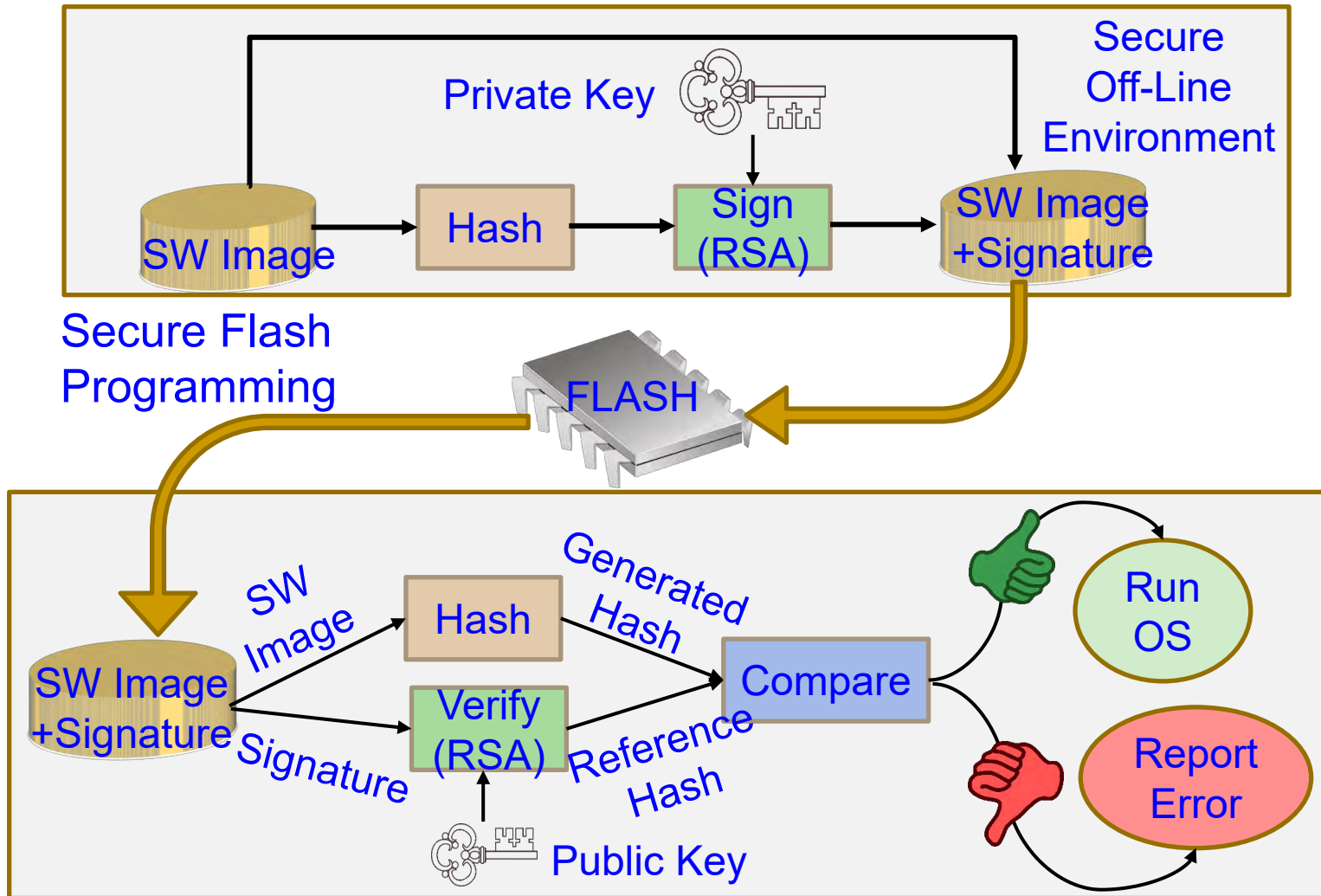


DPA Resilience Hardware - Synthesis Flow



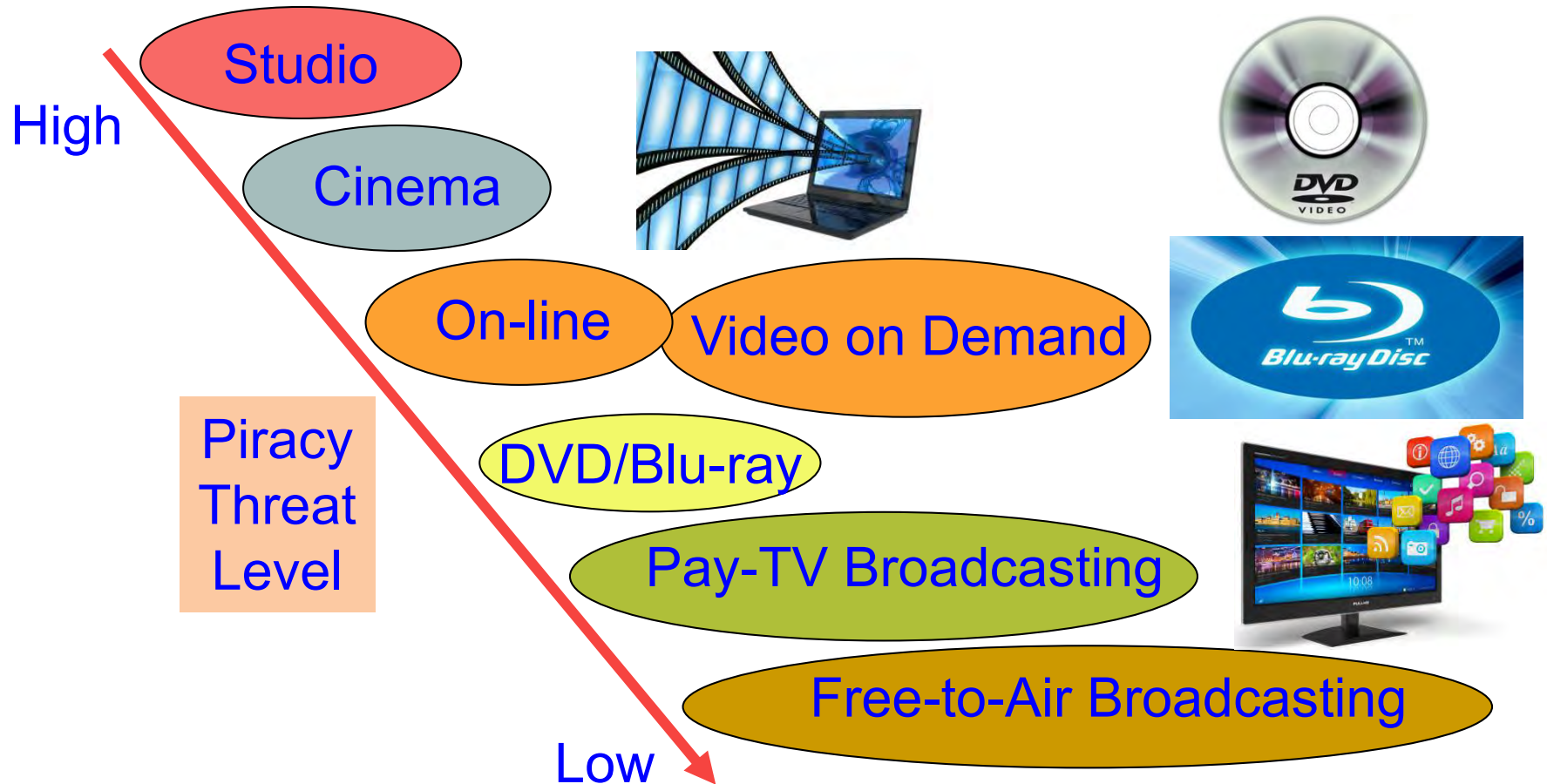
Source: Mohanty 2013, Elsevier CEE 2013

Firmware Security



Source: <https://www.nxp.com/docs/en/white-paper/AUTOSECURITYWP.pdf>

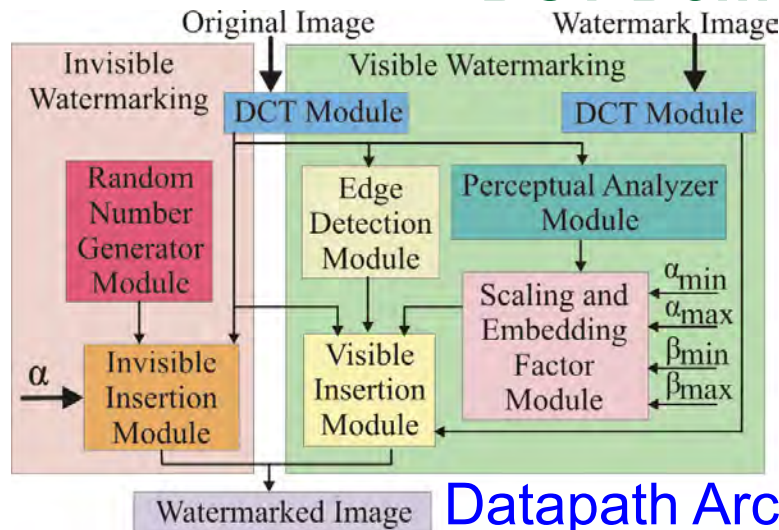
Multimedia Piracy – Movie/Video



“Film piracy cost the US economy \$20.5 billion annually.”

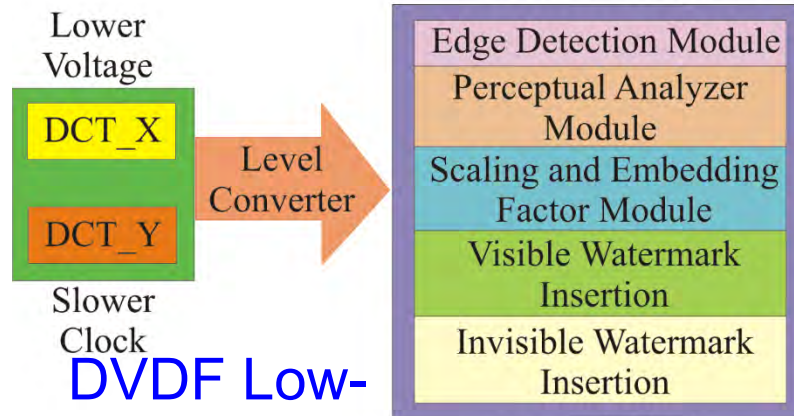
Source: http://www.ipi.org/ipi_issues/detail/illegal-streaming-is-dominating-online-piracy

Copyright Protection Hardwares – DCT Domain Watermarking



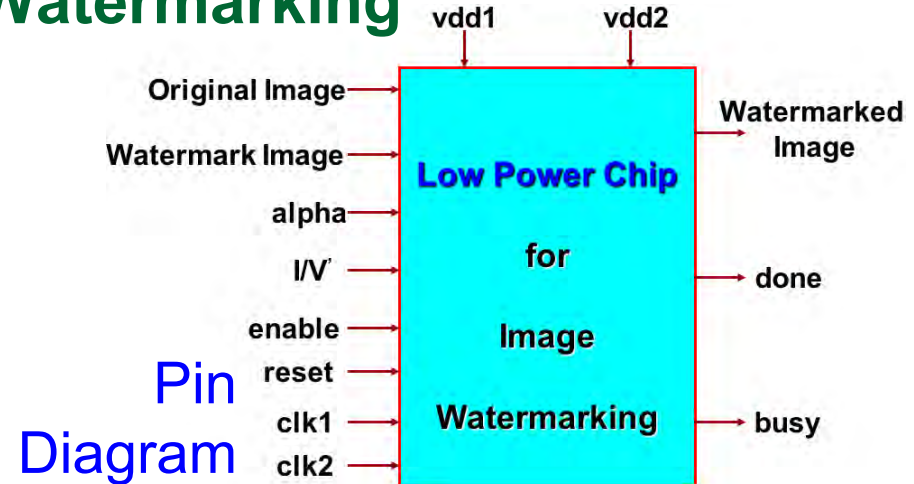
Datapath Architecture

Normal Voltage

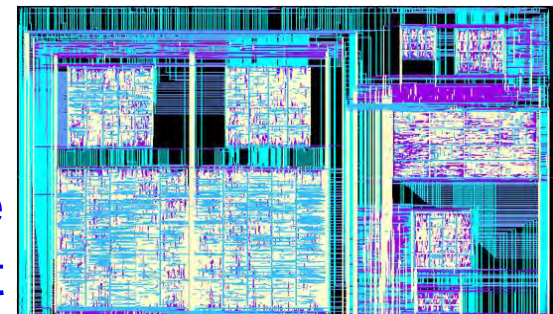


DVDF Low-Power Design

Normal Clock
Source: Mohanty 2006, TCASII May 2006



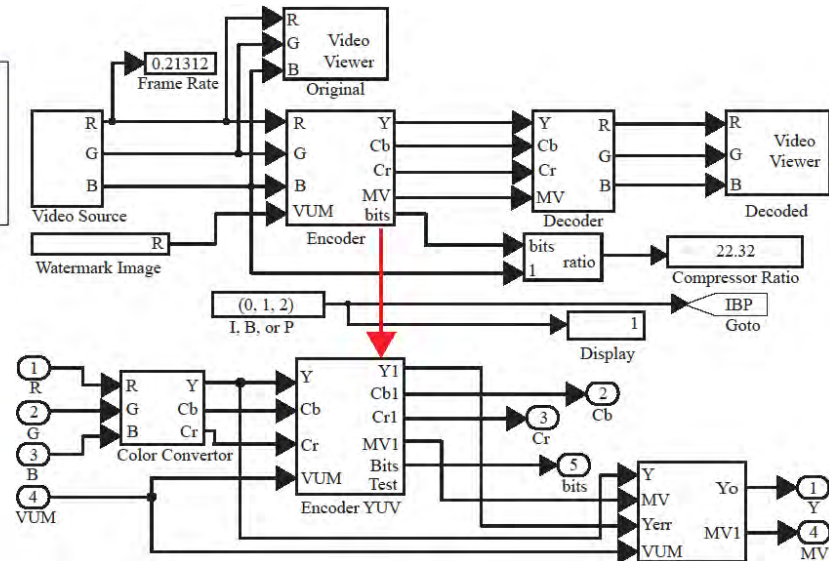
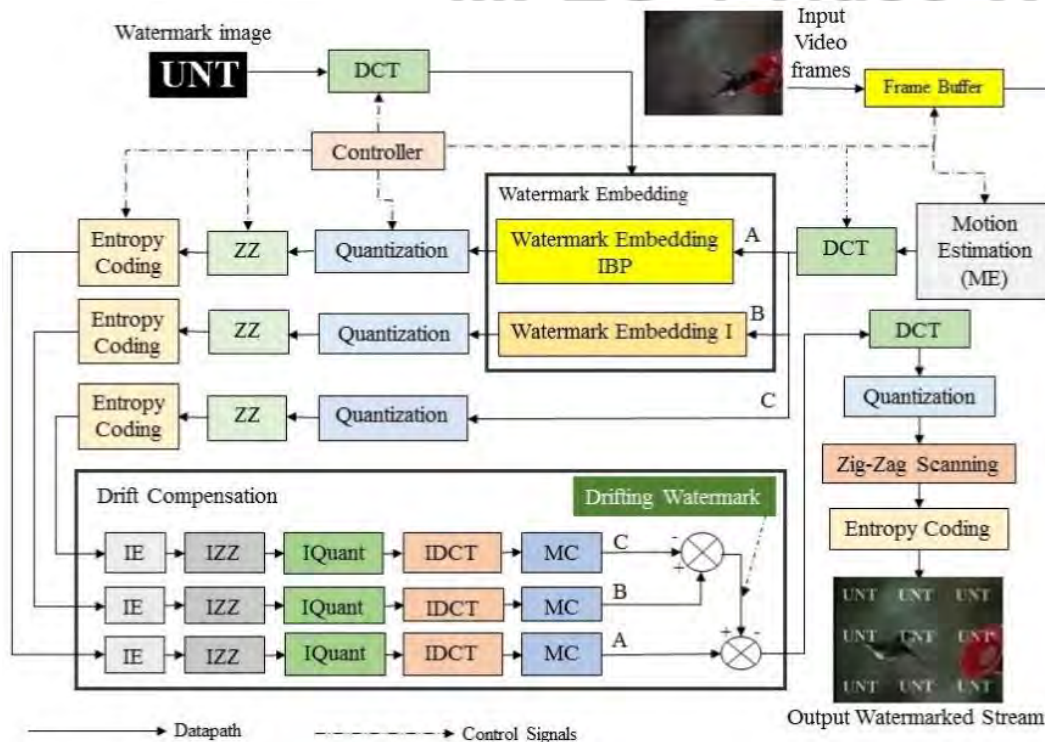
Pin Diagram



Hardware Layout

Physical Design Data
Total Area : 16.2 sq mm
No. of Transistors: 1.4 million
Power Consumption: 0.3 mW

Copyright Protection Hardware – MPEG-4 Video Watermarking



Video Watermarking Architecture: Simulink Model

Video Watermarking Architecture Datapath

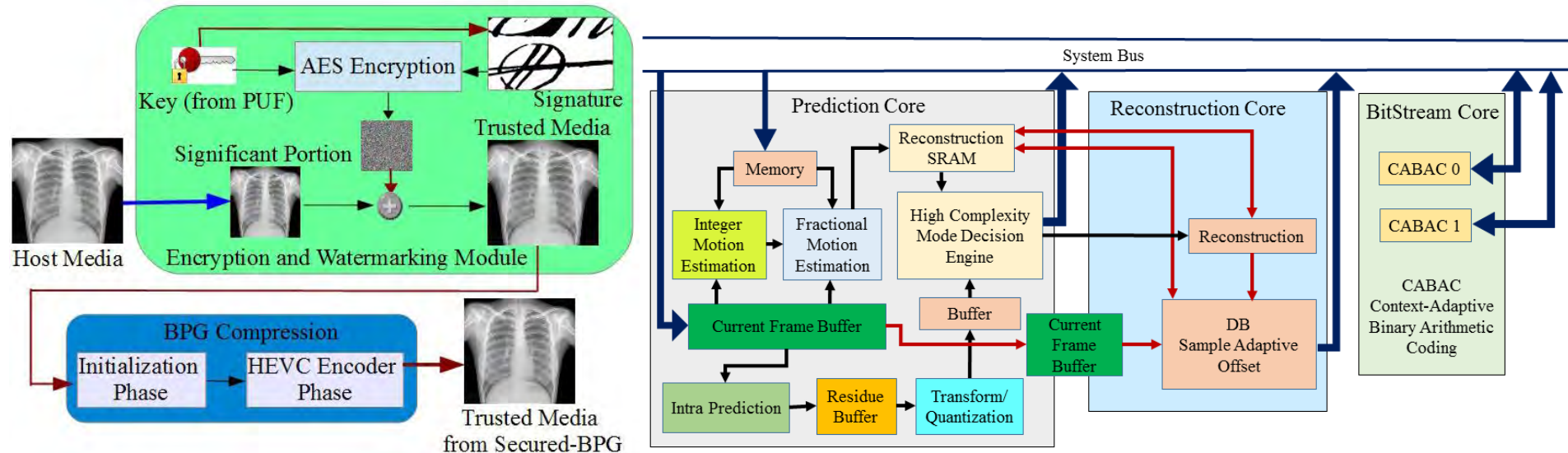
FPGA Prototyping

Throughput: 44 frames/sec

Logic Elements in FPGA Prototyping : 28322

Source: Mohanty 2011, JSS May 2011

DRM Hardware - Secure Better Portable Graphics (SBPG)

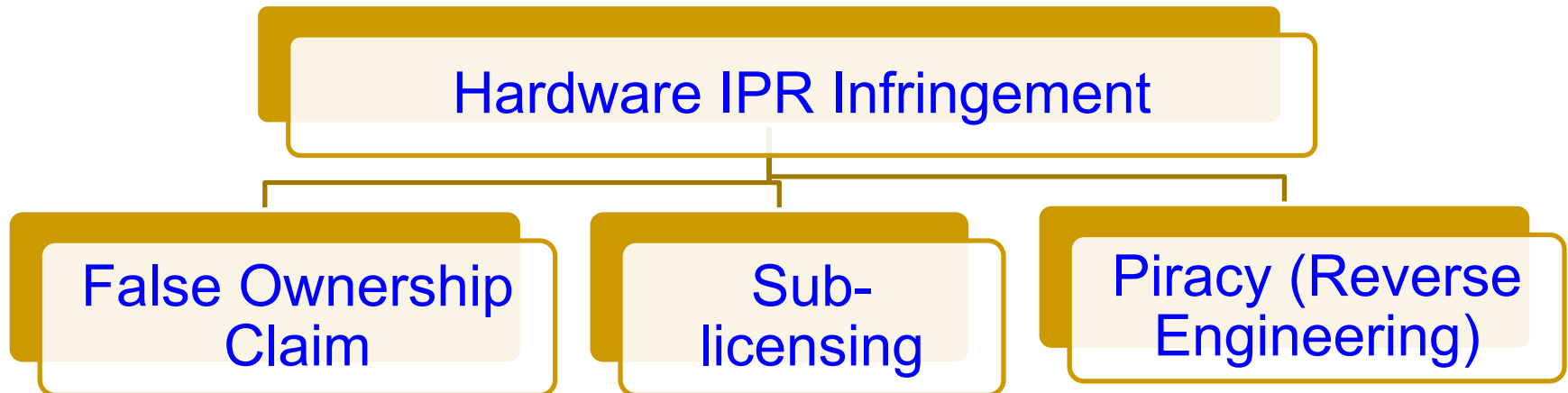
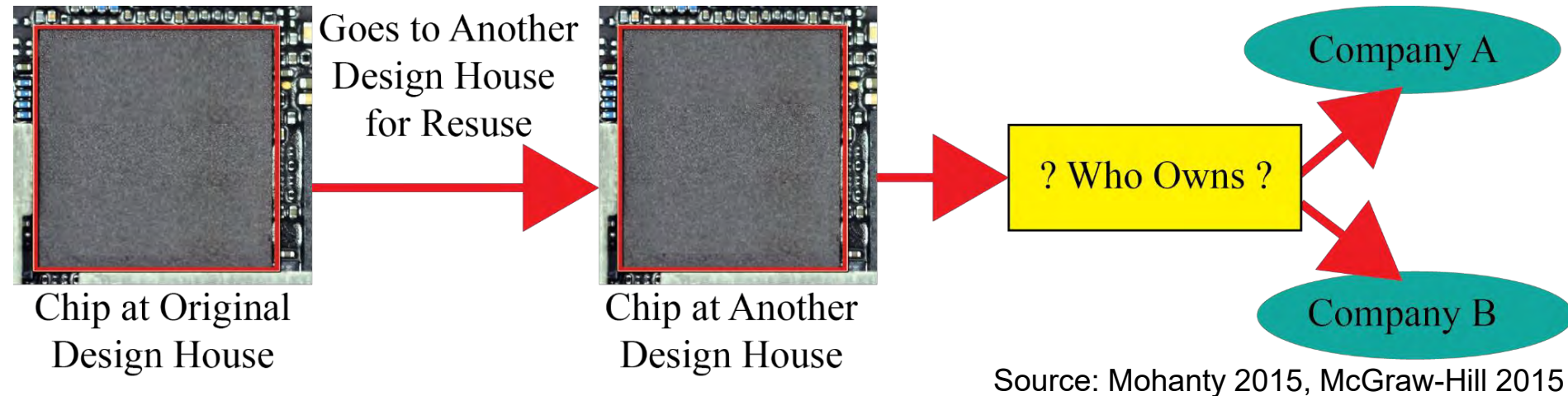


Idea of Secure BPG (SBPG) High-Efficiency Video Coding Architecture

Simulink Prototyping
Throughput: 44 frames/sec
Power Dissipation: 8 nW

Source: Mohanty 2016, ISVLSI 2016 and EuroSimE 2016

Hardware IP Right Infringement



Hardware Reverse Engineering



Source:
<http://legacy.lincolinteractive.org/html/CES%20Introduction%20to%20Engineering/Unit%203/u3l7.html>

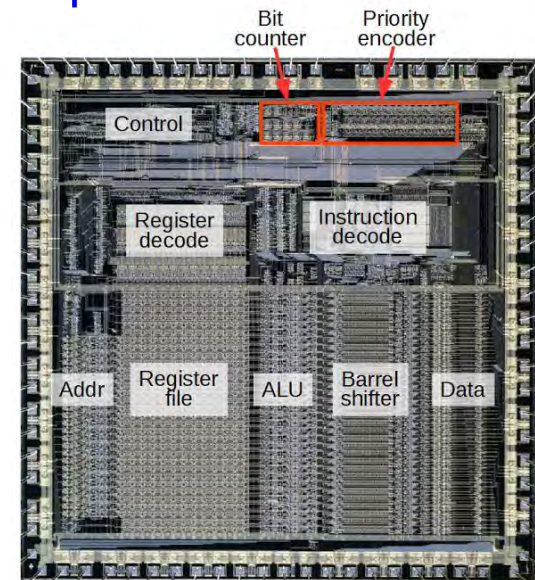
CE System disassembly
Subsystem identification,
modification



Source:
<https://www.slideshare.net/SOURCEConference/slicing-into-apple-iphone-reverse-engineering>

Source: http://grandideastudio.com/wp-content/uploads/current_state_of_hh_slides.pdf

Chip-Level Modification



Source: <http://pic-microcontroller.com/counting-bits-hardware-reverse-engineering-silicon-arm1-processor/>

Counterfeit Hardware

2014 Analog Hardware Market (Total Shipment Revenue US \$)



Wireless Market
\$18.9 billion (34.8%)



Consumer Electronics
\$9.0 billion (16.6%)



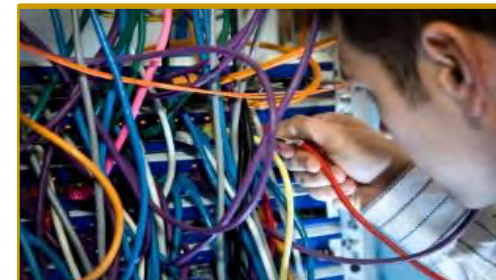
Industrial Electronics
\$8.9 billion (16.5%)



Automotive
\$8.5 billion (15.7%)



Data Processing
\$6.0 billion (11%)

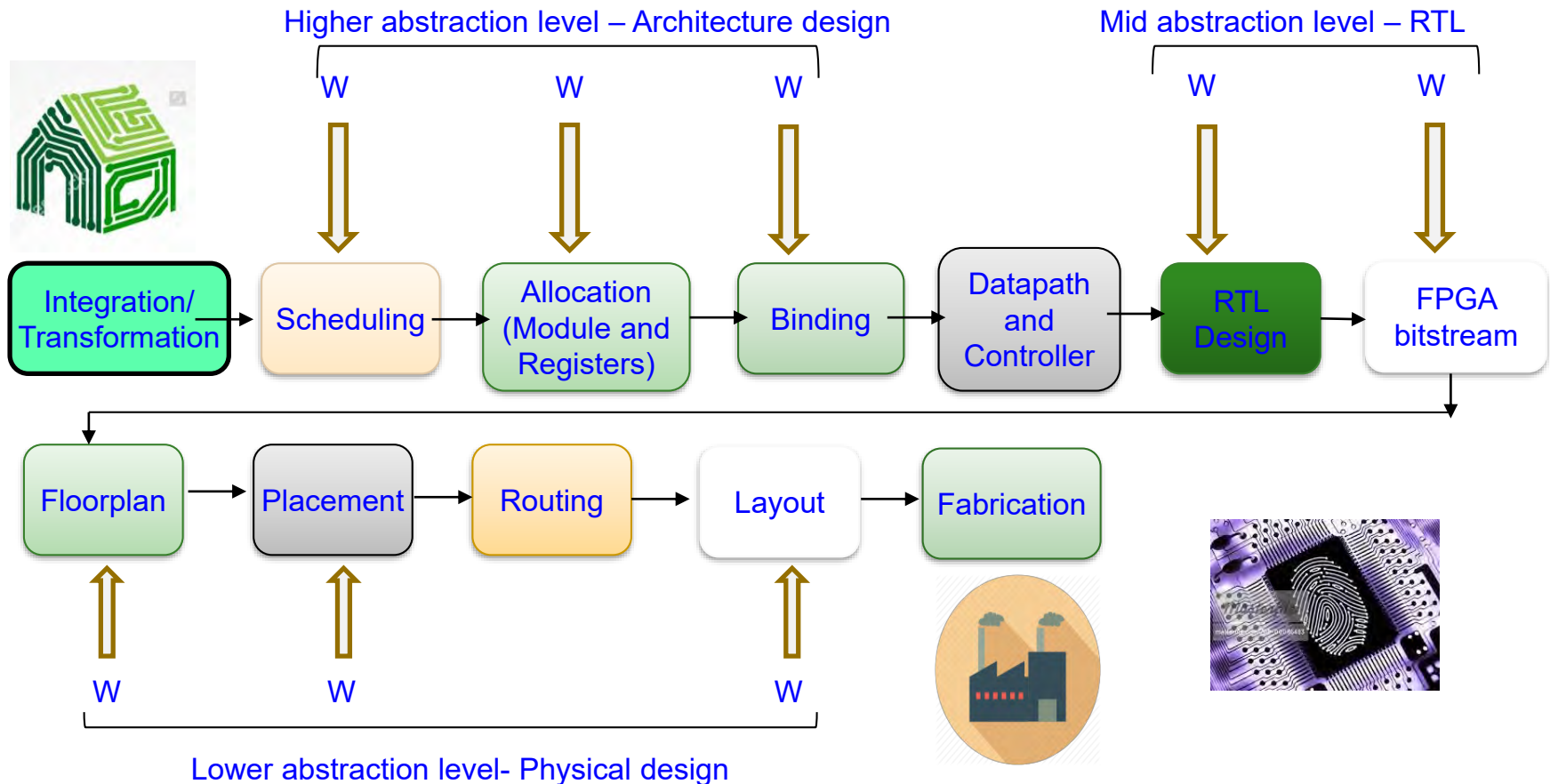


Wired Communications
\$2.9 billion (5.4%)

Source: <https://www.slideshare.net/rorykingihs/ihs-electronics-conference-rory-king-october>

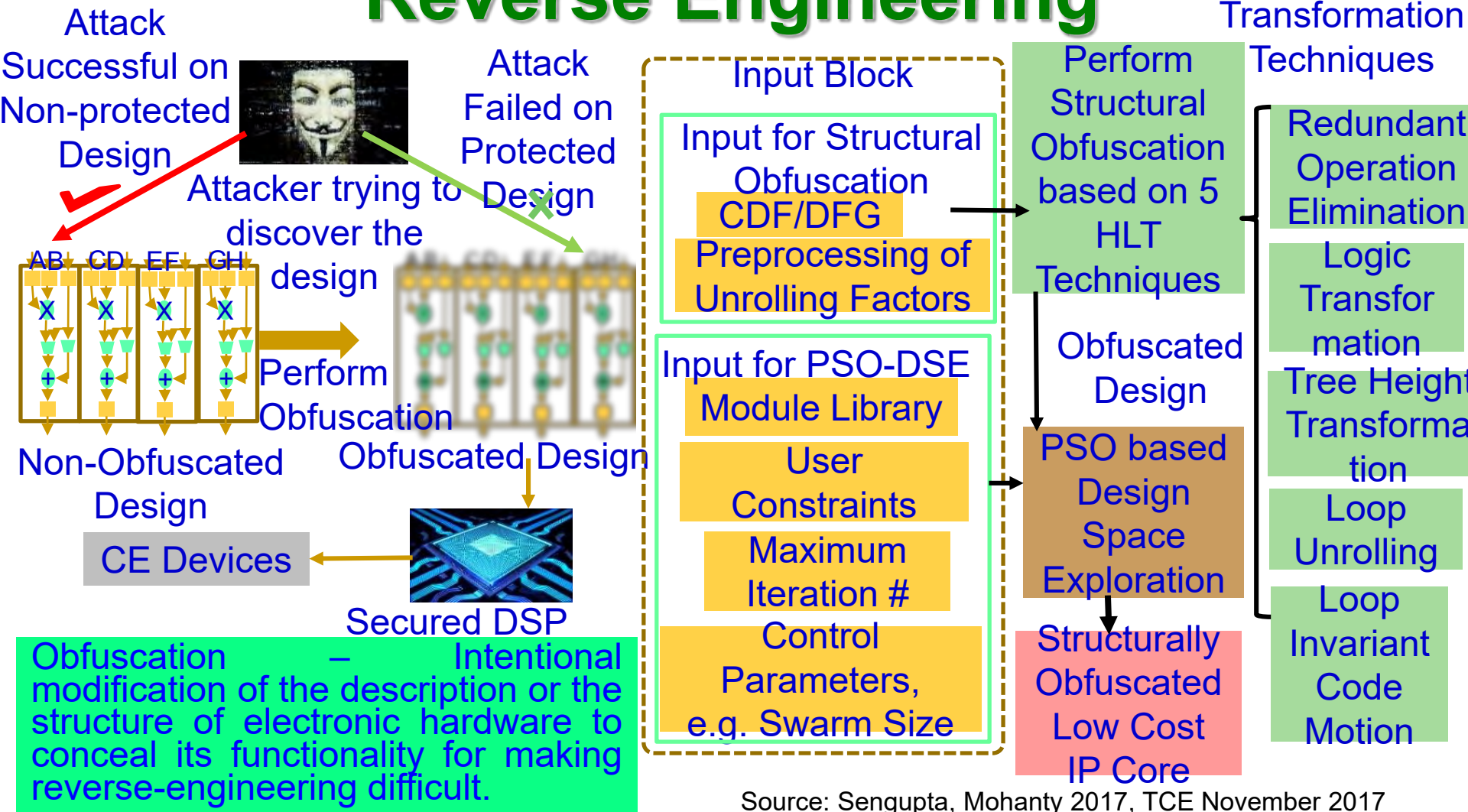
Top counterfeits could have impact of
\$300B on the semiconductor market.

Digital Hardware - Watermark



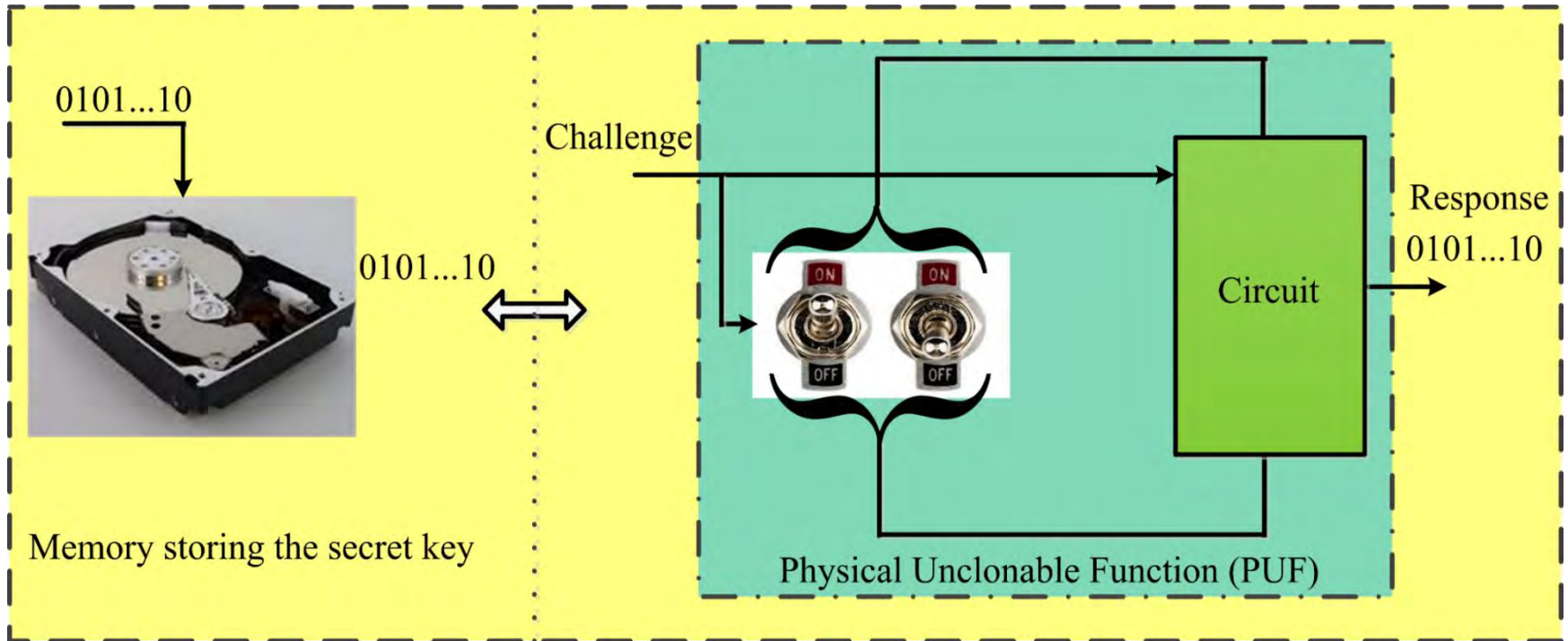
Source: Mohanty 2017: CE Magazine October 2017

Digital Hardware Synthesis to Prevent Reverse Engineering



Source: Sengupta, Mohanty 2017, TCE November 2017

PUF - Principle

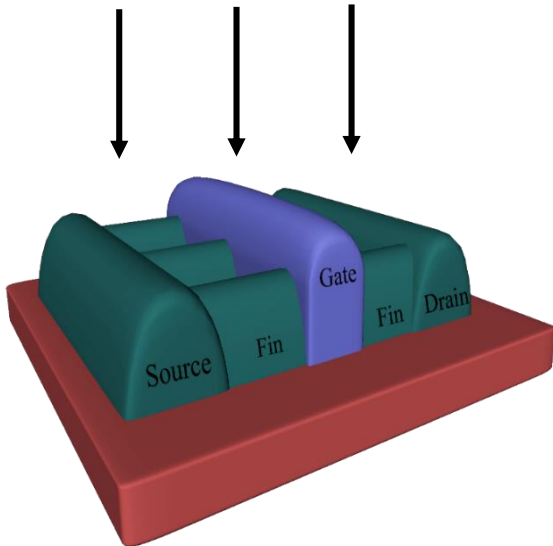


PUFs don't store keys in digital memory, rather derive a key based on the physical characteristics of the hardware; thus secure.

Source: Mohanty 2017, IEEE Potentials Nov-Dec 2017

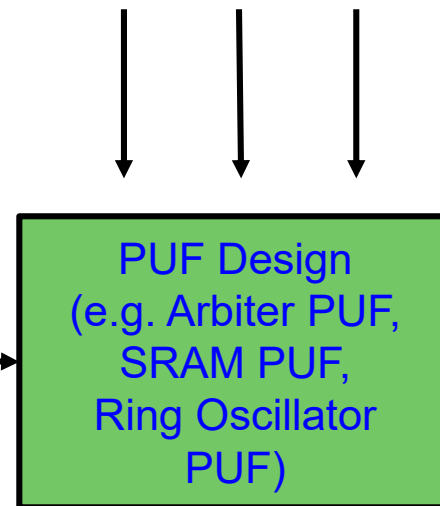
PUF - Principle

Manufacturing Variations
(e.g. Oxide Growth, Ion
Implantation, Lithography)



Parameters
Affected
Due to
Variations
(e.g. Length,
Gate-Oxide
Thickness,
Fin Height,
Fin Width)

Challenge Inputs
(Inputs given to PUF Module,
e.g. Select line of Multiplexer)



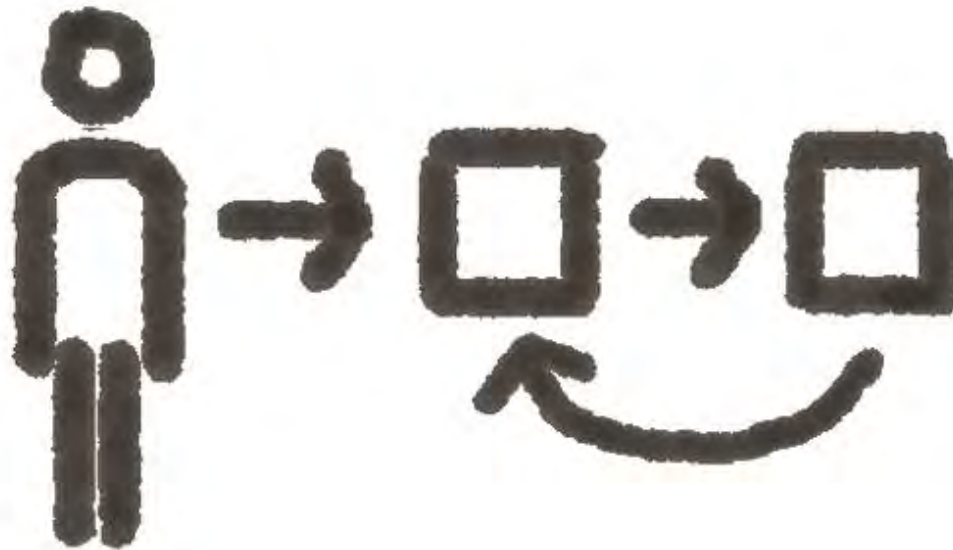
Challenge
Response
(Outputs from a
PUF Module)

Random
Binary Output
010101 ...

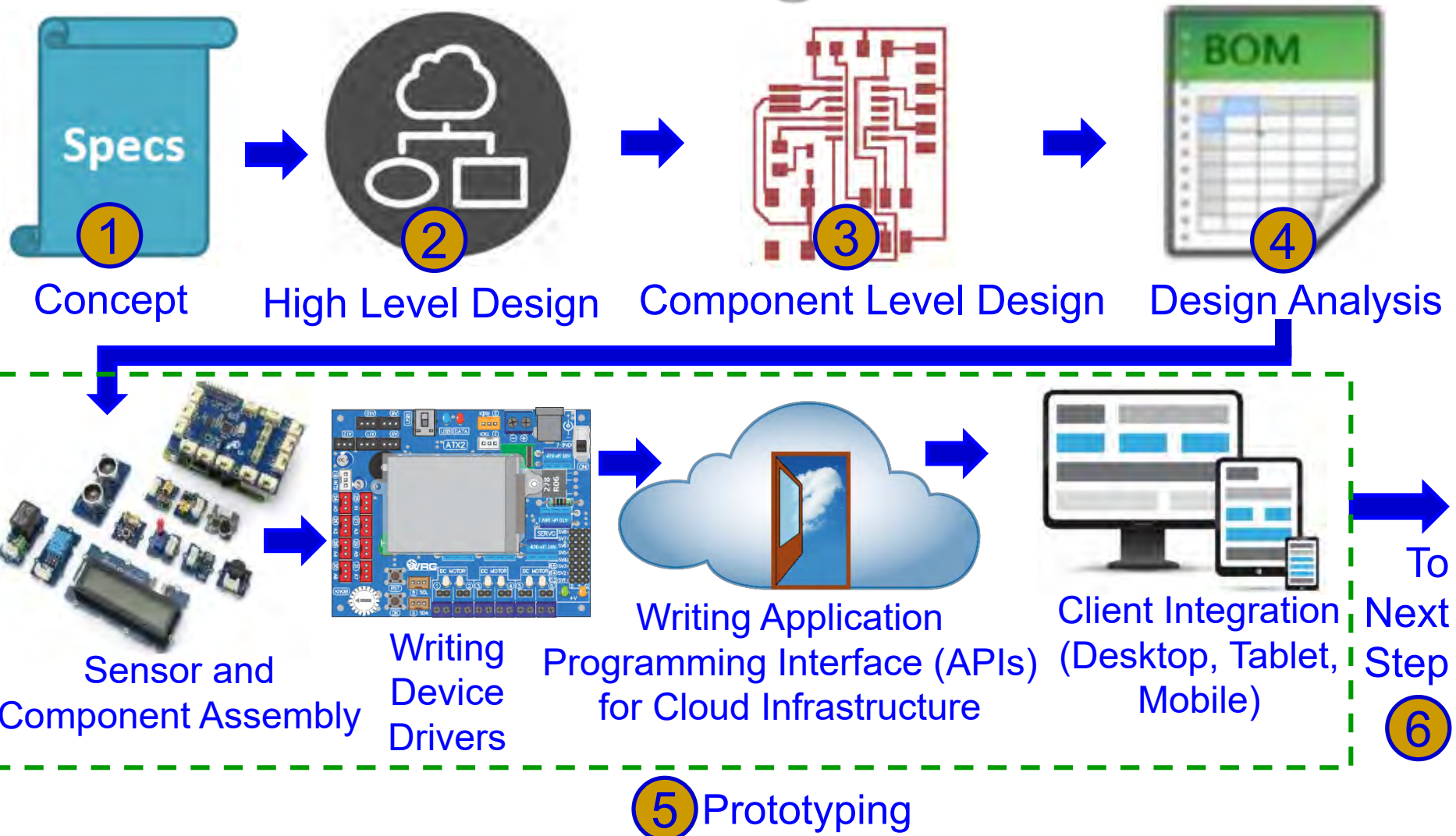
Silicon manufacturing process variations are
turned into a feature rather than a problem.

Source: Mohanty 2017, Springer ALOG 2017

Design Flow



IoT – Design Flow



Source: <http://events.linuxfoundation.org/sites/events/files/slides/Design%20-%20End-to-End%20-%20IoT%20Solution%20-%20Shivakumar%20Mathapathi.pdf>

IoT – Design Flow



⑥ Field Testing

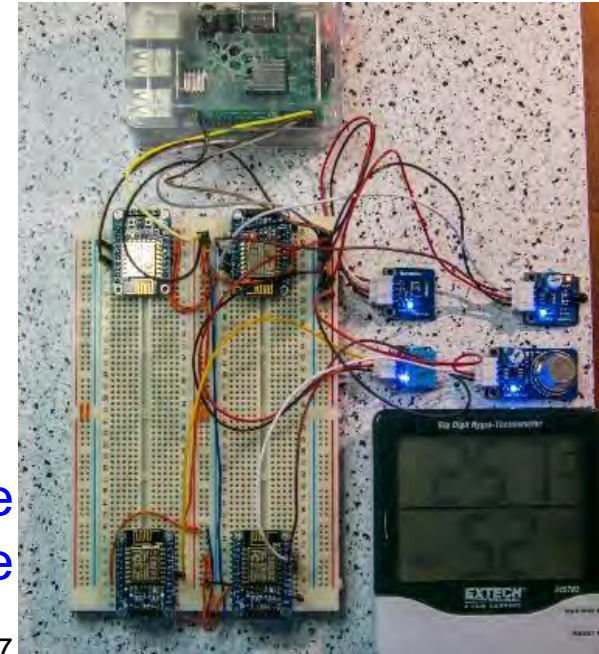
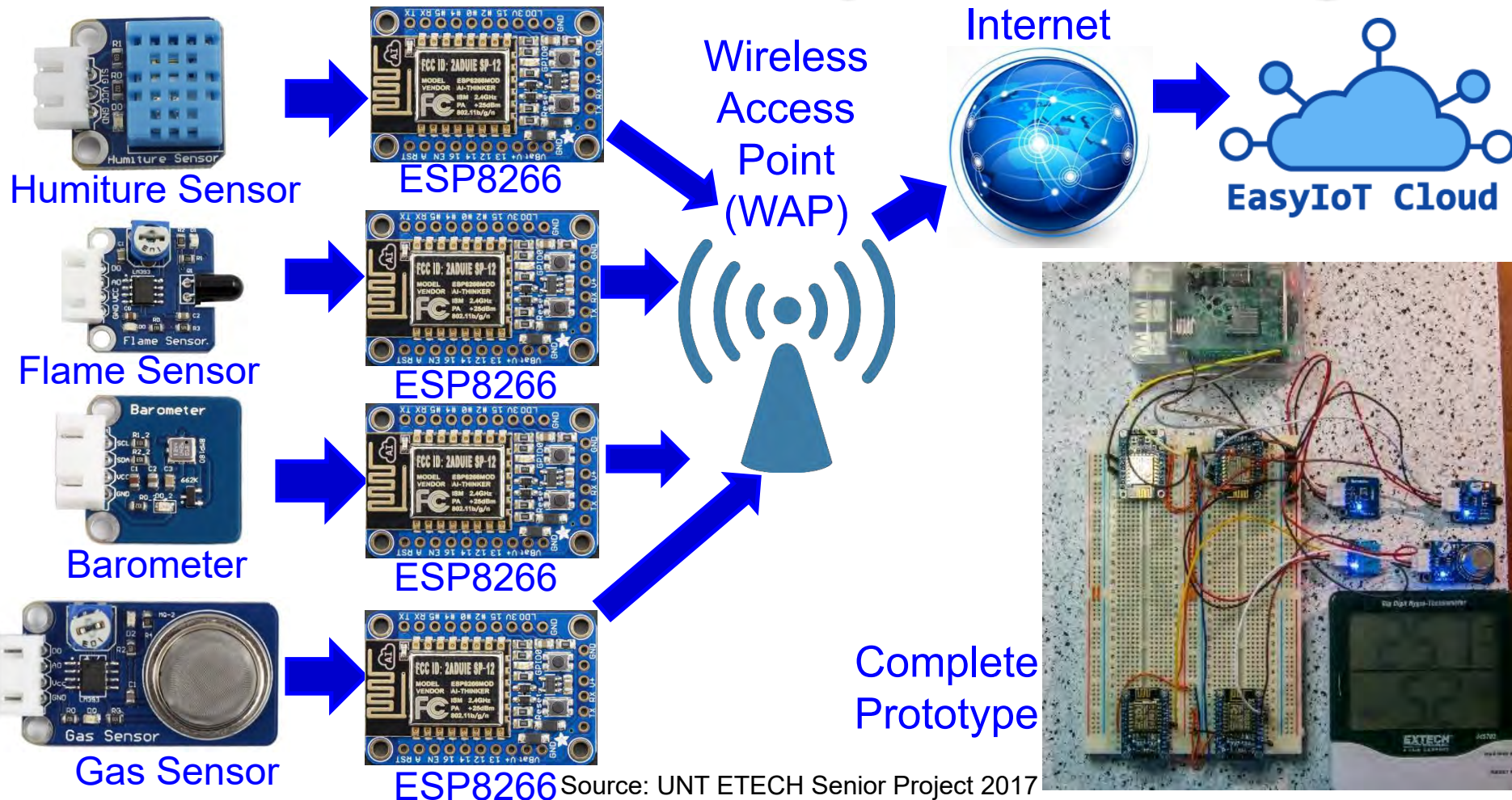
⑦ Release of Beta Version

⑧ Production

⑨ Release and Documentation

Source: <http://events.linuxfoundation.org/sites/events/files/slides/Design%20-%20End-to-End%20%20IoT%20Solution%20-%20Shivakumar%20Mathapathi.pdf>

IoT Design – Case Study – Indoor Air Quality Monitoring



Hardware for IoT

IoT
Hardware
Domains

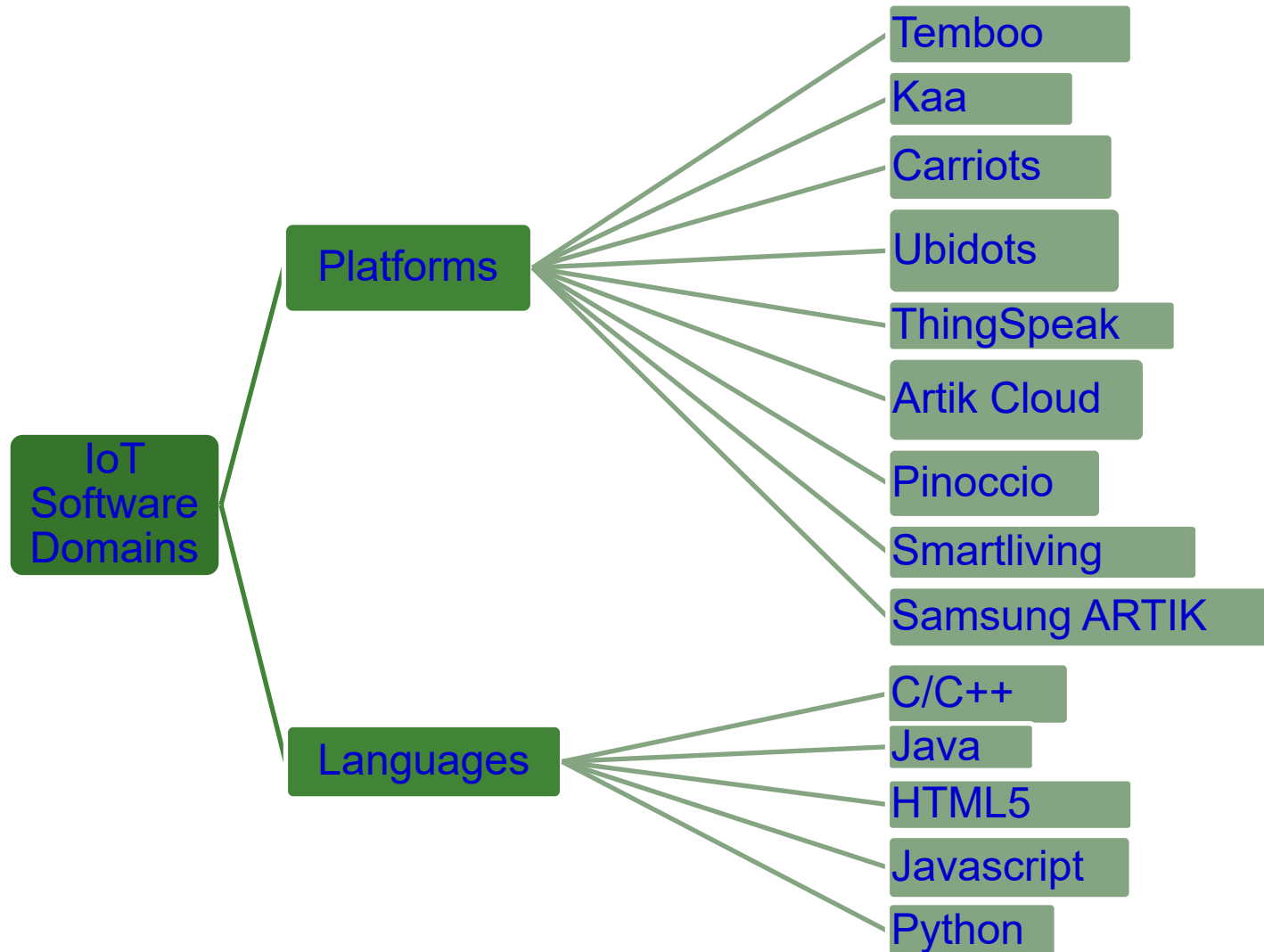
Embedded Systems and Boards (e.g. Arduino Yun, Raspberry Pi, BeagleBone, Samsung ARTIK)

Wearable Devices and Gadgets (e.g. Samsung Gear 2, FitBit Flex, FLORA, iWallet)

Features	Processor/Microcontroller	Graphics Processing Unit	Clock Speed	Size	Memory	RAM	Supply Voltage	Listed Price
SparkFun Blynk Board	Tensilica L106 32-b	No	26 MHz	51 mm x 42 mm	4 MB	128 KB	5 V via micro-USB/ Li-Po connector and charging circuit	US\$29.95
Arduino Yun	ATmega32u4 and Atheros AR9331 (for Linux)	No	16 MHz and 400 MHz	73 mm x 53 mm	32 KB and 16 MB + micro-SD	64 MB DDR2	5 V via micro-USB	US\$58
Raspberry Pi 3	Broadcom BCM2837 and ARM Cortex-A53 64-b Quad Core	VideoCore IV @ 300/400 MHz	1.2 GHz	85 mm x 56 mm	Micro-SD	1 GB LPDDR2	5 V via micro-USB	US\$35
cloudBit	Freescall i.MX233 (ARM926EJ-S core)	No	454 MHz	55 mm x 19 mm	Micro-SD slot with 4-GB micro-SD	64 MB	5 V via micro-USB	US\$59.95
Photon	STM32F205 120Mhz ARM Cortex M3	No	120 MHz	36.5 mm x 20.3 mm	1 MB	128 KB	5 V via micro-USB	US\$19
BeagleBone Black	AM335x ARM Cortex-A8	PowerVR SGX530	1 GHz	86 mm x 56 mm	4 GB 8-b eMMC, micro-SD	512 MB DDR3	5 V via mini-USB	US\$49
Pinoccio	ATmega256RFR2	No	16 MHz	70 mm x 25 mm	256 KB	32 KB	5 V via micro-USB/ Li-Po connector and charging circuit	US\$109
UDOO	Freescall i.MX 6 ARM Cortex-A9 and Atmel SAM3X8E ARM Cortex-M3	Vivante GC 2000 for 3-D + GC 355 for 2-D (vector graphics) + GC 320 for 2-D	1 GHz	110 mm x 85 mm	Micro-SD	1 GB DDR3	12 V	US\$135
Samsung Artik 10	ARM A15x4 and A7x4	Mali-T628 MP6 core	1.3 GHz and 1.0 GHz	39 mm x 29 mm	16 GB	2 GB LPDDR3	3.4-5 V	US\$100

Source: Singh 2017, CE Magazine, April 2017

Software for IoT



Source: Singh 2017, CE Magazine, April 2017

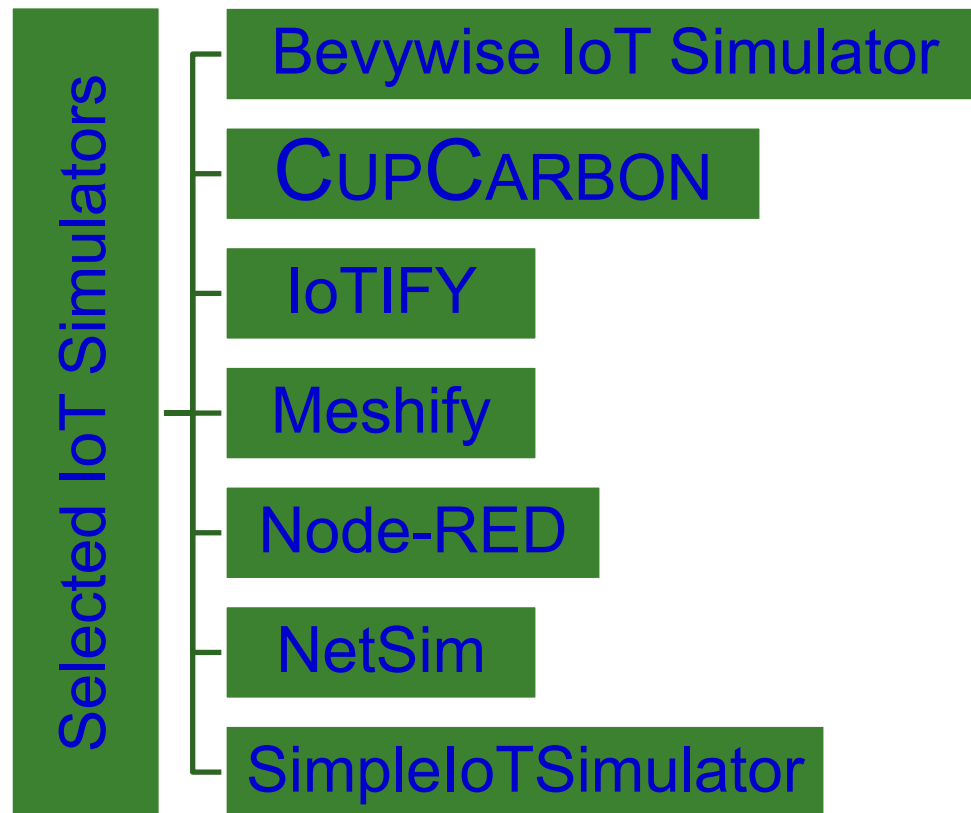
Tools and Solutions



IoT - Design & Simulation Challenges

- Traditional controllers and processors do not meet IoT requirements, such as multiple sensor, communication protocol, and security requirements.
- Existing tools are not enough to meet challenges such as time-to-market, complexity, cost of IoT.
- Can a framework be developed for simulation, verification, and optimization:
 - ❑ of individual (multidiscipline) “Things”
 - ❑ of IoT Components
 - ❑ of IoT Architecture

IoT Simulators



IoT Simulator - CUPCARBON

■ About

- ❑ CUPCARBON is a smart city and Internet of Things Wireless sensor network simulator (SCI-WSN)

■ Objective

- ❑ Design, Visualize, Debug
- ❑ Validate distributed algorithms
- ❑ Create environmental scenarios

■ Environments

- Design of mobility scenarios and the generation of natural events such as fires and gas as well as the simulation of mobiles such as vehicles and flying objects (e.g. UAVs, insects, etc.).
- A discrete event simulation of WSNs which takes into account the scenario designed on the basis of the first environment.



Source: <http://www.cupcarbon.com/>

IoT Simulators - Node-RED

■ About:

- ❑ Node-RED is a flow-based IoT Simulator.
- ❑ It is a programming tool for wiring together hardware devices, APIs and online services in new ways.
- ❑ The light-weight runtime is built on Node.js, taking full advantage of its event-driven, non-blocking model.

■ Editor:

- ❑ Browser-based editor.
- ❑ The flows created in Node-RED are stored using JSON which can be easily imported and exported for sharing with others.

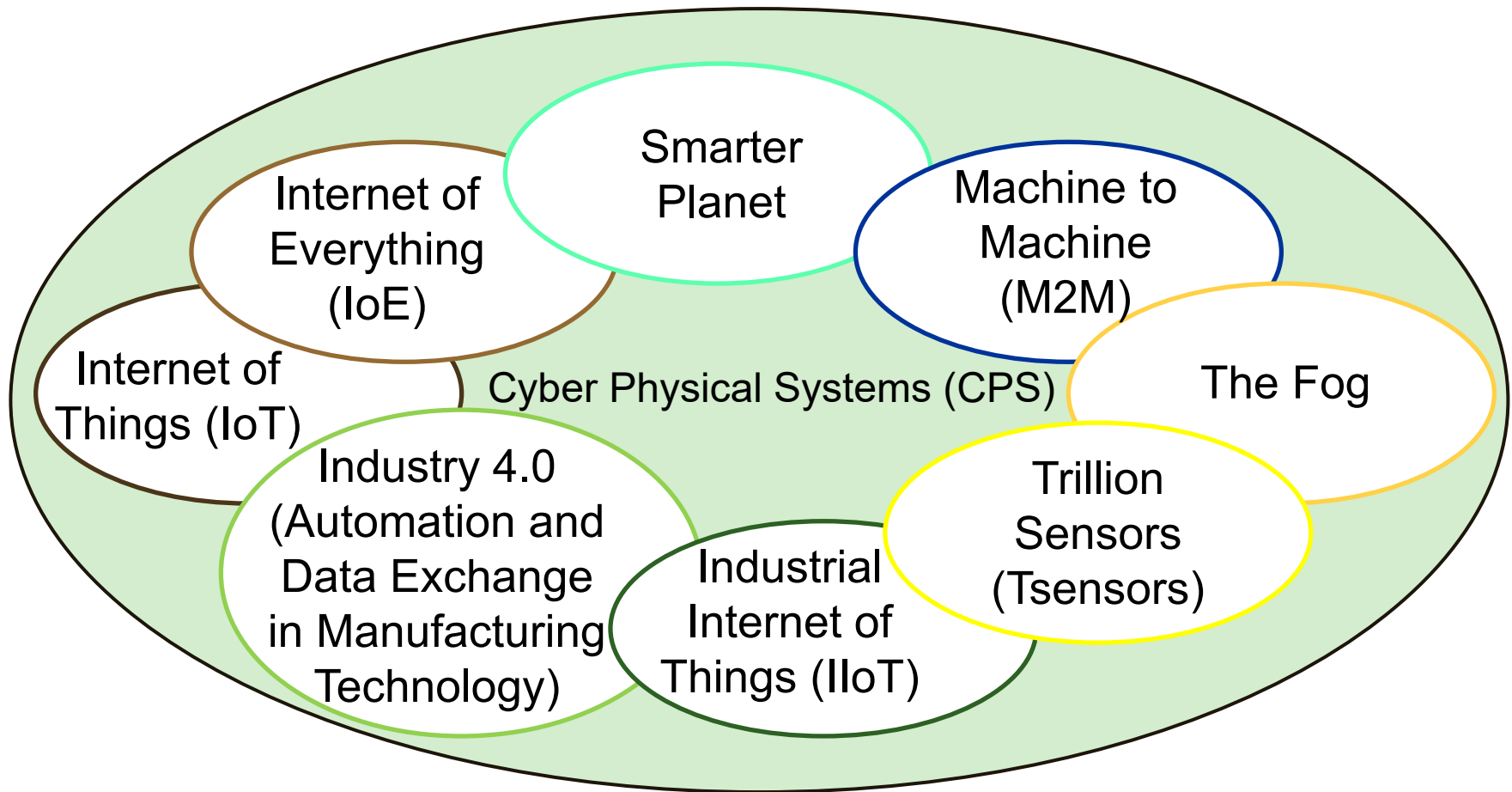
■ Advantages:

- ❑ Available for smaller computing devices such as Raspberry Pi.
- ❑ It takes moments to create cloud applications that combine services from across the platform.

Related Buzzwords



Some related Buzzwords



Source: Sangiovanni-Vincentelli 2016, ISC2 2016

IoT Vs Sensor Networks

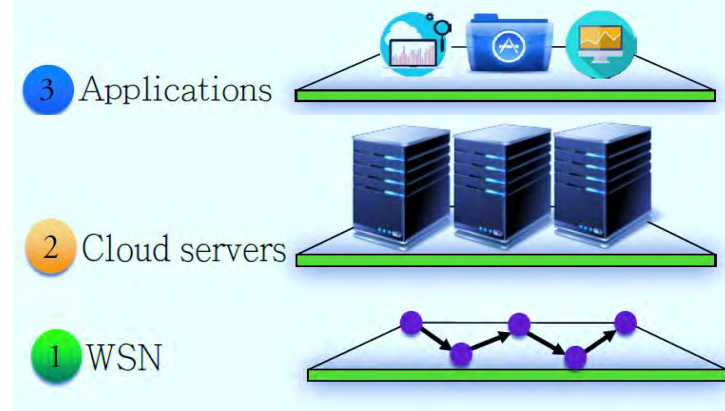
Wireless Sensor Networks (WSN)

- WSN is like the eyes and ears of the IoT.
- A network of small wireless electronic nodes which consists of different sensors.
- The purpose is to collect data from the environment.

IoT adds value to data!

IoT

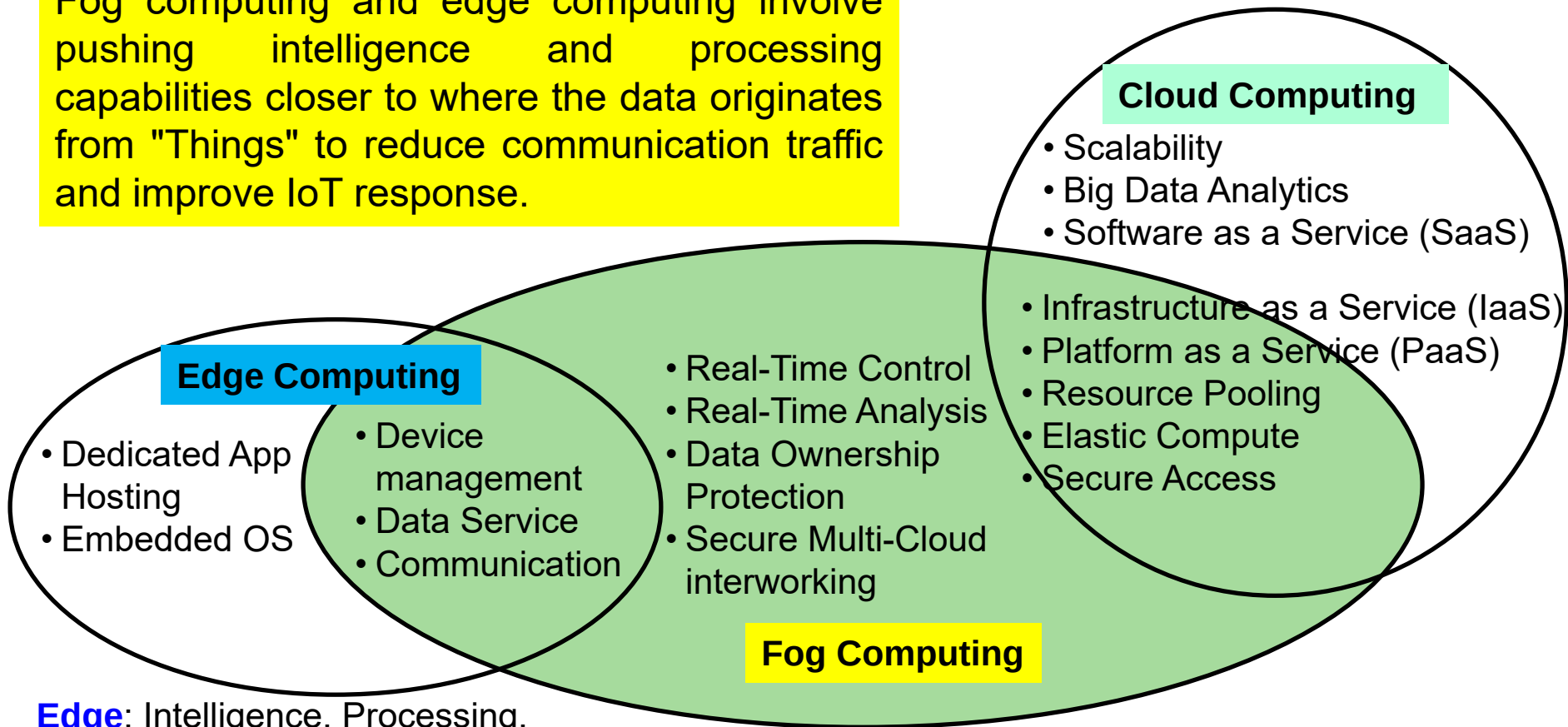
- IoT in a broad sense is like a brain.
- Store both real world data and can also be used to monitor the real world parameters and give meaningful interpretation.



Source: Nia 2017, IEEE TETC 2017

Fog Vs Edge Vs Cloud Computing

Fog computing and edge computing involve pushing intelligence and processing capabilities closer to where the data originates from "Things" to reduce communication traffic and improve IoT response.



Edge: Intelligence, Processing, and Communication - Devices like Programmable Automation Controllers (PACs)

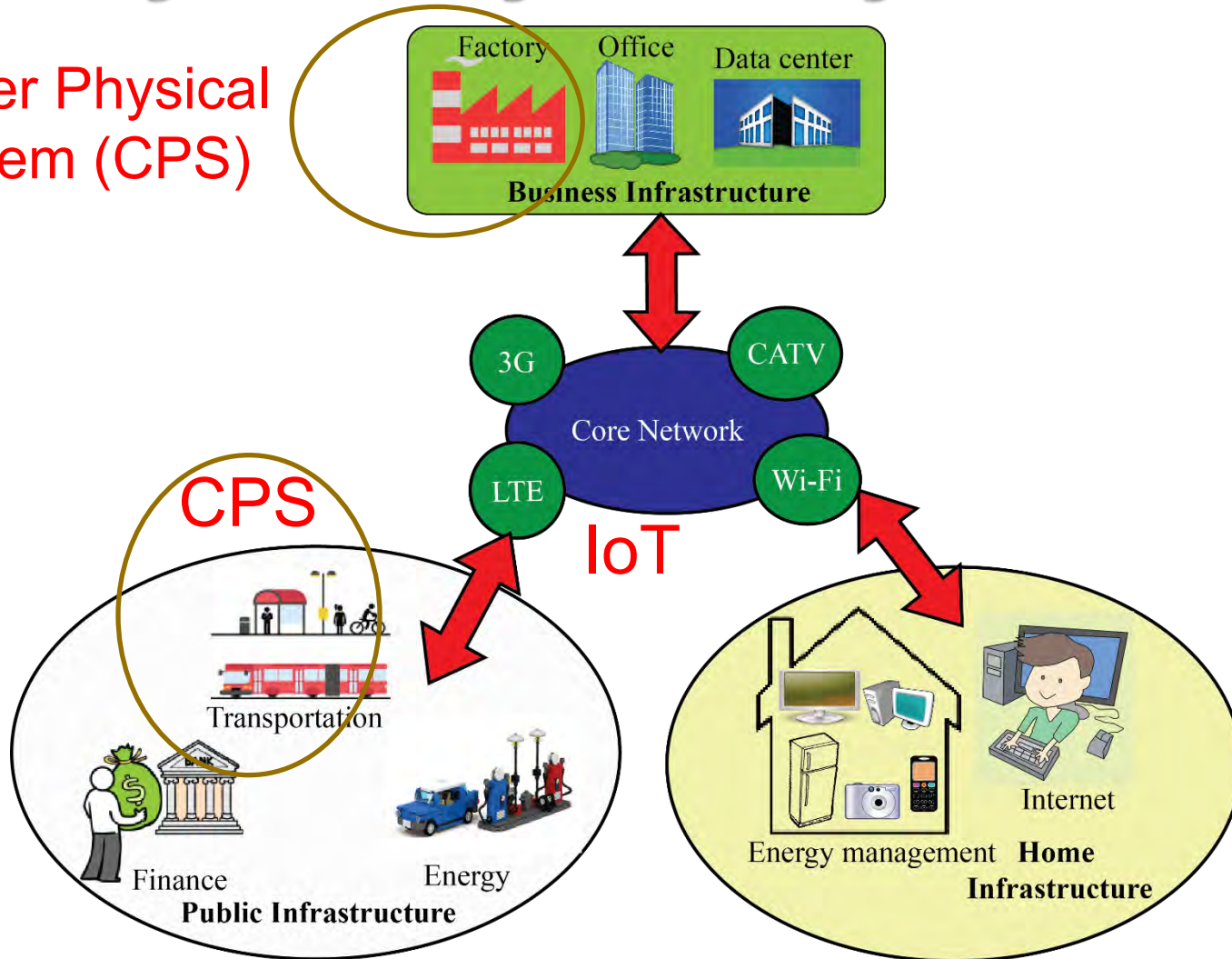
Fog: Intelligence - LAN, Processing - fog node or IoT gateway.

Source: <https://www.automationworld.com/fog-computing-vs-edge-computing-whats-difference>

Source: <https://www.nebbiolo.tech/wp-content/uploads/whitepaper-fog-vs-edge.pdf>

IoT Vs Cyber Physical Systems (CPS)

Cyber Physical
System (CPS)



Source: Mohanty 2016, CE Magazine July 2016

Internet of Every Things (IoE)

People
Connecting
people in
more relevant,
valuable ways



Process

Delivering the right
information to the right
person (or machine)
at the right time



Things

Physical devices and
objects connected to the
Internet and each other
for intelligent decision
making; often called
Internet of Things (IoT)



IoE

Source: http://iot.ieee.org/images/files/pdf/IEEE_IoT_Towards_Definition_Internet_of_Things_Revision1_27MAY15.pdf

Conclusions



Conclusions

- IoT has following components: Things, LAN, Cloud, Internet.
- IoT is backbone of smart cities.
- Scalability, Cost, Energy-consumption, Security are some important challenges of IoT.
- Security, Privacy, and Ownership Rights are critical for trustworthy IoT design.
- Physical Unclonable Functions (PUF) emerging as a good security solution.
- Coordination among the various researchers and design engineers is a challenge as IoT is multidisciplinary.

Future Directions

- Energy-Efficient “Thing” design is needed.
- Security and Privacy of Information need more research.
- Security of the CE systems (e.g. UAV, Smart Cars) needs research.
- Safer and efficient battery need research.
- IoT automatic design tool needs research.
- Some IoT simulators exist, but more needed for efficient, accurate, scalable, multi-discipline simulations.



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Hardwares are the drivers of the civilization, even softwares need them.

Thank You !!!

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